

5G RAN

Channel Management Feature Parameter Description

Issue	Draft B
Date	2026-01-30



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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Removed the impact relationship between "outer-loop robustness improvement for PDCCH aggregation level adaptation" and "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" from 5.3.2.3 Function Impacts because the Smooth Video Interaction feature is not recommended for commercial use in the current version.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added equivalent-CCE-usage-based PDCCH symbol adaptation. For details, see: • 5.1.2 PDCCH Time-Domain Resource Range • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Added the NRDUCellPdcchAlgo.PdcchEquivCceRatioThld parameter. Modified parameter: Added the EQUIV_CCE_RATIO_SYM_ADPT_SW option to the NRDUCellPdcch.PdcchAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added optimized PDCCH symbol adaptation. For details, see: • 5.1.2 PDCCH Time-Domain Resource Range • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the PDCCH_SYM_ADAPT_OPT_SW option to the NRDUCellPDCCH.PdcchAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added fast aggregation level reduction. For details, see: • 5.1.4.2.2 Scenario-based CCE Resource Configuration Optimization • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Added the NRDUCellPdcchAlgo.AggLvlFastDcrPwrThld parameter. Modified parameter: Added the AGG_LEVEL_FAST_DECREASE_SW option to the NRDUCellPdcch.PdcchAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added DTX filtering. For details, see: • 5.1.4.2.2 Scenario-based CCE Resource Configuration Optimization • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the DTX_FILTER_SW option to the NRDUCellPdcch.PdcchAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added CORESET 0 offset policy configuration. For details, see: • 5.1.4.2.2 Scenario-based CCE Resource Configuration Optimization • 5.2.2 Impacts • 5.3 Requirements • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Added the NRDUCellPdcch.Algo.Coreset0OffsetPol parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added multi-BWP1 ratio statistics optimization. For details, see: • 5.1.2 PDCCH Time-Domain Resource Range • 5.2.2 Impacts • 5.3.2 Functions • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the MULTI_BWP1_RATIO_STAT_OPT_SW option to the NRDUCellPdcch.PdcchAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Removed the cell bandwidth restriction on format-1 SR code channel allocation optimization from: • 6.1.3.3 Format-1 SR Code Channel Allocation Optimization (Low-Frequency TDD) • 6.4.1.1 Data Preparation	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added refined control of format-1 resources. For details, see: • 6.1.3.11 Refined Control of Format-1 Resources • 6.2.2 Impacts • 6.4.1 Data Configuration	Modified parameter: Added the F1_RES_REFINE_D_CTRL_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added HARQ resource increase. For details, see: • 6.1.3.14 HARQ Resource Increase • 6.2.2 Impacts • 6.3.3 Hardware • 6.4.1 Data Configuration	Modified parameter: Added the HARQ_SPEC_INCREASE_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added maximized search within HARQ resource set 1. For details, see: • 6.1.3.15 Maximized Search Within HARQ Resource Set 1 • 6.2.2 Impacts • 6.4.1 Data Configuration	Modified parameter: Added the HARQ_RES_SET1_MAX_SRCH_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added PUCCH symbol length number increase. For details, see: • 6.1.3.16 PUCCH Symbol Length Number Increase • 6.2.2 Impacts • 6.4.1 Data Configuration	Modified parameter: Added the PUCCH_SYM_TYPE_NUM_INCR_SW option to the NRDUCellPucch.PucchAlgoSwitch parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added CSI measurement optimization. For details, see: • 7.1.2.1 CSI-RS for CM Resource Management • 7.2.2 Impacts • 7.4.1 Data Configuration	Modified parameter: Added the CSI_MEAS_OPT_SW option to the NRDUCellCsirs.CsRsExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "outer-loop robustness improvement for PDCCH aggregation level adaptation" and "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation". For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between "downlink CCE allocation without ratio constraint" and "PDCCH uplink-to-downlink CCE ratio adaptation". For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added support for high-frequency functions by LampSite base stations. For details, see: 5.3.3 Hardware 6.3.3 Hardware	None	High-frequency TDD	DBS5900 LampSite
Added a mutually exclusive relationship between "SRS bandwidth adaptation" and "PUCCH resource reduction for heavy-traffic UEs". For details, see 6.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "inter-TRP format-4 resource sharing for HARQ feedback" and "format-3 and format-4 ACK RB resource adaptation". For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added SRS resource adjustment for 4T UEs. For details, see: 8.1.1 Overview 8.3.2.3 Function Impacts 8.3.5 Others 8.4.1 Data Configuration	Modified parameter: Added the UL_4T_SRS_RES_ADJ_SW option to the NRDUCellSrs.SrsAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added impact relationships with RF channel intelligent shutdown. For details, see 5.3.2.3 Function Impacts and 7.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added mmWave PDCCH coverage enhancement. For details, see: 5.1.4.2.2 Scenario-based CCE Resource Configuration Optimization 5.2.2 Impacts 5.3.2 Functions 5.4.1 Data Configuration	Modified parameter: Added the MMW_PDCCH_COVERAGE_ENH_SW option to the NRDUCellPdcch.PdcchAlgoExtSwitch parameter.	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Removed the mutually exclusive relationship between quadruple code division for format-4 resource allocation and High-speed Railway Superior Experience from: • 6.1.3.2 Quadruple Code Division for Format-4 Resource Allocation (Low-Frequency TDD) • 6.3.2.2 Mutually Exclusive Functions • 6.4.1.1 Data Preparation	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
 - Added descriptions of a prerequisite function of UE-specific CSI-RS reconfiguration reduction in low frequency bands. For details, see [7.3.2.1 Prerequisite Functions](#).
 - Added descriptions of the differences between high and low frequency bands for some functions. For details, see [2.3 Differences](#).
 - Added descriptions of the principles of setting the **NRDUCellPdcch.MaxCommonPdcchAggLvl** parameter (specifying the maximum aggregation level for the common PDCCH) to different values. For details, see [5.1.4.3 CCE Resource Configuration for the Common PDCCH](#).
 - Revised descriptions of the mutually exclusive relationship between "3CC aggregation" and "SRS interference coordination based on self-contained and uplink slots" in low-frequency TDD. For details, see [8.3.2.2 Mutually Exclusive Functions](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010009	Channel Management	4 PBCH Channel Management
FBFD-010009	Channel Management	5 PDCCH Channel Management
FBFD-010009	Channel Management	6 PUCCH Channel Management
FBFD-010009	Channel Management	7 CSI-RS Resource Management
FBFD-010009	Channel Management	8 SRS Resource Management

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
PBCH channel management	None	4 PBCH Channel Management
PDCCH channel management	Inward frequency shifting of the common PUCCH is supported only in SA networking. Other functions have no differences between NSA and SA.	5 PDCCH Channel Management
PUCCH channel management	None	6 PUCCH Channel Management
CSI-RS resource management	None	7 CSI-RS Resource Management
SRS resource management	None	8 SRS Resource Management

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
PBCH channel management	Supported in both low and high frequency bands, with the following differences: • Configuration of the time-domain position in the single SSB beam case is supported only in low frequency bands. • The supported time-domain SSB patterns vary between low and high frequency bands.	4 PBCH Channel Management

Function Name	Difference	Chapter/Section
PDCCH channel management	Supported in both low and high frequency bands, with the following differences: • In high frequency bands, the PDCCH always spreads over one or two symbols, or alternatively two symbols in some slots and one symbol in others. In low frequency bands, the PDCCH spreads over one, two, or three symbols. • Configuration of the maximum number of symbols allowed for PDCCH symbol number adaptation is supported only in low frequency bands. • PDCCH symbol number adaptation for multiple BWP1s is supported only in low frequency bands. • The multi-BWP1 PDCCH transmission algorithm is supported only in low frequency bands. • The enhanced multi-BWP1 PDCCH transmission algorithm is supported only in low frequency bands. • Configuration of the PDCCH aggregation level adaptation policy is supported only in low frequency bands. • PDCCH resource allocation enhancement is supported only in low frequency bands. • Aggregation level optimization based on PUSCH NACK/DTX is supported only in low frequency bands. • The PDCCH multi-symbol resource pool is supported only in low frequency bands. • Configuration of multiple PDCCH search spaces is supported only in low frequency bands. • PDCCH blind detection assignment optimization is	5 PDCCH Channel Management

Function Name	Difference	Chapter/Section
	supported only in low frequency bands. - Outer-loop robustness improvement for PDCCH aggregation level adaptation is supported only in low frequency bands. - The maximum CCE aggregation level for the UE-specific PDCCH is supported only in low frequency bands. - Transition from RRC reestablishment to RRC setup is supported only in low frequency bands. - The period for reserving PDCCH CCEs for uplink scheduling information is supported only in low frequency bands. - PDCCH uplink and downlink common CCE sharing is supported only in low frequency bands. - PDCCH antenna difference optimization is supported only in high frequency bands. - PDCCH uplink and downlink CCE sharing is supported only in low frequency bands. - Upper limit of the uplink CCE proportion is supported only in low frequency bands. - PDCCH blind detection number optimization is supported only in low frequency bands. - Centralized PDCCH CCE resource allocation is supported only in low frequency bands. - PDCCH CCE ratio adjustment in common slots is supported only in low frequency bands. - The maximum adjustment step for the PDCCH uplink CCE proportion is supported only in low frequency bands. - CCE resource optimization for low-priority scheduling based on	

Function Name	Difference	Chapter/Section
	uplink common DCI is supported only in low frequency bands. • Multi-BWP1 CCE allocation for top-priority UEs is supported only in low frequency bands. • mmWave PDCCH coverage enhancement is supported only in high frequency bands. • Multi-BWP1 ratio statistics optimization is supported only in low frequency bands. • Equivalent-CCE-usage-based PDCCH symbol adaptation is supported only in low frequency bands. • Optimized PDCCH symbol adaptation is supported only in low frequency bands. • Fast aggregation level reduction is supported only in low frequency bands. • DTX filtering is supported only in low frequency bands. • CORESET 0 offset policy configuration is supported only in low frequency bands.	

Function Name	Difference	Chapter/Section
Physical uplink control channel (PUCCH) channel management	Supported in both low and high frequency bands, with the following differences: • Cell-level PUCCH time-frequency formats include short formats and long formats. Both short formats and long formats are supported in low frequency bands, whereas only short formats are supported in high frequency bands. • PUCCH symbol number adaptation is supported only in high frequency bands. • PUCCH resource reduction for heavy-traffic UEs is supported only in high frequency bands. • Long PUCCH RB adaptation is supported only in low frequency bands. • Configuration of the cell-level long PUCCH RB number is supported only in low frequency bands. • Changing the RB number range for format-4 RB adaptation is supported only in low frequency bands. • Quadruple code division for format-4 resource allocation is supported only in low frequency bands. • Format-1 SR code channel allocation optimization is supported only in low frequency bands. • Format-1 HARQ-ACK resource scheduling in RB saving mode is supported only in low frequency bands. • Enhanced SR period adaptation is supported only in low frequency bands. • Configuration of the minimum SR period for adaptation is supported only in low frequency bands.	6 PUCCH Channel Management

Function Name	Difference	Chapter/Section
	• Configuration of the CSI reporting period is supported only in low frequency bands. • CSI reporting period adaptation is supported in low frequency bands, but not in high frequency bands. • PUCCH MRC/IRC adaptation is supported only in low frequency bands. • Format-4 PUCCH capability adaptation is supported only in low frequency bands. • Optimal resource utilization for PUCCH format 3 is supported only in low frequency bands. • Frequency-domain interference optimization for PUCCH format 1 is supported only in low frequency bands. • PUCCH RB adaptation is supported only in low frequency bands. • Format-1 SR code channel allocation in RB saving mode is supported only in low frequency bands. • Format-1 HARQ-ACK resource scheduling in RB saving mode is supported only in low frequency bands. • PUCCH resource reduction for heavy-traffic UEs in single-beam scenarios is supported only in high frequency bands. • PUCCH format-1 RB resource control is supported only in low frequency bands. • Inter-TRP RB resource sharing is supported only in low frequency bands. • PUCCH RB adaptation optimization is supported only in low frequency bands. • Format-3 and format-4 ACK RB resource adaptation is supported only in low frequency bands.	

Function Name	Difference	Chapter/Section
	• Inward frequency shifting of the common PUCCH is supported only in low frequency bands. • Disabling frequency hopping for format-4 resources is supported only in low frequency bands. • Refined control of format-1 resources is supported only in low frequency bands. • HARQ resource increase is supported only in low frequency bands. • Maximized search within HARQ resource set 1 is supported only in low frequency bands.	

Function Name	Difference	Chapter/Section
Channel state information-reference signal (CSI-RS) resource management	Supported in both low and high frequency bands, with the following differences: • Periodic and aperiodic CSI-RS for CM are supported in low frequency bands, whereas only aperiodic CSI-RS for CM is supported in high frequency bands. • Configuration of the maximum port number used for CSI-RS for CM is supported only in low frequency bands. • Port number adaptation for CSI-RS for CM is supported only in low frequency bands. • Inter-cell CSI-RS for CM interference avoidance is supported only in low frequency bands. • Full-buffer UE-specific CSI-RS for CM is supported only in low frequency bands. • Optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration is supported only in low frequency bands. • UE capability adaptation based on configurable CSI-RS for CM resources is supported only in low frequency bands. • Adaptation of CSI-RS for CM capabilities in codebooks is supported only in low frequency bands. • CSI-RS for CM reuse 3 is supported only in low frequency bands. • Configuration of the CSI-RS for BM measurement period is supported only in high frequency bands. • Full-buffer UE-specific CSI-RS for BM is supported only in high frequency bands.	7 CSI-RS Resource Management

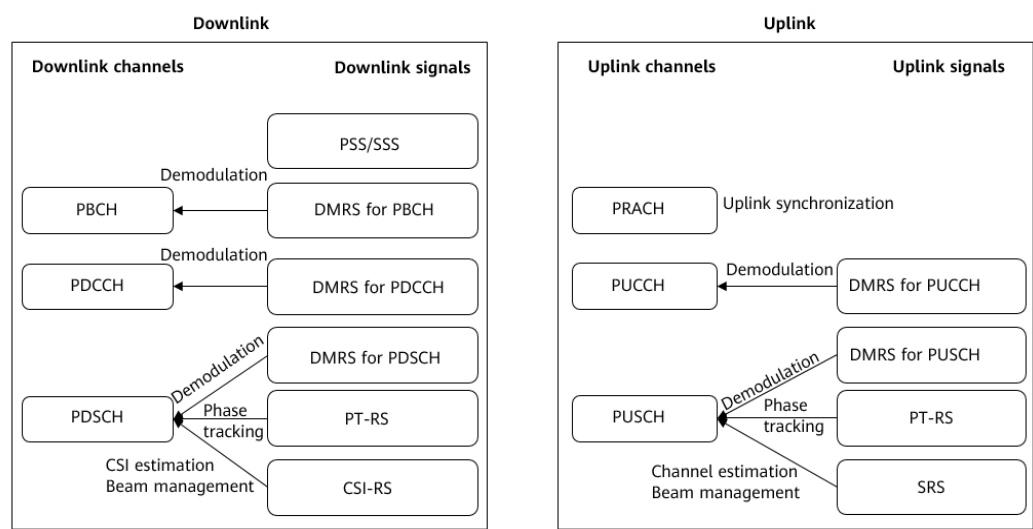
Function Name	Difference	Chapter/Section
	• Ignoring CSI-RS capabilities is supported only in high frequency bands. • UE-specific TRS resource management is supported only in high frequency bands. • QCL relationship management between TRS and SSB is supported only in low frequency bands. • QCL relationship management between TRS and periodic CSI-RS for CM is supported only in low frequency bands. • Interference measurement resource combination is supported only in low frequency bands. • CSI-RS resource allocation optimization is supported only in low frequency bands. • UE-specific CSI-RS reconfiguration reduction is supported only in low frequency bands. • CSI measurement optimization is supported only in low frequency bands.	

Function Name	Difference	Chapter/Section
SRS resource management	Supported in both low and high frequency bands, with the following differences: • The periodic mode is used in low frequency bands, whereas the aperiodic mode is used in high frequency bands. • Configuration of the transmission period for periodic SRS is supported only in low frequency bands. • SRS period adaptation is supported only in low frequency bands. • UE-characteristic-based SRS period adaptation is supported only in low frequency bands. • SRS interference coordination is supported only in low frequency bands. • SRS resource set adjustment is supported only in low frequency bands. • UE number increase for SRS is supported only in low frequency bands. • SRS transmission using common PUCCH resources is supported only in low frequency bands. • Coherent-codebook-based scheduling is supported only in low frequency bands. • SRS resource adjustment for 4T UEs is supported only in low frequency bands.	8 SRS Resource Management

3 Overview

To maximize data volume, management of physical channel or signal resources offers proper resource configurations that minimize the consumption of physical channel or signal resources and ensure demodulation performance for different channels. [Figure 3-1](#) lists the physical channels and signals available in an NR system.

Figure 3-1 Physical channels and signals



[Table 3-1](#) describes the functions of each physical channel or signal.

Table 3-1 Functions of each physical channel or signal

Type	Channel or Signal	Function	Reference
Downlink physical channels and signals	Primary synchronization signal (PSS)/Secondary synchronization signal (SSS)	Used for time-frequency synchronization and cell search.	4 PBCH Channel Management
	PBCH	Used for system information broadcast.	
	Demodulation reference signal (DMRS) for PBCH	Used for PBCH coherent demodulation.	
	PDCCH	Used for transmitting downlink control signaling.	5 PDCCH Channel Management
	DMRS for PDCCH	Used for PDCCH coherent demodulation.	
	PDSCH	Used for transmitting downlink user data.	For details, see <i>Downlink Scheduling</i> .
	DMRS for PDSCH	Used for PDSCH coherent demodulation.	
	Phase-tracking reference signal (PT-RS)	Used for downlink phase noise tracking and compensation.	
	CSI-RS	Used for downlink channel state measurement, beam management, time-frequency tracking, and other functions.	7 CSI-RS Resource Management
Uplink physical channels and signals	PRACH	Used for transmitting random access requests of UEs.	For details, see <i>Random Access Control</i> .
	PUCCH	Used for transmitting uplink control signaling.	6 PUCCH Channel Management
	DMRS for PUCCH	Used for PUCCH coherent demodulation.	
	PUSCH	Used for transmitting uplink user data.	For details, see <i>Uplink Scheduling</i> .
	DMRS for PUSCH	Used for PUSCH coherent demodulation.	

Type	Channel or Signal	Function	Reference
	PT-RS	Used for uplink phase noise tracking and compensation.	
	SRS	Used for uplink channel state measurement, beam management, and other functions.	8 SRS Resource Management

This document describes only the time-frequency resource allocation schemes for physical channels and signals. For definitions, see chapters 6 "Uplink" and 7 "Downlink" in 3GPP TS 38.211 V15.7.0.

For details about power configurations of each physical channel or signal, see *Power Control*.

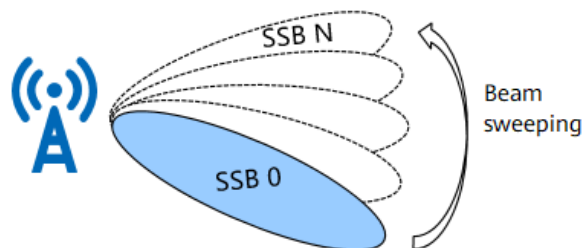
4 PBCH Channel Management

4.1 Principles

4.1.1 Overview

The NR system uses the multi-antenna beamforming technology to allow all types of channels and signals to use narrow beams with higher energy density and better directivity. Unlike a wide beam (such as an LTE beam), a narrow beam provides limited coverage and is unable to cover all UEs in a cell. As such, to cover all UEs in a cell, a beam sweeping mechanism is introduced in NR, which enables a base station to transmit beams in specific directions at different time to cover the entire cell. [Figure 4-1](#) illustrates how this mechanism works.

Figure 4-1 Beam sweeping implementation



For UEs to achieve downlink synchronization, the PSS, SSS, PBCH, and DMRS for PBCH are mandatory in each beam and transmission must be simultaneous. To ensure the simultaneous transmission of the PSS, SSS, PBCH, and DMRS for PBCH, they are packed together as a synchronization signal and PBCH block (SSB) in NR. Each beam that transmits an SSB is referred to as an SSB beam.

4.1.2 SSB Structure

As illustrated in [Figure 4-2](#), an SSB spreads over four consecutive symbols in the time domain and 20 RBs (240 subcarriers) in the frequency domain. The following occurs in the SSB:

- The PSS and SSS spread over symbols 0 and 2, respectively, in the time domain, and 127 subcarriers, separately, in the frequency domain.
- The PBCH (together with the DMRS for PBCH) spreads over symbols 1 and 3 in the time domain, and 240 subcarriers in the frequency domain. It also spreads over 48 subcarriers at both ends of symbol 2.

The master information block (MIB) is mapped to the REs of the PBCH and carried in an exclusive RRC message transmitted over the transport channel BCH. The BCH has a predefined transport format, which enables a UE to receive the MIB over the BCH to obtain information necessary for accessing the network without the need to obtain other information from the network. For details about MIB, see *5G Networking and Signaling*.

NOTE

For details about the PSS, SSS, PBCH, and DMRS for PBCH in an SSB, see section 7.4.3.1 "Time-frequency structure of an SS/PBCH block" in 3GPP TS 38.211 (Release 15).

Figure 4-2 SSB example

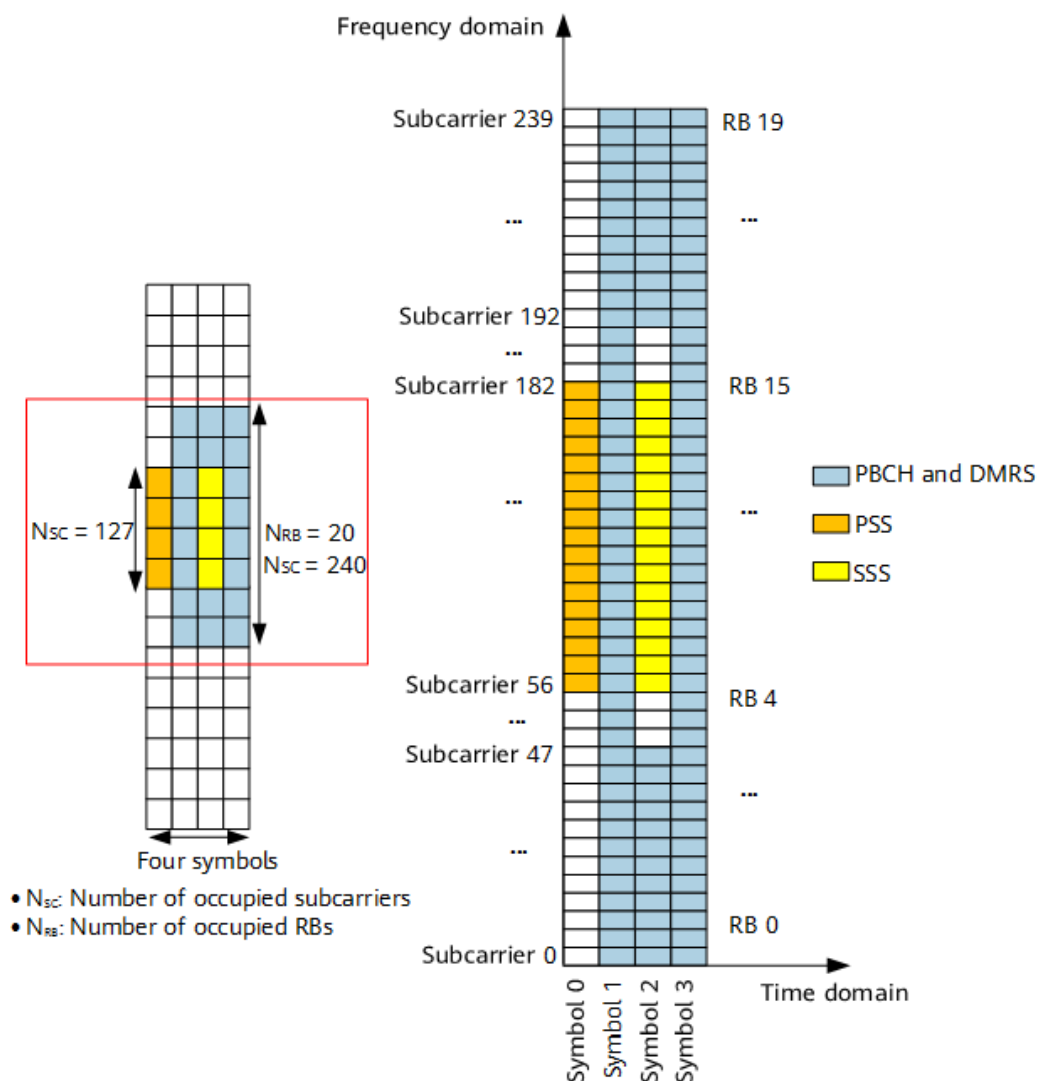


Table 4-1 OFDM symbols and subcarriers within an SSB for signals and the channel

Channel or Signal	OFDM Symbol Number	Subcarrier Number
PSS	0	56, 57, ..., 182
SSS	2	56, 57, ..., 182
Set to 0	0	0, 1, ..., 55, 183, 184, ..., 239
	2	48, 49, ..., 55, 183, 184, ..., 191
PBCH	1, 3	0, 1, ..., 239
	2	0, 1, ..., 47 192, 193, ..., 239
DMRS for PBCH	1, 3	0+v, 4+v, 8+v, ..., 236+v
	2	0+v, 4+v, 8+v, ..., 44+v 192+v, 196+v, ..., 236+v
Note: "v" indicates the value returned by cell PCI mod 4. For more information, see section 7.4.3.1 "Time-frequency structure of an SS/PBCH block" in 3GPP TS 38.211 V15.7.0.		

4.1.3 SSB Frequency-Domain Position

NR allows for flexible configurations of the SSB's frequency-domain position. The **NRDUCell.SsbFreqPos** parameter specifies the SSB's frequency-domain position in a cell, and the **NRDUCell.SsbDescMethod** parameter specifies the description method of the SSB's frequency-domain position in a cell. For details, see *Cell Management*.

4.1.4 SSB Time-Domain Position

NR introduces the following time-domain patterns of SSB transmissions: case A, case B, case C, case D, and case E, which suit different subcarrier spacing (SCS) values and frequency bands. For details, see section 4.1 "Cell search" in 3GPP TS 38.213 V15.12.0. Case A, case B, and case C are supported in low-frequency TDD. Case D and case E are supported in high-frequency TDD. The maximum number of SSBs and positions of the start symbols vary among the patterns, as shown in [Table 4-2](#).

In the current version: Case A and case C are supported in low-frequency TDD. Case D and case E are supported in high-frequency TDD. For details about the SCS values supported in the current version, see *Cell Management*.

Table 4-2 Time-domain SSB pattern

Pattern	SCS	Start Symbol Positions of SSBs	Frequency Within FR1 (≤ 3 GHz)		Frequency Within FR1 (> 3 GHz)		Frequency Within FR2	
			n	Maximum Number of SSBs	n	Maximum Number of SSBs	n	Maximum Number of SSBs
Case A	15 kHz	$\{2, 8\} + 14 \times n$	0, 1	4	0, 1, 2, 3	8	N/A	N/A
Case B	30 kHz	$\{4, 8, 16, 20\} + 28 \times n$	0	4	0, 1	8	N/A	N/A
Case C	30 kHz	$\{2, 8\} + 14 \times n$	$F < 1.88$ GHz: 0, 1 $F \geq 1.88$ GHz: 0, 1, 2, 3	$F < 1.88$ GHz: 4 $F \geq 1.88$ GHz: 8	0, 1, 2, 3	8	N/A	N/A
Case D	120 kHz	$\{4, 8, 16, 20\} + 28 \times n$	N/A	N/A	N/A	N/A	0, 1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17, 18	64
Case E	240 kHz	$\{8, 12, 16, 20, 32, 36, 40, 44\} + 56 \times n$	N/A	N/A	N/A	N/A	0, 1, 2, 3, 5, 6, 7, 8	64
Note 1: The maximum number of SSBs listed in this table is defined in 3GPP specifications. The actual number of transmitted SSBs can be less than or equal to the 3GPP-defined maximum number, or even only one SSB can be configured. If any SSB symbol positions defined in this table are unoccupied, they can be used for PDSCH transmission. Note 2: "F" in this table indicates working frequencies.								

For case A and case C supported in low-frequency TDD, when there is only one SSB, its time-domain position can be configured using the **NRDUCell.SsbTimePos** parameter.

- When this parameter is set to **SSB0**, SSB 0 conveyed in the corresponding case is transmitted.

- When this parameter is set to **SSB1**, SSB 1 conveyed in the corresponding case is transmitted.
- When this parameter is set to **DEFAULT**, SSB 0 conveyed in the corresponding case is transmitted.

Table 4-3 lists the maximum number of SSBs on different working frequencies.

Table 4-3 Maximum number of SSBs in different frequency bands

Frequency Band	Working Frequency (F)	Maximum Number of SSBs
FR1	$410\text{ MHz} \leq F \leq 3\text{ GHz}$	TDD ($F < 1.88\text{ GHz}$): 4 TDD ($F \geq 1.88\text{ GHz}$): 8
	$3\text{ GHz} < F \leq 7.125\text{ GHz}$	TDD: 8
FR2	$24.250\text{ GHz} \leq F \leq 52.600\text{ GHz}$	TDD: 64
Note: For details about the maximum number of SSBs in each frequency band, see section 4.1 "Cell search" in 3GPP TS 38.213 V15.12.0. Currently, TDD FR1 does not support working frequencies less than 2.3 GHz.		

4.1.5 SSB Beam

To facilitate reception on UEs and extend the coverage of a single SSB, beamforming is performed for the transmission of each SSB in NR. Each SSB has an SSB index, and each SSB beam corresponds to an SSB index. During the application of SSB beam sweeping, different SSBs are sequentially transmitted on separate beams in a period, resulting in varying transmission time between SSBs. SSB beam sweeping ensures that an entire cell is covered.

- Number of SSB beams
Each SSB has a unique number, known as an SSB index. In low frequency bands, the index is directly obtained from the PBCH DMRS. In high frequency bands, the least and the most significant three bits of the index are obtained from the PBCH DMRS and MIB, respectively.
The actual number of SSB beams (each corresponding to an SSB index) transmitted in a cell depends on configurations such as slot configuration and scenario-specific beam configurations. The number must be less than or equal to the maximum number of SSBs defined in 3GPP specifications. SIB1 or RRC signaling indicates which positions have no SSB transmissions and the PDSCH data can be transmitted in the corresponding vacant positions. For more information on slot configuration, see *Standards Compliance*. For more information on scenario-specific beams, see *Beam Management* in the case of low frequency bands and *mmWave Beam Management (High-Frequency TDD)* in the case of high frequency bands.

- SSB beam sweeping period

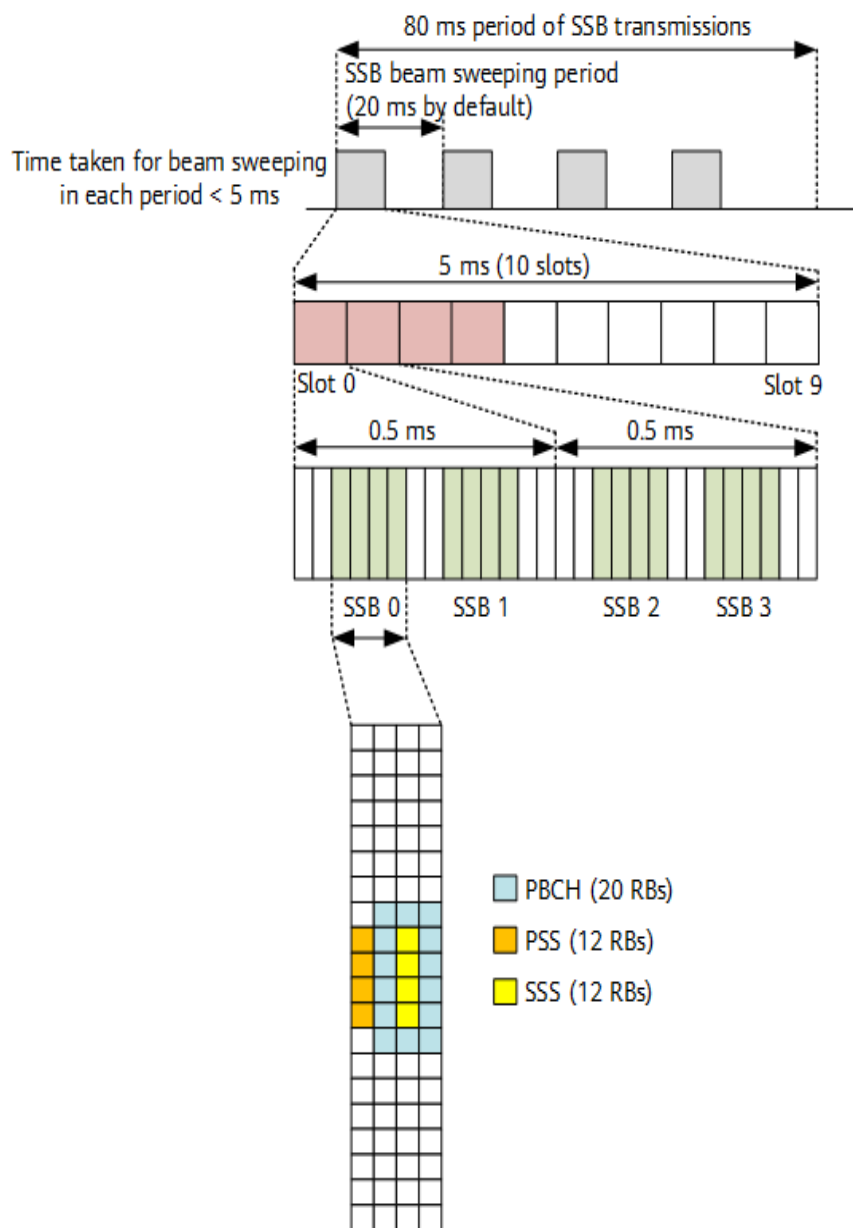
An 80 ms period of SSB transmission is applied. Within each 80 ms period, the gNodeB repeats SSB beam sweeping over the air interface in a specific case (SSB pattern). The SSB beam sweeping period can be specified by the **NRDUCell.SsbPeriod** parameter. The default SSB beam sweeping period is 20 ms, allowing for four repetitions of SSB beam sweeping within 80 ms. Each round of beam sweeping is completed within 5 ms. [Figure 4-3](#) shows an example SSB time sequence diagram.

According to 3GPP TS 38.213 (Release 15), UEs expect the gNodeB to perform SSB beam sweeping with a period of 20 ms during initial cell selection. If the actual SSB beam sweeping period is longer than 20 ms, the average duration for UEs to detect SSB may be prolonged, depending on SSB beam detection policies used by UEs.

In NR TDD low frequency bands, if the **NRDUCell.SsbPeriod** parameter (specifying the SSB beam sweeping period) is set to **MS5** and the **NRDUCell.SlotStructure** parameter (specifying the slot structure) for 4:1 slot configuration (4_1_DDDSU) is set to a value other than **SS1**, an SSB and SIB1 may be scheduled in the same slot. As a result, SIB1 scheduling may fail due to insufficient downlink resources. In addition, if the **NRDUCell.SsbPeriod** parameter is set to **MS5** or **MS10**, the time-domain positions of the SSB and other system information (OSI) may conflict with each other. As a result, OSI scheduling fails and UEs in the cell cannot receive system information.

In high-frequency TDD NSA networking, if the primary cell (PCell) of an NSA UE is an LTE FDD cell using frequency synchronization and the SSB beam sweeping period is longer than 20 ms, the UE may fail to detect any SSB and consequently fails to access the network. As such, you are advised to set the **NRDUCell.SsbPeriod** parameter to a value less than or equal to 20 ms in such a scenario.

Figure 4-3 Example SSB time sequence diagram (case C for 3.5 GHz as an example)



4.2 Network Analysis

4.2.1 Benefits

PBCH channel management improves network performance by properly utilizing PBCH resources.

Configuring the SSB time-domain position based on PCI modulo 3 can reduce the interference from neighboring cells on the SSB, increase the SSB signal to interference plus noise ratio (SINR) of the serving cell, and improve the UE access performance.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

When the SSBs of the local and neighboring cells are configured at the same time-domain position in the same slot, the UE access success rate decreases, the interference from SSBs to the PDSCH decreases, and the user-perceived throughput increases.

When the SSBs of the local and neighboring cells are configured at different time-domain positions in the same slot and the **NRDUCell.SsbTimePos** parameter is set to a value other than **DEFAULT** for the local cell, the duration UEs camp on 5G network increases. However, the number of cell edge users (CEUs) or number of UEs in the overlapping areas increase in the local cell, which may lead to deterioration of some counters related to access, handover, service drop, data transmission, and VoNR.

When the SSBs of the local and neighboring cells are configured at different time-domain positions in different slots and the **NRDUCell.SsbTimePos** parameter is set to a value other than **DEFAULT** for the local cell, the duration UEs camp on 5G network increases. However, the number of CEUs or number of UEs in the overlapping areas increase in the local cell, which may lead to deterioration of some counters related to access, handover, service drop, data transmission, and VoNR.

When the **NRDUCell.SsbTimePos** parameter is set to **SSB0_MOD3**, **SSB1_MOD3**, **SSB2_MOD3**, or **SSB_PCI_MOD3_STAGGER** and enhanced intra-frequency SSB interference avoidance (controlled by the **INTRA_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch.CommChnlIntrfAvoidSwitch** parameter) is disabled, transmission of the Remaining Minimum System Information (RMSI) of neighboring cells causes interference to PDSCH transmission of the local cell. For a UE experiencing strong interference from neighboring cells (for example, at the cell edge), the average downlink UE throughput (**User Downlink Average Throughput (DU)**) will decrease. For details about enhanced intra-frequency SSB interference avoidance, see *3D Networking Experience Improvement*.

4.3 Requirements

4.3.1 Licenses

This basic function is not under license control.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	SSB beam sweeping period	NRDUCell.SsbPeriod	<i>Channel Management</i>	- When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected) and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT160, the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) can only be set to MS20. - When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected) and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT320, the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) can only be set to MS20 or MS40. - When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch

RAT	Function Name	Function Switch	Reference	Description
				<i>h</i> parameter selected) and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT640 , the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) cannot be set to MS160 .

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	R16 posSIB	gNBSibConfig.SibType set to SIB_TYPE_POS SIB	None	When the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) is set to MS5 or MS10 , the time-domain positions of SSBs and positioning SIBs (posSIBs) may conflict, leading to posSIB transmission failures.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Except for the RAT, the sub-functions of this function have no other requirements on main control boards and baseband processing units. All main control boards

and baseband processing units of the corresponding RAT support these sub-functions. For details about the RATs supported by different main control boards and baseband processing units, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Except for the RAT, the sub-functions of this function have no other requirements on RF modules. All RF modules of the corresponding RAT support these sub-functions. For details about the RATs supported by different RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 4-4](#) describes the parameters used for function optimization.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
SSB Period	NRDUCell.SsbPeriod	Retain the default value. A shorter SSB beam sweeping period results in more overheads. An SSB beam sweeping period longer than 20 ms increases access delay.

Parameter Name	Parameter ID	Setting Notes
SSB Time Position	NRDUCell.SsbTimePos	Retain the default value. When the SSBs of the local and neighboring cells are configured at different time-domain positions, the SSB SINRs of these cells increase, the number of UEs in overlapping areas and at cell edges increases, the overlapping coverage rate and the number of UEs in these cells increase, and the UE access and handover success rates decrease. In addition, staggered SSB time-domain positions among cells increase the interference to the PDSCHs for neighboring cells and decrease the user-perceived throughput.

 NOTE

For details about data configuration of the SSB's frequency-domain position, see *Cell Management*.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

Low-frequency TDD command examples

```
//Setting the SSB period  
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20;
```

```
//Configuring the time-domain position in the single SSB beam case  
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, SsbTimePos=SSB0;
```

High-frequency TDD command examples

```
//Setting the SSB period  
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This basic function takes effect by default and does not require activation verification.

4.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

5 PDCCH Channel Management

5.1 Principles

5.1.1 Overview

The PDCCH transmits the following three types of downlink control information (DCI):

- Downlink grants
These include information such as the PDSCH resource indication, modulation and coding scheme (MCS), and HARQ process. DCI formats 1_0 and 1_1 are supported.
- Uplink grants
These include information such as the PUSCH resource indication and MCS. DCI formats 0_0 and 0_1 are supported.
- Transmit power control (TPC) commands
These include the TPC commands for the PUSCH for a group of UEs, supplementing the TPC commands for the PUSCH and PUCCH in uplink grants. DCI format 2_2 is supported.

NOTE

For more information on DCI formats, see section 7.3 "Downlink control information" in 3GPP TS 38.212 (Release 15).

The PDCCH time-frequency resource range, control channel element (CCE) resource configuration, and other PDCCH-related configurations will be described in the following sections.

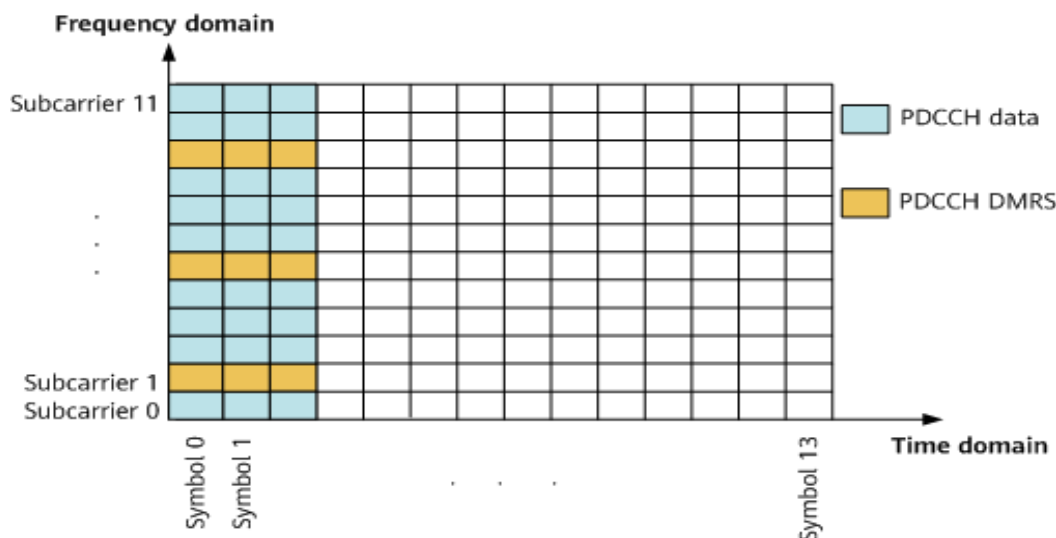
5.1.2 PDCCH Time-Domain Resource Range

Each slot is 14 symbols (numbered 0 to 13) long. In accordance with NR protocols, the PDCCH of a cell spreads over the first one to three symbols of a slot, as shown in [Figure 5-1](#), where each cell in the grid represents an RE. In the current version:

NR TDD allows the PDCCH to spread over one, two, or three symbols in low frequency bands.

NR TDD allows the PDCCH to always spread over one or two symbols, or alternatively two symbols in some slots and one symbol in others in high frequency bands.

Figure 5-1 Position of the PDCCH



PDCCH Symbol Number Configuration

The number of PDCCH symbols can be either statically or adaptively configured.

- Static configuration: Set the **NRDUCellPdcch.OccupiedSymbolNum** parameter to specify the number of occupied symbols.
- Adaptive configuration: Select the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter to enable PDCCH symbol number adaptation. Once this function is enabled, the number of PDCCH symbols is adaptively adjusted based on required CCEs. All UEs are then notified of the new number of symbols through an RRCReconfiguration message. This type of adjustment occurs in the following scenarios:
 - When the PDCCH requires a small number of CCEs, the number of PDCCH symbols decreases, and spared time-frequency resources are allocated to the PDSCH.
 - When the PDCCH requires a large number of CCEs, the number of PDCCH symbols increases until the number reaches the maximum value allowed in static configuration.

When PDCCH symbol number adaptation is enabled in low frequency bands:

- The maximum number of symbols allowed for PDCCH symbol number adaptation can be specified by the **NRDUCellPdcch.UePdcchAdaptMaxSymNum** parameter, which must be set to **3SYM** if the **NRDUCellPdcch.OccupiedSymbolNum** parameter is set to **3SYM**.
- The multi-BWP1 PDCCH transmission algorithm can be enabled by selecting the **PDCCH_MULTI_BWP1_ALGO_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. This algorithm adaptively adjusts the number of PDCCH symbols based on the CCE usages corresponding to individual BWP1s. To prevent UE data scheduling

failures in a cell serving a small number of UEs with limited service types (for example, only uplink or downlink services), multi-BWP1 ratio statistics optimization can be enabled by selecting the **MULTI_BWP1_RATIO_STAT_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function optimizes the method of collecting statistics for PDCCH CCE ratio adaptation in multi-BWP1 scenarios.

 NOTE

Multiple BWP1s are involved when UE bandwidth adaptation is enabled for a cell and the cell serves both UEs that support the cell bandwidth (using one BWP1 equal to the cell bandwidth) and UEs that do not support the cell bandwidth (using another BWP1 equal to the bandwidth specified by the **NRDUCellBwp.DlMinCarrierBw** parameter).

For more information on UE bandwidth adaptation, see *Scalable Bandwidth*.

- If CCE resources are insufficient, equivalent-CCE-usage-based PDCCH symbol adaptation can be enabled by selecting the **EQUIV_CCE_RATIO_SYM_ADPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter. This function adaptively adjusts the number of PDCCH symbols based on the equivalent CCE usage. If the equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted is greater than or equal to the equivalent PDCCH CCE usage threshold, PDCCH symbol adaptation is triggered to increase the number of PDCCH symbols. The equivalent PDCCH CCE usage threshold is specified by the **NRDUCellPdcchAlgo.PdcchEquivCceRatioThld** parameter, and the equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted is equal to **N.CCE.UL.Used.Equivalent** divided by **N.CCE.UL.Avail.Equivalent**.
- In extreme CCE allocation scenarios, it may be difficult to adaptively adjust the number of PDCCH symbols. In such cases, optimized PDCCH symbol adaptation can be enabled by selecting the **PDCCH_SYM_ADAPT_OPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter. This function optimizes the triggering conditions for PDCCH symbol number adaptation to increase the probability that PDCCH symbol number adaptation takes effect in preceding extreme scenarios under light load.

PDCCH Multi-Symbol Resource Pool (Low-Frequency TDD)

In low-frequency TDD, when the cell bandwidth ranges from 80 MHz to 100 MHz and the PDCCH occupies two symbols, PDCCH resource allocation currently allows each UE to use only one PDCCH symbol. The PDCCH multi-symbol resource pool allows each UE to be allocated two PDCCH symbols in a large-bandwidth system.

This function is controlled by the **MULTI_SYM_CORESET_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

 NOTE

The PDCCH multi-symbol resource pool is inapplicable to FWA scenarios.

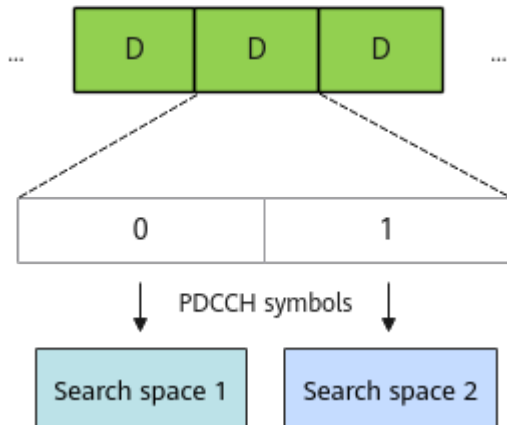
Configuration of Multiple PDCCH Search Spaces (Low-Frequency TDD)

In low-frequency TDD, if the cell-level PDCCH occupies two or more symbols but the UE-specific PDCCH occupies only one symbol, the CCE allocation success rate is low. To address this issue, configuration of multiple PDCCH search spaces is introduced and is controlled by the **MULTI_SEARCH_SPACE_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

When this function is enabled, the base station configures two search spaces for individual UEs, as shown in [Figure 5-2](#), which uses a cell-level PDCCH occupying two symbols as an example. If CCE resource conflicts occur in the first symbol, the base station attempts to allocate CCE resources in the second symbol to prevent a low CCE allocation success rate. A search space consists of a set of PDCCH candidates of the same CCE aggregation level in a control resource set (CORESET). For details about search space, see section 6.3.2 Radio resource control information elements in 3GPP TS 38.331 V15.5.1.

With configuration of multiple PDCCH search spaces in effect, if the assignment of blind detection times to each PDCCH CCE aggregation level is uneven, uplink and downlink CCE allocation may fail due to insufficient blind detection times. To address this issue, PDCCH blind detection assignment optimization is introduced. This function is controlled by the **PDCCH_BLIND_DET_ASSIGN_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. It can balance blind detection times across different PDCCH CCE aggregation levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times.

Figure 5-2 Configuration of multiple PDCCH search spaces



5.1.3 PDCCH Frequency-Domain Resource Range

When rate-matching-pattern-configuration-based PDCCH rate matching is disabled, each PDCCH symbol in a cell spreads over a full bandwidth.

When rate-matching-pattern-configuration-based PDCCH rate matching is enabled, the frequency-domain range occupied by each PDCCH symbol in a cell is specified by the **NRDUCellPdcch.OccupiedRbNum** parameter.

- When the value of the **NRDUCellPdcch.OccupiedRbNum** parameter is **0** or greater than or equal to the number of RBs corresponding to the downlink system bandwidth specified by the **NRDUCell.DlBandwidth** parameter, the PDCCH spreads over a full bandwidth.

- When the value of the **NRDUCellPdcch.OccupiedRbNum** parameter is greater than 0 and less than the number of RBs corresponding to the downlink system bandwidth specified by the **NRDUCell.DlBandwidth** parameter, the frequency-domain range occupied by the PDCCH is specified by the **NRDUCellPdcch.OccupiedRbNum** parameter.

The **NRDUCellPdcch.OccupiedRbNum** parameter needs to be configured based on UE distribution in a cell, to adapt to scheduling of uplink- and downlink-specific DCI and prevent access failures or service drops caused by scheduling failures. For details, see *Downlink Scheduling*.

NOTE

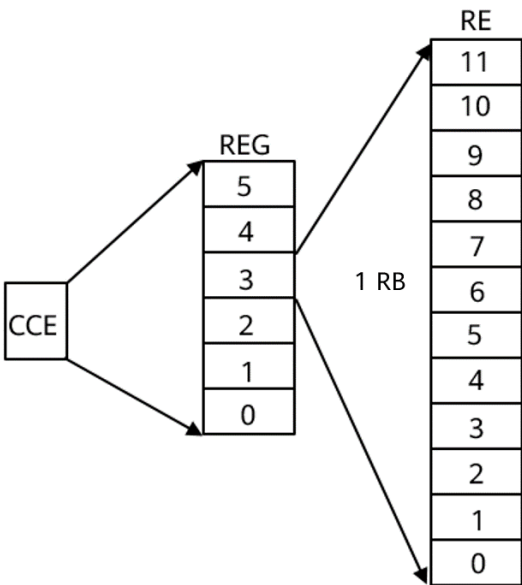
For more information on rate-matching-pattern-configuration-based PDCCH rate matching, see *Downlink Scheduling*.

5.1.4 PDCCH CCE Resource Configuration

5.1.4.1 Overview

A CCE is the smallest resource unit for PDCCH transmission. One CCE contains six resource element groups (REGs) and each REG corresponds to a single RB. [Figure 5-3](#) shows the relationships among the CCE, REG, and RE.

Figure 5-3 Relationships among the CCE, REG, and RE



The gNodeB allocates 1, 2, 4, 8, or 16 CCEs for the PDCCH. The number of CCEs allocated to the PDCCH corresponds to aggregation level 1, 2, 4, 8, or 16 stipulated in 3GPP specifications. [Table 5-1](#) lists the mapping between aggregation levels and the number of CCEs.

Table 5-1 Mapping between aggregation levels and the number of CCEs

Aggregation Level	Number of CCEs
1	1
2	2
4	4
8	8
16	16

CCE aggregation level 1 indicates that the PDCCH spreads over one CCE, while CCE aggregation level 2 indicates that the PDCCH spreads over two CCEs. The same rules apply to other aggregation levels.

- CCE aggregation level 16 provides the lowest bit rate and offers the best demodulation performance.
In most cases, as cell center users (CCUs) experience good signal quality, applying CCE aggregation level 16 to all UEs in a cell is a waste of PDCCH CCEs.
- CCE aggregation level 1 provides the highest bit rate and offers the worst demodulation performance.
Non-CCUs experience relatively poor signal quality. If CCE aggregation level 1 is applied to all UEs in a cell, the PDCCH may not be correctly demodulated for non-CCUs.

To maximize CCE resource utilization, Huawei gNodeBs support CCE resource and aggregation level adjustment, including CCE resource configuration for the UE-specific PDCCH and common PDCCH.

5.1.4.2 CCE Resource Configuration for the UE-Specific PDCCH

CCE resources for the UE-specific PDCCH are dynamically adjusted. The gNodeB dynamically selects an appropriate CCE aggregation level and allocates proper CCE resources based on the PDCCH SINR to enable the PDCCH block error rate (BLER) to converge on the target BLER specified by the **NRDUCellPdcch.PdcchBlrTarget** parameter.

CCE resource configuration for the UE-specific PDCCH includes SINR-based CCE resource configuration and scenario-based CCE resource configuration optimization.

5.1.4.2.1 SINR-based CCE Resource Configuration

The gNodeB estimates the PDCCH SINR based on information such as UE measurement results and cell load to select a CCE aggregation level. It also supports configuration of a PDCCH SINR offset for the estimated PDCCH SINR and dynamically adjusts the estimated PDCCH SINR based on information such as PUSCH discontinuous transmission (DTX) to obtain the most accurate PDCCH SINR, thereby selecting a more appropriate CCE aggregation level.

PDCCH SINR Estimation

The gNodeB estimates the PDCCH SINR and selects a CCE aggregation level in different UE access phases as follows:

- When a UE is initially accessing a cell and initial CCE aggregation level selection optimization is enabled (by selecting the **AGG_LVL_INIT_SELECT_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter), the gNodeB estimates the PDCCH SINR based on uplink measurement information and cell load information to select an appropriate CCE aggregation level for the UE. Under favorable channel conditions and heavy cell load, the CCE aggregation level will be lowered to increase the number of UEs allowed by PDCCH resources. When the **PDCCH_MULTI_BWP1_ALGO_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter is selected, initial CCE aggregation level selection optimization for multiple BWP1s is supported. When UE bandwidth adaptation is enabled, the gNodeB adjusts the CCE aggregation level based on the CCE usages corresponding to individual BWP1s.

NOTE

Multiple BWP1s are involved when UE bandwidth adaptation is enabled for a cell and the cell serves both UEs that support the cell bandwidth (using one BWP1 equal to the cell bandwidth) and UEs that do not support the cell bandwidth (using another BWP1 equal to the bandwidth specified by the **NRDUCellBwp.DlMinCarrierBw** parameter).

For details about UE bandwidth adaptation, see *Scalable Bandwidth*.

- After a UE initially accesses a cell, the gNodeB estimates the PDCCH SINR based on the channel quality indicator (CQI) information reported by the UE by default to select an appropriate CCE aggregation level for the UE. During PDCCH SINR estimation, a PDCCH aggregation level adaptation policy can be configured (by setting the **NRDUCellPdcch.PdcchAggLvlAdaptPol** parameter) to control whether the gNodeB considers information such as the MCS index for PDSCH scheduling when estimating the PDCCH SINR. If this parameter is set to **JOINT_ADAPT**, the gNodeB considers information such as the MCS index for PDSCH scheduling when estimating the PDCCH SINR. If this parameter is set to **SEPARATE_ADAPT**, the gNodeB does not consider information such as the MCS index for PDSCH scheduling when estimating the PDCCH SINR. In the current version, you are advised to set this parameter to **SEPARATE_ADAPT** because the channel characteristics of the PDCCH and PDSCH differ greatly.

PDCCH SINR Offset Configuration

The PDCCH SINR offset is specified by the **NRDUCellPdcch.IblerPdcchSinrOffset** parameter. This parameter should be configured based on the target PDSCH initial block error rate (IBLER) (specified by the **NRDUCellPdsch.DlTargetIbler** parameter) of a cell.

- When the target PDSCH IBLER of a cell is set to a value greater than 10%, it is recommended that the PDCCH SINR offset be set to a small value. A smaller value of the PDCCH SINR offset results in a higher CCE aggregation level being selected by the gNodeB.
- When the target PDSCH IBLER of a cell is set to a value less than 10%, it is recommended that the PDCCH SINR offset be set to a large value. A larger

value of the PDCCH SINR offset results in a lower CCE aggregation level being selected by the gNodeB.

- When the target PDSCH IBLER of a cell is set to 10%, it is recommended that the PDCCH SINR offset retains the default value, which has no impact on CCE aggregation level selection by the gNodeB.

 NOTE

The PDSCH IBLER (also known as downlink IBLER) of a cell is calculated using the following formula: Downlink IBLER = Number of TBs failing to be initially transmitted on the DL-SCH/ Number of TBs initially transmitted on the DL-SCH.

Dynamic PDCCH SINR Adjustment Based on PUSCH DTX

PUSCH-DTX-based aggregation level adaptation (controlled by the **PUSCH_DTX_AGG_LVL_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter) is supported to determine whether the gNodeB dynamically adjusts the estimated PDCCH SINR based on PUSCH DTX information.

- When the **PUSCH_DTX_AGG_LVL_ADAPT_SW** option is selected, PUSCH-DTX-based aggregation level adaptation is enabled. The gNodeB dynamically adjusts the estimated PDCCH SINR based on PUSCH DTX information to select a more appropriate CCE aggregation level.
- When the **PUSCH_DTX_AGG_LVL_ADAPT_SW** option is deselected, PUSCH-DTX-based aggregation level adaptation is disabled. The gNodeB does not dynamically adjust the estimated PDCCH SINR based on PUSCH DTX information.

5.1.4.2.2 Scenario-based CCE Resource Configuration Optimization

The gNodeB can select a more appropriate CCE aggregation level for UEs based on different scenarios.

Aggregation Level Optimization Based on PUSCH NACK/DTX (Low-Frequency TDD)

Aggregation level optimization based on PUSCH NACK/DTX (controlled by the **PUSCH_NACK_DTX_AGG_LVL_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter) is supported to determine whether the gNodeB adjusts the PDCCH aggregation level when PUSCH NACK/DTX confusions occur.

- When the **PUSCH_NACK_DTX_AGG_LVL_OPT_SW** option is selected, this function is enabled. The gNodeB uses information about PUSCH NACK/DTX confusions to adjust the aggregation level.
- When the **PUSCH_NACK_DTX_AGG_LVL_OPT_SW** option is deselected, this function is disabled. The gNodeB will not use information about PUSCH NACK/DTX confusions to adjust the aggregation level.

PDCCH Aggregation Level Optimization in the Case of Consecutive DTXs

PDCCH aggregation level optimization in the case of consecutive DTXs is controlled by the **PDCCH_AGG_LVL_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

- When the **PDCCH_AGG_LVL_OPT_SW** option is selected, PDCCH aggregation level optimization is enabled in the case of consecutive DTXs to slow down PDCCH aggregation level adjustment.
- When the **PDCCH_AGG_LVL_OPT_SW** option is deselected, PDCCH aggregation level optimization is disabled in the case of consecutive DTXs.

PDCCH Aggregation Level Optimization in the Case of Consecutive Non-DTXs

PDCCH aggregation level optimization in the case of consecutive non-DTXs is controlled by the **NRDUCellPdcch.NonDtxSinrIncreaseStep** parameter.

- When this parameter is set to a value other than **0**, PDCCH aggregation level optimization is enabled in the case of consecutive non-DTXs. The gNodeB increases the SINR of the PDCCH by the value of this parameter.
- When this parameter is set to **0**, PDCCH aggregation level optimization is disabled in the case of consecutive non-DTXs.

Outer-Loop Robustness Improvement for PDCCH Aggregation Level Adaptation (Low-Frequency TDD)

Outer-loop robustness improvement for PDCCH aggregation level adaptation (controlled by the **PUSCH_HARQ_DTX_AGG_LVL_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter) is supported. This option takes effect only when the **PUSCH_ACK_DTX_DETECT_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter is selected.

- When the **PUSCH_HARQ_DTX_AGG_LVL_OPT_SW** option is selected, this function is enabled to identify DTXs over the PUSCH after downlink scheduling. During downlink retransmissions, DTXs over the PUSCH can be used for outer-loop adjustment of the PDCCH aggregation level, accelerating aggregation level convergence and decreasing the downlink IBLER and residual block error rate (RBLER).
- When the **PUSCH_HARQ_DTX_AGG_LVL_OPT_SW** option is deselected, this function is disabled.

Maximum PDCCH CCE Aggregation Level (Low-Frequency TDD)

To reduce PDCCH CCE resource consumption for different services, a maximum CCE aggregation level for the UE-specific PDCCH can be configured using the **NRDUCellPdcchAlgo.PdcchMaxAggLevel** parameter, which allows the gNodeB to select a CCE aggregation level not exceeding the configured value.

- When this parameter is set to **AGG_LVL_16**, the maximum aggregation level is 16. This value is recommended when the KPIs related to accessibility, retainability, and mobility need to be improved.
- When this parameter is set to **AGG_LVL_8**, the maximum aggregation level is 8. This value is recommended when the success rate of PDCCH CCE allocation for uplink and downlink scheduling needs to be increased.
- When this parameter is set to **AGG_LVL_8_16_ADAPT**, the maximum aggregation level is adaptively switched between 8 and 16. When adjusting the PDCCH CCE aggregation level of a UE, the base station considers the CSI

report from the UE to determine whether the maximum aggregation level is 8 or 16. This value is recommended when the success rate of PDCCH CCE allocation for uplink and downlink scheduling as well as the KPIs related to accessibility, retainability, and mobility need to be considered.

- When this parameter is set to **AGG_LVL_4**, the maximum aggregation level is 4. This value is recommended when the channel quality of all UEs served by the base station is favorable, the CCE resource consumption for single UEs needs to be further reduced, and the success rate of PDCCH CCE allocation for uplink and downlink scheduling needs to be further increased. The channel quality can be determined by monitoring the ChMeas.Pwr.DuCell function subset.

PDCCH Resource Allocation Enhancement (Low-Frequency TDD)

PDCCH resource allocation enhancement is introduced to reduce power consumption in PDCCH resource allocation under heavy PDCCH CCE load. This function is controlled by the **PDCCH_RES_ALLOC_ENH_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. When this option is selected, the gNodeB adjusts the PDCCH transmit power based on the current PDCCH CCE load. This also affects CCE aggregation level selection for the UE-specific PDCCH. Under heavy PDCCH CCE load, the gNodeB preferentially selects a low CCE aggregation level and increases the PDCCH transmit power.

PDCCH CCE Ratio Adjustment in Common Slots (Low-Frequency TDD)

In low-frequency NR TDD, PDCCH CCE ratio adjustment in common slots is controlled by the **PDCCH_COMM_SLOT_RATIO_ADJ_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. With this function, the CCE ratio is adjusted in the slots with PDCCH common DCI. This increases the number of available downlink CCEs and the downlink CCE allocation success rate.

Transition from RRC Reestablishment to RRC Setup in Common PDCCH Congestion Scenarios (Low-Frequency TDD)

In common PDCCH congestion scenarios, transition from RRC reestablishment to RRC setup for UEs involved in RRC reestablishments is controlled by the **RRC_REESTAB_TO_SETUP_SW** option of the **NRCellAlgoSwitch.ProcessSwitch** parameter.

- If this option is selected, transition from RRC reestablishment to RRC setup is enabled, and the base station checks whether both the CCE usage and CCE allocation failure rate of the common PDCCH exceed the threshold specified by the **NRDUCellRac.RaCtrlCommPdcchCceUsageThd** parameter.
 - If they exceed the threshold, UEs involved in RRC reestablishments enter the RRC setup procedure. The gNodeB sends the RRC setup message to the UEs over the UE-specific PDCCH.
 - If they do not exceed the threshold, UEs involved in RRC reestablishments retain the original procedure. The gNodeB sends the RRC reestablishment message to the UEs over the common PDCCH.
- If this option is deselected, UEs involved in RRC reestablishments retain the original procedure. The gNodeB sends the RRC reestablishment message to the UEs over the UE-specific PDCCH.

mmWave PDCCH Coverage Enhancement (High-Frequency TDD)

If the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU** for a high-frequency TDD cell under weak coverage and light load, UEs will experience PDCCH miss-detection, resulting in limited PDCCH coverage. To address this issue, mmWave PDCCH coverage enhancement is introduced and is controlled by the **MMW_PDCCH_COVERAGE_ENH_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter. With this function enabled, the number of configured blind detection times for the maximum PDCCH CCE aggregation level of 8 is increased, enhancing PDCCH coverage performance.

Fast Aggregation Level Reduction

In low frequency bands, if CCE resources are insufficient, fast aggregation level reduction can be enabled by selecting the **AGG_LEVEL_FAST_DECREASE_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function quickly reduces the aggregation levels when the increase in the PDCCH power is less than or equal to the power threshold for fast aggregation level reduction. This allows lower aggregation levels to be selected for UEs in a cell, improving the PDCCH capacity of the cell.

The power threshold for fast aggregation level reduction is specified by the **NRDUCellPdcchAlgo.AggLvlFastDcrPwrThld** parameter, and the increase in the PDCCH power is determined by the PDCCH CCE aggregation level and PDCCH detection result.

DTX Filtering

In low frequency bands, if CCE resources are insufficient, DTX filtering can be enabled by selecting the **DTX_FILTER_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function excludes certain DTX occasions from aggregation level adjustment, allowing lower aggregation levels to be selected for UEs in a cell and improving the PDCCH capacity of the cell.

CORESET 0 Offset Policy Configuration

In low-frequency TDD, to flexibly configure the frequency-domain position of CORESET 0 relative to the SSB when uplink RBs are insufficient, the CORESET 0 offset policy can be configured by setting the **NRDUCellPdcchAlgo.Coreset0OffsetPol** parameter to **EDGE_FIRST**. With this parameter setting, the offset of the frequency-domain position of CORESET 0 relative to the center frequency of the SSB is adjusted to a protocol-defined value that makes CORESET 0 closest to the frequency-domain edge of the cell.

5.1.4.3 CCE Resource Configuration for the Common PDCCH

CCE resources for the common PDCCH are statically adjusted. The maximum aggregation level for the common PDCCH can be specified by the **NRDUCellPdcch.MaxCommonPdcchAggLvl** parameter.

- If the **NRDUCellPdcch.MaxCommonPdcchAggLvl** parameter is set to **AGGLVL4**, the CCE aggregation level used by the common PDCCH is equal to the parameter value.

- If the **NRDUCellPdcch.MaxCommonPdcchAggLvl** parameter is set to **AGGLVL8** or **AGGLVL16**, the CCE aggregation level used by the common PDCCH is less than or equal to the parameter value.

5.1.5 PDCCH Uplink and Downlink CCE Resource Allocation

PDCCH uplink and downlink CCE resource allocation includes CCE resource allocation for the UE-specific PDCCH and common PDCCH.

5.1.5.1 Uplink and Downlink CCE Resource Allocation for the UE-Specific PDCCH

For a slot with DCI transmissions for both uplink and downlink scheduling, the ratio of available CCEs for uplink scheduling information to those for downlink scheduling information can be statically or adaptively configured.

- Static configuration: Set the **NRDUCellPdcch.UlMaxCcePct** parameter.
 - A larger value of this parameter results in a higher proportion of available CCEs for uplink scheduling information and a lower proportion of available CCEs for downlink scheduling information, decreasing the chances of downlink scheduling and even causing service drops.
 - A smaller value of this parameter results in a lower proportion of available CCEs for uplink scheduling information and a higher proportion of available CCEs for downlink scheduling information, decreasing the chances of uplink scheduling and even causing service drops.
- Adaptive configuration: Select the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter to enable adaptive configuration. After this function is enabled, the ratio of available CCEs for uplink scheduling information to those for downlink scheduling information is adaptively adjusted based on factors such as the CCEs required for uplink and downlink scheduling information and CCE usage, and the setting of the **NRDUCellPdcch.UlMaxCcePct** parameter no longer takes effect.

When adaptive configuration is enabled, the restrictions on resource adjustment are as follows:

- In NR TDD, the adaptively configured resource ratio takes effect in each individual slot with DCI transmissions for both uplink and downlink scheduling.
- In low frequency bands, an upper limit of the uplink CCE proportion can be set using the **NRDUCellPdcch.UlCceRatioUpperLimit** parameter, which ensures that the adaptively configured uplink CCE proportion does not exceed the value of this parameter. In cells other than FDD 5 MHz cells, this parameter takes effect only when PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter) is enabled.
- In low-frequency NR TDD, a maximum adjustment step for the PDCCH uplink CCE proportion can be set using the **NRDUCellPdcch.UlCcePctMaxAdjStep** parameter, which ensures that each adjustment to the uplink CCE proportion does not exceed the value of this parameter.
- In low frequency bands, the multi-BWP1 PDCCH transmission algorithm can be enabled using the **PDCCH_MULTI_BWP1_ALGO_SW** option of the

NRDUCellPdcch.PdcchAlgoEnhSwitch parameter. This algorithm supports PDCCH uplink-to-downlink CCE ratio adaptation for multiple BWP1s, which adaptively adjusts the ratio of available CCEs for uplink scheduling information to those for downlink scheduling information based on the factors, such as the CCEs required for uplink and downlink scheduling information and CCE usages, corresponding to individual BWP1s. This function increases the access success rate of small-bandwidth UEs but may decrease the CCE allocation success rate for CEUs.

 NOTE

Multiple BWP1s are involved when UE bandwidth adaptation is enabled for a cell and the cell serves both UEs that support the cell bandwidth (using one BWP1 equal to the cell bandwidth) and UEs that do not support the cell bandwidth (using another BWP1 equal to the bandwidth specified by the **NRDUCellBwp.DlMinCarrierBw** parameter).

For details about UE bandwidth adaptation, see *Scalable Bandwidth*.

After the multi-BWP1 PDCCH transmission algorithm is enabled, the enhanced multi-BWP1 PDCCH transmission algorithm can be enabled using the **PDCCH_MULTI_BWP1_ALGO_ENH_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. This algorithm supports enhanced PDCCH uplink-to-downlink CCE ratio adaptation for multiple BWP1s, which further optimizes allocation of available CCEs for uplink and downlink scheduling information to increase the CCE allocation success rate for UEs.

In high-frequency NR TDD, when the **NRDUCellPdcch.OccupiedSymbolNum** parameter is set to **PART_SLOT_2SYM**, **PART_SLOT_3SYM**, or **2SYM**, or when the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter is selected, neither static nor adaptive configuration of the ratio of available CCEs for uplink scheduling information to those for downlink scheduling information takes effect. CCEs are allocated based on the "first come, first served" principle without a preset PDCCH uplink-to-downlink CCE ratio.

 NOTE

In low frequency bands, a CPE supports blind detection of a maximum 96 PDCCH RBs. In high frequency bands, a CPE supports blind detection of a maximum 48 PDCCH RBs. For CPE users, the effective resource ratio is subject to the CPE's blind detection capability.

PDCCH Uplink and Downlink CCE Sharing (Low-Frequency TDD)

In low-frequency NR TDD, PDCCH uplink and downlink CCE sharing can be enabled using the **UL_DL_CCE_SHARING_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function allows the downlink CCE resources available after being allocated for PDSCH scheduling to be used as uplink CCE resources for PUSCH scheduling.

Period for Reserving PDCCH CCEs for Uplink Scheduling Information (Low-Frequency TDD)

In low-frequency NR TDD, a dedicated slot is used to schedule aperiodic CSI reports (that is, CSI-RS measurement results) for each slot configuration, which

may waste PDCCH CCE resources. To reduce resource waste, the **NRDUCellPdcchAlgo.PdcchUlCceRsvPeriod** parameter can be used to configure the period for reserving PDCCH CCEs for uplink scheduling information in the slot for aperiodic CSI report scheduling.

- If there are no aperiodic CSI reports scheduled in this dedicated slot during the current reservation period, the PDCCH CCEs in the slot will be used for downlink scheduling in the next reservation period.
- If there is an aperiodic CSI report scheduled in this dedicated slot during the current reservation period, the PDCCH CCEs in the slot will still be reserved for aperiodic CSI report scheduling in the next reservation period.

Enhanced PDCCH Resource Adaptation for Low-Priority Scheduling

When the proportion of CCEs for downlink scheduling information is excessively low due to a high proportion of uplink preallocation or aperiodic CSI scheduling times, you are advised to enable enhanced PDCCH resource adaptation for low-priority scheduling, which includes the following sub-functions:

- Whether uplink preallocation is considered in the PDCCH scheduling resource adaptation algorithm is controlled by the **UL_PREALLOCATION_CCE_RES** option of the **NRDUCellPdcch.PDCCHAdjRefResType** parameter. If this option is deselected, uplink preallocation is not considered in the PDCCH scheduling resource adaptation algorithm. In this case, PDCCH resource adaptation for low-priority uplink preallocation takes effect, increasing available downlink CCE resources.
- Whether aperiodic CSI scheduling is considered in the PDCCH scheduling resource adaptation algorithm is controlled by the **APERIODIC_CSI_CCE_RES** option of the **NRDUCellPdcch.PDCCHAdjRefResType** parameter. If this option is deselected, aperiodic CSI scheduling is not considered in the PDCCH scheduling resource adaptation algorithm. In this case, PDCCH resource adaptation for low-priority aperiodic CSI scheduling takes effect, increasing available downlink CCE resources.

NOTE

The PDCCH scheduling resource adaptation algorithm involves PDCCH symbol number adaptation (controlled by the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter) and PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter).

Multi-BWP1 CCE Allocation for Top-Priority UEs

To optimize CCE allocation for UEs with the highest priority when multiple BWP1s are configured, multi-BWP1 CCE allocation for top-priority UEs is introduced. This function is controlled by the **MULTI_BWP1_TOP_UE_CCE_ALLOC_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. If this option is selected (that is, multi-BWP1 CCE allocation for top-priority UEs is enabled) and the CCE aggregation level is greater than or equal to 8 for a UE with the highest uplink or downlink scheduling priority, resource allocation to the UE is restricted by the uplink-to-downlink CCE ratio of the BWP1 used by the UE but not those of other BWP1s.

Centralized PDCCH CCE Resource Allocation (Low-Frequency TDD)

Centralized PDCCH CCE resource allocation is controlled by the **PDCCH_CCE_CENTRAL_ALLOC_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. After this function takes effect, the gNodeB determines the start frequency-domain positions of uplink and downlink CCE resources based on PCIs (for non-hyper cells) or TRP IDs (for hyper cells) to increase the CCE usage. If uplink and downlink CCEs are insufficient, the average uplink and downlink UE throughputs can be increased.

5.1.5.2 Uplink and Downlink CCE Resource Allocation for the Common PDCCH (Low-Frequency TDD)

PDCCH Uplink and Downlink Common CCE Sharing

In low frequency bands, PDCCH uplink and downlink common CCE sharing is controlled by the **UL_DL_COMM_CCE_SHARING_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function allows the downlink common CCE resources available after being allocated for PDSCH scheduling to be used as uplink common CCE resources for PUSCH scheduling, regardless of the percentage of reserved uplink common resources specified by the **NRDUCellPdcch.CommULRsvPct** parameter.

CCE Resource Optimization for Low-Priority Scheduling Based on Uplink Common DCI

This function is introduced to optimize CCE resources for low-priority scheduling based on uplink common DCI. It is controlled by the **UL_COMM_LOW_PRI_CCE_UNRSV_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. If this option is selected, CCE resource optimization for low-priority scheduling based on uplink common DCI takes effect and CCE resource reservation for uplink common DCI is not triggered for low-priority scheduling based on uplink common DCI.

5.1.6 Other PDCCH Configurations

This section describes other PDCCH-related configurations.

PDCCH Blind Detection Number Optimization (Low-Frequency TDD)

During PDCCH resource allocation, uplink and downlink CCEs may fail to be allocated because of insufficient blind detections. As such, an increased number of blind detections can raise the PDCCH resource allocation success rate.

PDCCH blind detection number optimization is controlled by the **PDCCH_BLIND_DETECT_CNT_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter. When this function takes effect, the number of PDCCH blind detections increases, raising the uplink and downlink CCE allocation success rates. If uplink and downlink CCEs are insufficient, the average uplink and downlink UE throughputs can be increased.

PDCCH Antenna Difference Optimization (High-Frequency TDD)

This function uses the antenna difference calculated based on SRS measurements for PDCCH link adaptation, and is controlled by the **PDCCH_ANT_DIFF_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

- When the **PDCCH_ANT_DIFF_OPT_SW** option is selected, PDCCH antenna difference optimization is enabled. The gNodeB uses the antenna difference calculated based on SRS measurements for PDCCH link adaptation.
- When the **PDCCH_ANT_DIFF_OPT_SW** option is deselected, PDCCH antenna difference optimization is disabled. The gNodeB uses a fixed antenna difference value for PDCCH link adaptation.

5.2 Network Analysis

5.2.1 Benefits

PDCCH channel management improves network performance by properly utilizing PDCCH resources.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

[Table 5-2](#) describes the network impacts of each sub-function involved in PDCCH channel management. For the definitions of related indicators, see [5.4.2 Activation Verification](#).

Table 5-2 Network impacts of the sub-functions involved in PDCCH channel management

Sub-Function Name	Network Impacts
PDCCH symbol number configuration	If the number of PDCCH symbols is statically configured using the NRDUCellPdcch.OccupiedSymbolNum parameter, changing the value of this parameter will cause the cell to restart and all UEs served by the cell to re-access the network. In high frequency bands, enabling the PDCCH to spread over two symbols in some slots and one symbol in others (by setting the NRDUCellPdcch.OccupiedSymbolNum parameter to PART_SLOT_2SYM) instead of enabling the PDCCH to occupy one or two symbols results in the following scenario-specific impacts: • The uplink and downlink cell throughputs increase when the cell serves a large number of UEs with a large proportion of CEUs and has heavy traffic requirements. • The uplink and downlink cell throughputs may slightly decrease when the cell serves a small number of UEs with a small proportion of CEUs and has light traffic requirements. • The average downlink and uplink CCE aggregation levels may decrease when the cell serves a large proportion of CCUs. • The average downlink and uplink CCE aggregation levels may increase when the cell serves a large proportion of CEUs.

Sub-Function Name	Network Impacts
PDCCH symbol number adaptation	In low frequency bands, after this function takes effect, the changes in the number of PDCCH symbols trigger RRC reconfiguration for all UEs, which may increase the service drop rate. In low frequency bands, an increase in the number of PDCCH symbols (for example, from 1 to 2) results in the following impacts: • The number of available CCEs increases, the CCE usage may decrease, and the numbers of UEs scheduled in the uplink and downlink in each TTI may increase. In enhanced Mobile Broadband (eMBB), the uplink and downlink PRB usages, UE throughputs, and cell throughputs may increase, and packet delay may decrease. • The number of PDSCH symbols decreases. In scenarios where PDSCH congestion is not accompanied by PDCCH congestion, the uplink and downlink UE and cell throughputs may decrease, and packet delay may increase. In low frequency bands, a decrease in the number of PDCCH symbols (for example, from 2 to 1) results in the following impacts: • The number of available CCEs decreases, the CCE usage may increase, and the numbers of UEs scheduled in the uplink and downlink in each TTI may decrease. In eMBB, the uplink and downlink PRB usages, UE throughputs, and cell throughputs may decrease, and packet delay may increase. • The number of PDSCH symbols increases. In scenarios where PDSCH congestion is not accompanied by PDCCH congestion, the uplink and downlink UE and cell throughputs may increase, and packet delay may decrease. In low frequency bands, when the NRDUCellPdcch.UePdcchAdaptMaxSymNum parameter (specifying the maximum number of symbols allowed for UE-specific PDCCH symbol number adaptation) is set to 3SYM, the maximum number of PDCCH symbols increases from 2 to 3, resulting in the following impacts: • The number of available CCEs increases, the CCE usage may decrease, and the numbers of UEs scheduled in the uplink and downlink in each TTI may increase. In eMBB, the uplink and downlink PRB usages, UE throughputs, and cell throughputs may increase, and packet delay may decrease. • The number of PDSCH symbols decreases. In scenarios where PDSCH congestion is not accompanied by PDCCH congestion, the uplink and downlink UE and cell throughputs may decrease, and packet delay may increase. • The number of PDCCH symbols may be more likely to change. The changes in the number of PDCCH symbols

Sub-Function Name	Network Impacts
	trigger RRC reconfiguration for all UEs, which may further increase the service drop rate.
Multi-BWP1 ratio statistics optimization	After this function takes effect, the uplink and downlink CCE allocation success rates and the average uplink and downlink UE throughputs increase in cells with limited service types but may fluctuate in other scenarios.
Equivalent-CCE-usage-based PDCCH symbol adaptation	After this function takes effect, an increase in PDCCH symbols allows more UEs to be scheduled, increasing the uplink and downlink CCE allocation success rates. However, this comes at the cost of fewer PDSCH symbols, which may lead to fluctuations in the average uplink and downlink cell throughputs.
Optimized PDCCH symbol adaptation	After this function takes effect in extreme CCE allocation scenarios under light load, the probability of reducing the number of PDCCH symbols increases. This reduction frees up more PDSCH symbols, which may lead to decreased CCE allocation success rate, increased CCE usage, and higher average downlink UE throughput.

Sub-Function Name	Network Impacts
PDCCH symbol number adaptation	In high-frequency TDD, after this function takes effect, the changes in the number of PDCCH symbols trigger RRC reconfiguration for half of the UEs, which may increase the service drop rate. In high-frequency TDD, an increase in the number of PDCCH symbols results in the following impacts: • If the number of available PDCCH symbols for uplink or downlink scheduling information increases, then the number of available CCEs increases, the CCE usage may decrease, and the numbers of UEs scheduled in the uplink and downlink in each TTI may increase. As a result, the uplink and downlink PRB usages, UE throughputs, and cell throughputs may increase, and packet delay may decrease. • If the number of available PDCCH symbols for downlink scheduling information remains unchanged while the number of available PDCCH symbols for uplink scheduling information increases, then the number of available PDSCH symbols decreases, the number of available CCEs increases, the CCE usage may decrease, and the number of UEs scheduled in the downlink in each TTI may increase. As a result, the downlink PRB usage may increase, the downlink UE and cell throughputs may decrease, and packet delay may increase. • If the number of available PDCCH symbols for uplink scheduling information increases, uplink scheduling accelerates. In this case, both the uplink cell traffic volume and uplink transmission duration in the cell increase. If they increase by different percentages, the average uplink cell throughput may decrease. In high-frequency TDD, a decrease in the number of PDCCH symbols results in the following impacts: • If the number of available PDCCH symbols for uplink or downlink scheduling information decreases, then the number of available CCEs decreases, the CCE usage may increase, and the numbers of UEs scheduled in the uplink and downlink in each TTI may decrease. As a result, the uplink and downlink PRB usages and the downlink UE and cell throughputs may decrease, and packet delay may increase. • If the number of available PDCCH symbols for downlink scheduling information remains unchanged while the number of available PDCCH symbols for uplink scheduling information decreases, then the number of available PDSCH symbols increases, the number of available CCEs decreases, the CCE usage may increase, and the number of UEs scheduled in the downlink in each TTI may decrease. As a result, the downlink PRB usage may decrease, the downlink

Sub-Function Name	Network Impacts
	UE and cell throughputs may increase, and packet delay may decrease. If the number of available PDCCH symbols for uplink scheduling information decreases, uplink scheduling slows down. In this case, both the uplink cell traffic volume and uplink transmission duration in the cell decrease. If they decrease by different percentages, the average uplink cell throughput may increase.
PDCCH multi-symbol resource pool	In low-frequency TDD, after this function takes effect: - The average downlink and uplink cell throughputs increase when all of the following conditions are met: (1) The NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM or 3SYM, or the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is selected. (2) The number of RRC_CONNECTED UEs in the cell is greater than 100. (3) The cell bandwidth is 80 MHz or higher. (4) The equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted is greater than 60%. (5) Both the downlink and uplink PRB usages are less than 80%. - The PDCCH multi-symbol resource pool has high requirements on UE demodulation performance. An increase in the BLER for UEs with poor demodulation performance may decrease the average downlink and uplink cell throughputs. - The CCE allocation success rate increases and the difference between the number of Msg3 receptions and the number of Msg4 responses decreases. As a result, the RRC reestablishment rate may decrease and the contention-based access success rate may increase.
Configuration of multiple PDCCH search spaces	In low-frequency TDD, after this function takes effect: The CCE allocation success rate increases for UEs under favorable network coverage but decreases for UEs under weak network coverage. As a result, the average throughput of UEs under weak network coverage decreases. If PDCCH blind detection assignment optimization is enabled, the uplink and downlink CCE allocation success rates and average uplink and downlink UE throughputs increase, except in scenarios with a large number of CEUs, where these indicators may decrease.

Sub-Function Name	Network Impacts
Configuration of the PDCCH aggregation level adaptation policy	After this function takes effect: - When the NRDUCellPdcch.PdcchAggLvlAdaptPol parameter is set to SEPARATE_ADAPT for a cell where the channel quality changes rapidly, a more accurate aggregation level can be selected for the UE-specific PDCCH. - The PDCCH aggregation level may decrease if the PDSCH MCS outer-loop adjustment value is small before this setting takes effect, and may increase if the PDSCH MCS outer-loop adjustment value is large before this setting takes effect. - This increases the number of UEs that can be scheduled in the cell, average downlink and uplink cell throughputs, and average downlink and uplink UE throughputs while ensuring the coverage performance of the UE-specific PDCCH. - When the NRDUCellPdcch.PdcchAggLvlAdaptPol parameter is set to JOINT_ADAPT, network performance is not affected.
PDCCH SINR offset configuration	After this function takes effect: - A smaller PDCCH SINR offset results in a higher CCE aggregation level, which improves PDCCH reliability and reduces the probability of DCI miss-detection. - A larger PDCCH SINR offset results in a lower CCE aggregation level, which prevents insufficient PDCCH resources from impacting cell throughput.
PUSCH-DTX-based aggregation level adaptation	After this function takes effect: - The PDCCH coverage at the cell edge may improve. - UEs are unlikely to experience PDCCH miss-detection. - The PDCCH resource consumption for UEs decreases. - The CCE usage increases. - The average downlink and uplink cell throughputs increase. - The average downlink and uplink UE throughputs increase. - The PUSCH IBLER of CEUs decreases.
Aggregation level optimization based on PUSCH NACK/DTX	After this function takes effect: - The PDCCH coverage at the cell edge may improve. - The probability of PDCCH miss-detection by UEs decreases when uplink large-packet services are performed. - The CCE usage of the cell increases. - The average downlink and uplink cell throughputs increase. - The average downlink and uplink UE throughputs increase. - The PUSCH IBLER of CEUs decreases.

Sub-Function Name	Network Impacts
PDCCH aggregation level optimization in the case of consecutive DTXs	After this function takes effect: The PDCCH aggregation level decreases. When CCEs are insufficient, the average downlink and uplink cell throughputs increase.
PDCCH aggregation level optimization in the case of consecutive non-DTXs	After this function (controlled by the NRDUCellPdcch.NonDtxSinrIncreaseStep parameter) takes effect: • A larger value of this parameter results in a faster increase in the SINR of the PDCCH, a lower aggregation level, and fewer PDCCH resources to be consumed. However, when there is burst interference, a larger value of this parameter results in an increased PDCCH BLER and decreased average uplink and downlink UE throughputs. • A smaller value of this parameter results in a slower increase in the SINR of the PDCCH, a higher aggregation level, and more PDCCH resources to be consumed. However, when there is burst interference, a smaller value of this parameter results in a decreased PDCCH BLER and increased average uplink and downlink UE throughputs.
Outer-loop robustness improvement for PDCCH aggregation level adaptation	After this function takes effect: • If the average aggregation level is low before this function takes effect, the average aggregation level increases and the uplink and downlink BLERs decrease after this function takes effect. This may decrease the uplink and downlink CCE allocation success rates, which in turn decreases the average downlink and uplink cell throughputs and the average downlink and uplink UE throughputs. • If the average aggregation level is high before this function takes effect, the average aggregation level decreases after this function takes effect. This may increase the uplink and downlink CCE allocation success rates, which in turn increases the average downlink and uplink cell throughputs and the average downlink and uplink UE throughputs.

Sub-Function Name	Network Impacts
Maximum CCE aggregation level for the UE-specific PDCCH	Configuring a maximum CCE aggregation level for the UE-specific PDCCH has the following impacts: • If the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter is set to AGG_LVL_16, there is no impact on the network. • If the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter is set to AGG_LVL_8 or AGG_LVL_4, the average downlink and uplink cell throughputs will increase. However, the PDCCH coverage performance will deteriorate, which may increase the service drop rate in the cell. Setting this parameter to AGG_LVL_8 or AGG_LVL_4 when not under an extremely heavy load will not bring the preceding gains but will still negatively affect the PDCCH coverage performance, which may increase the service drop rate in the cell. You are advised to set this parameter to AGG_LVL_8 or AGG_LVL_4 under an extremely heavy load (for example, when there are more than 600 UEs accessing a cell per second). The value AGG_LVL_4 results in further reductions in the CCE resource consumption for single UEs and greater preceding network impacts. • If the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter is set to AGG_LVL_8_16_ADAPT, the average PDCCH downlink and uplink aggregation levels and the CCE usage will decrease, while the uplink and downlink CCE allocation success rates will increase. When PDCCH CCEs are insufficient, the average uplink and downlink UE throughputs and the average uplink and downlink cell throughputs increase.
PDCCH resource allocation enhancement	After this function takes effect: Under heavy PDCCH CCE load, the gNodeB preferentially selects a low CCE aggregation level. In eMBB scenarios, the numbers of UEs scheduled in the uplink and downlink in a cell and the average downlink and uplink cell throughputs may increase. A low CCE aggregation level may cause PDCCH demodulation performance to deteriorate and the service drop rate to increase for some CEUs. Enabling this function in heavy-load scenarios with high PDCCH CCE usage will lead to higher uplink and downlink RBLERs and fluctuations in average uplink and downlink UE throughputs.

Sub-Function Name	Network Impacts
PDCCH CCE ratio adjustment in common slots	In low-frequency TDD, after this function takes effect, the number of available downlink CCEs increases, the downlink CCE allocation failure rate decreases, and the downlink user-perceived rate increases in the cell. After the downlink CCE allocation failure rate decreases in a cell, the number of UEs scheduled on the PDCCH in common slots increases, the duration of downlink data transmission at the MAC layer in the cell increases, and the downlink cell throughput may decrease. In addition, when the number of available downlink CCEs in a cell increases, the number of available uplink CCEs decreases because the total number of CCEs is fixed. This results in an increase in the uplink CCE allocation failure rate and a potential decrease in the uplink user-perceived rate in the cell. You are advised to select the UL_DL_CCE_SHARING_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter when this function is enabled.
Transition from RRC reestablishment to RRC setup	After this function takes effect: • It triggers subsequent signaling procedures, decreasing the total number of RRC reestablishment requests and successful RRC reestablishments, increasing the total number of initial context, PDU session, and QoS flow setup requests, and possibly decreasing their setup success rates. • It causes NG interface reestablishment, prolonging the service interruption duration. When voice service UEs involved in RRC reestablishments enter the RRC setup procedure, voice services are temporarily interrupted. • It requires more UE-specific PDCCH resources. In extreme scenarios, if the UE-specific PDCCH resource usage is very high, the UE-specific CCE allocation failure rate increases. However, the number of UEs accessing the cell and the number of UEs in the cell increase.
mmWave PDCCH coverage enhancement	After this function takes effect in weak-coverage and light-load scenarios, the performance of UEs under weak coverage improves, but the uplink and downlink CCE allocation success rates and the average uplink and downlink cell throughputs may decrease.
Fast aggregation level reduction	After this function takes effect, UEs are more likely to be assigned lower aggregation levels. This increases the uplink and downlink CCE allocation success rates, but may also increase the uplink and downlink RBLERs and decrease the average uplink and downlink cell throughputs.
DTX filtering	After this function takes effect, UEs are more likely to be assigned lower aggregation levels. This increases the uplink and downlink CCE allocation success rates, but may also increase the uplink and downlink RBLERs and decrease the average uplink and downlink cell throughputs.

Sub-Function Name	Network Impacts
CORESET 0 offset policy configuration	After this function takes effect, the offset of the frequency-domain position of CORESET 0 relative to the center frequency of the SSB is adjusted to a protocol-defined value that makes CORESET 0 closest to the frequency-domain edge of the cell. This enables PUCCH position multiplexing, eliminating the need for extra PUCCH resources and preserving available PUSCH resources. As a result, the average uplink and downlink cell throughputs may fluctuate.
PDCCH uplink-to-downlink CCE ratio adaptation	- After this function (controlled by the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter) takes effect, the uplink-to-downlink CCE ratio is adjusted based on services. If the cell is reset, the proportion of available CCEs for uplink scheduling information is restored to the initial value (specified by the NRDUCellPdcch.UlMaxCcePct parameter) and then adjusted based on services. As such, the uplink-to-downlink CCE ratios before and after the cell reset may be inconsistent, causing changes in the value of N.CCE.ULDCI.Avail.Equivalent. - In low frequency bands, changing the upper limit of the uplink CCE proportion for PDCCH uplink-to-downlink CCE ratio adaptation (specified by the NRDUCellPdcch.UlCceRatioUpperLimit parameter) has the following impacts: - A smaller value of this parameter results in a smaller uplink CCE proportion, a lower uplink CCE allocation success rate in CCE congestion scenarios, and a higher downlink CCE allocation success rate. - A larger value of this parameter results in a larger uplink CCE proportion, a higher uplink CCE allocation success rate in CCE congestion scenarios, and a lower downlink CCE allocation success rate.
Maximum adjustment step for the PDCCH uplink CCE proportion	In low-frequency TDD, after this function (specified by the NRDUCellPdcch.UlCcePctMaxAdjStep parameter) takes effect: - The smaller the value of this parameter, the slower the adjustment of the uplink CCE proportion. As a result, the uplink and downlink CCE allocation success rates decrease when the numbers of required uplink and downlink CCEs change rapidly. - The larger the value of this parameter, the faster the adjustment of the uplink CCE proportion. As a result, the uplink and downlink CCE allocation success rates increase when the numbers of required uplink and downlink CCEs change rapidly.

Sub-Function Name	Network Impacts
PDCCH uplink and downlink CCE sharing	In low-frequency TDD, after this function takes effect: - The average uplink cell throughput and average uplink UE throughput may increase. - The values of PDCCH-related indicators for the uplink, such as the uplink CCE allocation success rate, number of equivalent CCEs used for the uplink, number of UEs scheduled in the uplink, uplink PRB usage, and equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted, may increase. - Networking-related indicators may improve. For example, the RRC reestablishment rate decreases.
Period for reserving PDCCH CCEs for uplink scheduling information	In low-frequency TDD, you are advised to configure the period for reserving PDCCH CCEs for uplink scheduling information (by setting the NRDUCellPdcchAlgo.PdcchUlCceRsvPeriod parameter to a value other than NOT_CONFIG) when PDCCH downlink CCE resources are insufficient and there are no aperiodic CSI reports (for example, when the CCE usage is less than or equal to 70%, and features or functions involving aperiodic CSI reporting, such as DL CoMP, are not enabled). - Configuring this period in recommended scenarios can increase the CCE usage and average downlink cell throughput. - Configuring this period in non-recommended scenarios may decrease the average downlink cell throughput.
Enhanced PDCCH resource adaptation for low-priority scheduling	After this function takes effect (with the UL_PREALLOCATION_CCE_RES or APERIODIC_CSI_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter deselected): When uplink preallocation or aperiodic CSI scheduling is not considered in the PDCCH scheduling resource adaptation algorithm, the number of available downlink CCEs increases, improving downlink experience. Meanwhile, the number of available uplink CCEs decreases, and the CCE allocation failure rate for UEs with low uplink scheduling priorities increases.
Multi-BWP1 CCE allocation for top-priority UEs	After this function takes effect, the delay decreases and the CCE usage and CCE allocation success rate increase for CEUs with the highest priority. However, other UEs for which this function does not take effect may fail to be scheduled, possibly decreasing the uplink and downlink throughputs of these UEs.
Centralized PDCCH CCE resource allocation	In low-frequency TDD, after this function takes effect, the uplink and downlink CCE allocation success rates and the average uplink and downlink UE throughputs increase when uplink and downlink CCEs are insufficient. If CCE allocation failures are mainly caused by insufficient available CCEs, the average uplink and downlink throughputs will not increase.

Sub-Function Name	Network Impacts
PDCCH uplink and downlink common CCE sharing	After this function takes effect in a cell serving a large number of RRC_CONNECTED UEs, the common CCE allocation success rate, the access success rate, and the number of RRC_CONNECTED UEs in the cell increase, but the user-perceived rate may decrease.
CCE resource optimization for low-priority scheduling based on uplink common DCI	After this function takes effect in a cell where low-priority scheduling accounts for a large proportion in the scheduling based on uplink common DCI, resource reservation for uplink common DCI can hardly be triggered and therefore the values of the following indicators increase: the CCE allocation failure rate for uplink common DCI, CCE allocation success rate for UEs, access success rate, average uplink and downlink UE throughputs, and CCE usage.
PDCCH blind detection number optimization	In low-frequency TDD, after this function takes effect in a cell with insufficient uplink and downlink CCEs, the uplink and downlink CCE allocation success rates and the average uplink and downlink UE throughputs increase, and the downlink retransmission rate may fluctuate. If CCE allocation failures are mainly caused by insufficient available CCEs, the average uplink and downlink throughputs will not increase.

Sub-Function Name	Network Impacts
PDCCH antenna difference optimization	In high-frequency 4T and 8T scenarios, after this function takes effect: If the gNodeB uses more than two ports to transmit CSI-RS for CM, or if the gNodeB uses two ports to transmit CSI-RS for CM and the 2-port CSI-RS coverage enhancement function (controlled by the CSIRS_2PORT_COV_ENH_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter) is enabled and takes effect: • When UEs are distributed among multiple beams: – The average number of antennas used by the PDCCH may decrease, possibly increasing the numbers of UEs scheduled in the uplink and downlink in each TTI, the CCE usage, the uplink and downlink PRB usages, the average uplink and downlink UE throughputs, and the average uplink and downlink cell throughputs. – The average number of antennas used by the PUSCH and the average uplink MCS index may decrease when there is a large proportion of UEs with continuous uplink service requirements on the network. As a result, the average uplink UE and cell throughputs may decrease. In this case, it is recommended that the PUSCH_BEAM_ALLOC_OPT_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter be selected to enable the base station to consider the RB resource requirements of UEs at different beams during PUSCH beam allocation, achieving better PUSCH beam allocation results and improving uplink performance. • When UEs are distributed under a single beam: – The average number of antennas used by the PDCCH may decrease, possibly increasing the average aggregation level, decreasing the numbers of UEs scheduled in the uplink and downlink in each TTI, changing the CCE usage, and decreasing the uplink and downlink PRB usages, the average uplink and downlink UE throughputs, and the average uplink and downlink cell throughputs. – The average number of antennas used by the PUSCH and the average uplink MCS index may decrease, possibly decreasing the uplink PRB usage, average uplink UE throughput, and average uplink cell throughput. If the gNodeB uses two ports to transmit CSI-RS for CM and the 2-port CSI-RS coverage enhancement function (controlled by the CSIRS_2PORT_COV_ENH_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter) is either disabled or enabled but has not taken effect: • When UEs are distributed among multiple beams:

Sub-Function Name	Network Impacts
	- The average number of antennas used by the PDCCH may increase, possibly decreasing the numbers of UEs scheduled in the uplink and downlink in each TTI, the CCE usage, the uplink and downlink PRB usages, the average uplink and downlink UE throughputs, and the average uplink and downlink cell throughputs. - The average number of antennas used by the PUSCH and the average uplink MCS index may increase, possibly increasing the uplink PRB usage, average uplink UE throughput, and average uplink cell throughput. • When UEs are distributed under a single beam: - The average number of antennas used by the PDCCH may increase, possibly decreasing the average aggregation level, increasing the numbers of UEs scheduled in the uplink and downlink in each TTI, changing the CCE usage, and increasing the uplink and downlink PRB usages, the average uplink and downlink UE throughputs, and the average uplink and downlink cell throughputs. - The average number of antennas used by the PUSCH and the average uplink MCS index may increase, possibly increasing the uplink PRB usage, average uplink UE throughput, and average uplink cell throughput.

5.3 Requirements

5.3.1 Licenses

This basic function is not under license control.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH-DTX-based aggregation level adaptation	PUSCH_DTX_AGG_LVL_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Aggregation level optimization based on PUSCH NACK/DTX (controlled by the PUSCH_NACK_DTX_AGG_LVL_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) can be enabled only when PUSCH-DTX-based aggregation level adaptation (controlled by the PUSCH_DTX_AGG_LVL_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) is enabled.
Low-frequency TDD	Minimum downlink carrier bandwidth	NRDUCellBwp.DlMinCarrierBw	<i>Channel Management</i>	The multi-BWP1 PDCCH transmission algorithm (controlled by the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) takes effect only when the NRDUCellBwp.DlMinCarrierBw parameter is set to a value other than NOT_CONFIG or the REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is selected.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink cell bandwidth	NRDUCell.DlBandwidth	<i>Scalable Bandwidth</i>	The PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) can be enabled only when all of the following conditions are met: • The cell bandwidth is 80 MHz, 90 MHz, or 100 MHz (with the NRDUCell.DlBandwidth parameter set to CELL_BW_80M, CELL_BW_90M, or CELL_BW_100M). • The number of PDCCH symbols is greater than or equal to 2 (with the NRDUCellPdcch.OccupiedSymbolNum parameter set to 2SYM or 3SYM, or with the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter selected).
	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	
	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	
Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	The enhanced multi-BWP1 PDCCH transmission algorithm (controlled by the PDCCH_MULTI_BWP1_ALGO_ENH_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) takes effect only when the multi-BWP1 PDCCH transmission algorithm is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Multi-BWP1 ratio statistics optimization (controlled by the MULTI_BWP1_RATIO_STAT_OPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter) takes effect only when the multi-BWP1 PDCCH transmission algorithm is enabled.
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	PDCCH uplink-to-downlink CCE ratio adaptation takes effect only when the PDCCH is not configured to spread over two symbols or spread over two symbols in some slots and one symbol in others (by setting NRDUCellPdcch.OccupiedSymbolNum to a value other than 2SYM and PART_SLOT_2SYM).
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	Static configuration of the proportion of available CCEs for uplink scheduling information takes effect only when the PDCCH is not configured to spread over two symbols or spread over two symbols in some slots and one symbol in others (by setting NRDUCellPdcch.OccupiedSymbolNum to a value other than 2SYM and PART_SLOT_2SYM).
Low-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) can be enabled only when the NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM or 3SYM .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced PDCCH resource adaptation based on uplink preallocation	UL_PREALLOCATION_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter	<i>Channel Management</i>	CCE resource optimization for low-priority scheduling based on uplink common DCI takes effect only when enhanced PDCCH resource adaptation based on uplink preallocation is disabled. This is because uplink preallocation is not considered in the PDCCH scheduling resource adaptation algorithm and can be identified as low-priority scheduling when enhanced PDCCH resource adaptation based on uplink preallocation is disabled.
Low-frequency TDD	Enhanced PDCCH resource adaptation based on aperiodic CSI scheduling	APERIODIC_CSI_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter	<i>Channel Management</i>	CCE resource optimization for low-priority scheduling based on uplink common DCI takes effect only when enhanced PDCCH resource adaptation based on aperiodic CSI scheduling is disabled. This is because aperiodic CSI scheduling is not considered in the PDCCH scheduling resource adaptation algorithm and can be identified as low-priority scheduling when enhanced PDCCH resource adaptation based on aperiodic CSI scheduling is disabled.
Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Multi-BWP1 CCE allocation for top-priority UEs takes effect only when the multi-BWP1 PDCCH transmission algorithm is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	The maximum adjustment step for the PDCCH uplink CCE proportion can take effect only when PDCCH uplink-to-downlink CCE ratio adaptation is in effect.
Low-frequency TDD	UE-specific PDCCH Aggregation Level	NRDUCellPdcch.UeSpecificPdcchAggLvl	None	Configuration of the PDCCH aggregation level adaptation policy (specified by the NRDUCellPdcch.PdcchAggLvlAdaptPol parameter) takes effect only when the NRDUCellPdcch.UeSpecificPdcchAggLvl parameter is set to ADAPTIVE .
High-frequency TDD	Number of RBs for common control resources	NRDUCellCoreset.CommonCtrlResRbNum	None	mmWave PDCCH coverage enhancement (controlled by the MMW_PDCCH_COVERAGE_ENH_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) takes effect only when the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24 or RB48 .

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When PDCCH symbol number adaptation (controlled by the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) is enabled, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled. When rate-matching-pattern-configuration-based PDCCH rate matching is enabled, restrictions are placed on the number of PDCCH symbols. For details, see <i>Downlink Scheduling</i> .
High-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In high-frequency NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when either of the following conditions is met: - The NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM. - The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24 or RB96.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In low-frequency NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when any of the following conditions is met: - The NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM or 3SYM and the NRDUCell.DlBandwidth parameter is not set to CELL_BW_30M or CELL_BW_20M. - The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24 or RB96 and the NRDUCell.DlBandwidth parameter is not set to CELL_BW_30M or CELL_BW_20M.
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with rate-matching-pattern-configuration-based PDCCH rate matching.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When both the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter and the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter are selected, the multi-BWP1 PDCCH transmission algorithm (controlled by the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) must be disabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In low-frequency NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when any of the following conditions is met: - The NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM or 3SYM (indicating two or three UE-specific PDCCH symbols), and the NRDUCell.DlBandwidth parameter is not set to CELL_BW_30M or CELL_BW_20M (indicating that the downlink cell bandwidth is not 30 MHz or 20 MHz). - The NRDUCellCoreset.CommonCtrlResRbNum parameter (indicating the number of RBs for common control resources in a cell) is set to RB24 or RB96, and the NRDUCell.DlBandwidth parameter is not set to CELL_BW_30M or CELL_BW_20M (indicating that the downlink cell bandwidth is not 30 MHz or 20 MHz). - The NRDUCellPdcch.OccupiedRbNum parameter is set to a value less than 18 and the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAl

RAT	Function Name	Function Switch	Reference	Description
				<i>goEnhSwitch</i> parameter) is enabled.
Low-frequency TDD	Uplink enhanced scheduling of aperiodic CSI reports	APER_CSI_RPT_ENHANCE_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Uplink Scheduling</i>	Aperiodic CSI CCE resources (controlled by the APERIODIC_CSI_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter) cannot be enabled together with uplink enhanced scheduling of aperiodic CSI reports.
Low-frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with PDCCH symbol number adaptation.
Low-frequency TDD	PDCCH multi-symbol resource pool	MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with the PDCCH multi-symbol resource pool.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	CSI-RS beam measurement period	NRDUCellCsirs.CsirsBeamMeasPeriod	<i>Channel Management</i>	The NRDUCellCsirs.CsirsBeamMeasPeriod parameter must be set to SLOT320 when all of the following conditions are met: (1) The PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter or the PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter is selected. (2) The UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is selected. (3) The NRDUCellTrp.TxRxMode parameter is set to 2T2R . (4) The NRDUCell.SlotStructure parameter is not set to SS202 .
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>MIMO (TDD)</i>	When CCE resource optimization is enabled, the NRDUCellCoreset.CommonCtrlResRbNum parameter (specifying the number of RBs for common control resources) cannot be set to RB96 or RB24 .
Low-frequency TDD	Smart PDCCH symbol allocation	PDCCH_SYM_SMART_ALLOC_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When smart PDCCH symbol allocation is enabled, the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB96 and the SCHEDULE_FIRST_HALF_SLOT_SW option of the NRDUCellCollabServ.SulAlgorithmSwitch parameter is selected, the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter) cannot be enabled together with super uplink.
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	When the NRDUCellBwp.DlMinCarrierBw parameter is set to a value other than 80M , 90M , and NOT_CONFIG , the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter) cannot be enabled together with UE bandwidth adaptation.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter) cannot be enabled together with single DCI.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	The PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with single DCI.
Low-frequency TDD	Maximum number of RBs supported by power saving BWP	NRDUCellUePwrSaving.Bwp2MaxRbNum	<i>UE Power Saving</i>	When both configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) and power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter) are enabled, the NRDUCellUePwrSaving.Bwp2MaxRbNum parameter (specifying the maximum number of RBs supported by power saving BWP) can only be set to RB51 .
Low-frequency TDD	Scheduling in common PDCCH symbols for RedCap UEs	REDCAP_COMMON_PDCCH_SYMBOL_SWITCH option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Scheduling in common PDCCH symbols for RedCap UEs cannot be enabled when the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Scheduling in common PDCCH symbols for RedCap UEs	REDCAP_COMMON_PDCCH_SYM_SCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Scheduling in common PDCCH symbols for RedCap UEs and configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together when the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB96 .
Low-frequency TDD	Scheduling in common PDCCH symbols for RedCap UEs	REDCAP_COMMON_PDCCH_SYM_SCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	The NRDUCellPdcch.OccupiedSymbolNum parameter can only be set to 2SYM or 3SYM when the following conditions are met: (1) The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB96 . (2) The UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is deselected. (3) Scheduling in common PDCCH symbols for RedCap UEs is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level PUSCH interference avoidance	DL_UE_LEVEL_INTRF_AVOID and UL_CELL_LEVEL_INTRF_AVOID options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	<i>Interference Avoidance</i>	When the CORESET 0 offset policy is configured (with the NRDUCellPdcchAlgo.Coreset0OffsetPolicy parameter set to EDGE_FIRST), neither of the following functions can be enabled: • Uplink cell-level interference avoidance (controlled by the UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) • Downlink UE-level interference avoidance (controlled by the DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter)
Low-frequency TDD	PDCCH symbol optimization	PDCCH_SYMBOL_OPT_SW option of the NRDUCellMntPara.MntResourceOptSw parameter	None	Configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with PDCCH symbol optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH rate matching enhancement	PDCCH_RATE_MATCH_ENH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	When the NRDUCellPdcch.OccupiedSymbolNum parameter (specifying the number of PDCCH symbols) is set to 1SYM , PDCCH symbol number adaptation (controlled by the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) cannot be enabled together with PDCCH rate matching enhancement.
Low-frequency TDD High-frequency TDD	RB number configuration for common control resources	NRDUCellCoreset.CommonCtrlResRbNum	None	When the NRDUCellPdcch.MaxCommonPdcchAggLvl parameter is set to AGGLVL16 , the NRDUCellCoreset.CommonCtrlResRbNum parameter (specifying the number of RBs for common control resources in the cell) can only be set to RB96 .

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When rate-matching-pattern-configuration-based PDCCH rate matching is enabled, it is recommended that the NRDUCellPdcch.UlMaxCcePct parameter (specifying the maximum percentage of CCEs available for uplink scheduling information) be set to 50. • If this parameter is set to a value less than 50, the CCEs used for DCI containing uplink scheduling information may be insufficient. This can cause transmission of the DCI containing uplink scheduling information to fail, resulting in resynchronization failures. As a result, RRC reconfiguration is repeatedly triggered over the air interface. • If this parameter is set to a value greater than 50, the CCEs used for DCI containing downlink scheduling information may be insufficient, thereby decreasing downlink throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When PDCCH symbol number adaptation (controlled by the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) is enabled and the number of PDCCH symbols differs between the serving and cooperating cells, intra-base-station DL CoMP does not take effect. When the number of PDCCH symbols in the serving cell is greater than that in a cooperating cell, you are advised to select the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter. In this case, intra-base-station DL CoMP can take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM requires that the cooperating and serving cells have the same time-domain resources for PDSCH data transmission. Therefore, the number of symbols occupied by the PDCCH in the cooperating cell must be less than or equal to that in the serving cell. When PDCCH symbol number adaptation is enabled, the number of symbols occupied by the PDCCH may differ between the serving and cooperating cells. • With the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAllgoEnhSwitch parameter deselected, Co-MM can take effect only if the number of symbols occupied by the PDCCH in the cooperating cell is the same as that in the serving cell. • With the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAllgoEnhSwitch parameter selected, Co-MM can take effect if the number of symbols occupied by the PDCCH in the cooperating cell is less than or equal to that in the serving cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DCI scheduling optimization for switching from BWP2 to BWP1	BWP2_SWITCHING_DCI_SCH_OPT_SW option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When DCI scheduling optimization for switching from BWP2 to BWP1 is enabled, CCEs allocated to the DCI for switching from BWP2 to BWP1 are not restricted by the uplink CCE proportion during PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter).
Low-frequency TDD	Resource-based adaptive scheduling for massive MIMO	NRDUCellPdsch.DISchOptTimeThld not set to 0	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When both PDCCH uplink-to-downlink CCE ratio adaptation and resource-based adaptive scheduling for massive MIMO are enabled (the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter is selected and the NRDUCellPdsch.DISchOptTimeThld parameter is not set to 0), the number of active UEs in the uplink in the cell may increase. As a result, the uplink cell throughput will decrease when there are uplink services in the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>MIMO (TDD)</i>	Assume that one of the following options is selected: • AVERAGE_3DEIRP_CONTROL_SW option of the NRDUCellSmartPowerLock.SmartPowerLockSw parameter • INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter • INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter • INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter When CCE resource optimization is enabled, CCE resource allocation optimization in the slot for aperiodic CSI report scheduling (with NRDUCellPdcchAlgo.PdcchULCceRsvPeriod set to a value other than NOT_CONFIG) does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When the load is heavy in the uplink, the uplink interference increases and the number of PUSCH DTXs increases. As a result, CCE aggregation level selection is inaccurate. When packet-length-based scheduling optimization is enabled, you are advised to enable dynamic PDCCH SINR adjustment based on PUSCH DTX (controlled by the PUSCH_DTX_AGG_LVL_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter). This allows the estimated PDCCH SINR to be dynamically adjusted based on the PUSCH DTX information so that a more appropriate CCE aggregation level can be selected.
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, you are advised to enable initial CCE aggregation level selection optimization (controlled by the AGG_LVL_INIT_SELECT_OPTION_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter). This reduces the CCE aggregation level, increases the UE capacity allowed by PDCCH resources, and reduces the CCE allocation failure rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, you are advised to set the NRDUCellPdcch.PdcchAggLvlAdaptPol parameter (specifying the policy of PDCCH aggregation level adaptation) to SEPARATE_ADAPT . This allows the aggregation level selected for the UE-specific PDCCH to be more accurate and reduces the CCE allocation failure rate, when the channel quality changes rapidly.
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, you are advised to enable PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter). This increases the proportion of PDCCH resources used for uplink scheduling but may decrease the proportion of PDCCH resources used for downlink scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, you are advised to enable PDCCH symbol number adaptation (controlled by the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) to increase PDCCH resources. In addition, it is recommended that the NRDUCellPdcch.HeavyLoadCceUsageThld parameter be set to 50.
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	In 2T, 4T, and 8T scenarios, enabling PDCCH resource allocation enhancement (controlled by the PDCCH_RES_ALLOC_ENH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter) will decrease the PDCCH power for some UEs and may cause a decrease in the cell downlink rate. Therefore, you are not advised to enable symbol power saving together with PDCCH resource allocation enhancement in 2T, 4T, and 8T scenarios. If both functions are enabled, selecting the SYM_SHUTDOWN_UE_WEIGHT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter can reduce the impact on cell throughput.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When RF channel intelligent shutdown is enabled, you are advised to enable PDCCH symbol number adaptation to reduce the impact on the user-perceived rate.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapabilityBasedFunctionSw parameter	<i>RedCap</i>	In low-frequency TDD, when RedCap Introduction, power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter), and the enhanced multi-BWP1 PDCCH transmission algorithm (controlled by the PDCCH_MULTI_BWP1_ALGO_ENH_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter) are all enabled, the DCI allocation success rate, CCE usage, cell throughput, and user-perceived rate increase.
Low-frequency TDD	RedCap PDCCH symbol number adjustment	REDCAP_PDCCH_SYM_NUM_ADJ_SW option of the NRDUCellRedCap.RedCapAlgorithmSwitch parameter	<i>RedCap</i>	When configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgorithmEnhSwitch parameter) is enabled, RedCap PDCCH symbol number adjustment does not take effect.
Low-frequency TDD	RedCap power aggregation	REDCAP_POWER_AGGREGATION_SW option of the NRDUCellPdcch.PdcchAlgorithmExtSwitch parameter	<i>RedCap</i>	When both PDCCH resource allocation enhancement (controlled by the PDCCH_RES_ALLOC_ENH_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch) and RedCap power aggregation are enabled, RedCap power aggregation does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink CCE allocation without ratio constraint	DL_CCE_ALLO_NO_CONSTR_AIN_SW option of the gNBDUQciAlgoParamGrp.QciAlgoSwitch parameter	<i>VoNR</i>	When downlink CCE allocation without ratio constraint takes effect, PDCCH uplink-to-downlink CCE ratio adaptation does not take effect for voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs.
Low-frequency TDD	Minimum PDCCH aggregation level	gNBDUSchParamGroup.PdcchMinAggLvl	<i>URLLC</i>	For UEs for which the configured minimum PDCCH aggregation level takes effect, if the configured maximum PDCCH CCE aggregation level is lower than the configured minimum aggregation level, the maximum PDCCH CCE aggregation level takes effect instead of the configured value. That is, for these UEs, if the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter is set to AGG_LVL_4 and the gNBDUSchParamGroup.PdcchMinAggLvl parameter is set to AGG_LVL_8 , the value AGG_LVL_16 of the NRDUCellPdcchAlgo.PdcchMaxAggLevel parameter takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Minimum CCE aggregation level	NRDUCellServExp.MinCceAggLevel	None	For UEs for which the configured minimum CCE aggregation level takes effect, if the configured maximum PDCCH CCE aggregation level is lower than the configured minimum aggregation level, the maximum PDCCH CCE aggregation level takes effect instead of the configured value. That is, for these UEs, if the NRDUCellPdcchAlgorithm.PdcchMaxAggLevel parameter is set to AGG_LVL_4 and the NRDUCellServExp.MinCceAggLevel parameter is set to AGG_LVL_8 , the value AGG_LVL_16 of the NRDUCellPdcchAlgorithm.PdcchMaxAggLevel parameter takes effect.
High-frequency TDD	CSI-RS for BM in SSB slots	SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected) and the PDCCH spreads over two symbols or spreads over two symbols in some slots and one symbol in others (specified by the NRDUCellPdcch.OccupiedSymbolNum parameter) in high frequency bands, the average downlink UE throughput is slightly lower than that when the PDCCH spreads over only one symbol.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RB number configuration for common control resources	NRDUCellCoreset.CommonCtrlResRbNum	None	When the NRDUCellCoreset.CommonCtrlResRbNum parameter (specifying the number of RBs for common control resources in a cell) is set to RB24 and the NRDUCellPdcch.OccupiedSymbolNum parameter (specifying the number of PDCCH symbols) is set to 1SYM , PDCCH CCEs are wasted.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

 NOTE

3900 series base stations do not support high-frequency TDD functions.

Boards

Except for the RAT, the sub-functions of this function have no other requirements on main control boards and baseband processing units. All main control boards and baseband processing units of the corresponding RAT support these sub-functions. For details about the RATs supported by different main control boards and baseband processing units, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Except for the RAT, the sub-functions of this function have no other requirements on RF modules. All RF modules of the corresponding RAT support these sub-functions. For details about the RATs supported by different RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

For high-frequency TDD cells, when the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**:

- If the **NRCCell.CellActiveState** parameter value is **CELL_ACTIVE**, the **NRDUCellPdcch.OccupiedSymbolNum** parameter (specifying the number of PDCCH symbols) must be set to **PART_SLOT_3SYM**.
- The **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter (controlling PDCCH symbol number adaptation) must be deselected.
- The **MMW_PDCCH_COVERAGE_ENH_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter (controlling mmWave PDCCH coverage enhancement) can be selected.

For low-frequency TDD cells, when the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**, it is recommended that the **NRDUCellPdcch.OccupiedSymbolNum** parameter (specifying the number of PDCCH symbols) be set to **2SYM**.

When the CORESET 0 offset policy is configured (with the **NRDUCellPdcchAlgo.Coreset0OffsetPol** parameter set to **EDGE_FIRST**), the **NRDUCell.DlBandwidth** parameter cannot be set to **CELL_BW_90M**.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 5-3](#) describes the parameters used for function optimization.

Table 5-3 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Occupied Symbol Number	NRDUCellPdcch.OccupiedSymbolNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Maximum CCE Percentage	NRDUCellPdcch.UlMaxCcePct	None	Set this parameter based on the proportions of UEs running uplink services and UEs running downlink services.
PDCCH UL CCE Reserve Period	NRDUCellPdcchAlgo.PdcchULCceRsvPeriod	None	Use the recommended value.
Number of Occupied RBs	NRDUCellPdcch.OccupiedRbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter is configurable when the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is selected. Set this parameter as required.
Max Common PDCCH Aggregation Level	NRDUCellPdcch.MaxCommonPdcchAggLvl	None	Set this parameter as required. - When the downlink bandwidth (specified by NRDUCell.DlBandwidth) of a cell is less than or equal to 40 MHz, you are advised to set this parameter to AGGLVL4. - When the downlink bandwidth (specified by NRDUCell.DlBandwidth) of a cell is greater than 40 MHz, you are advised to set this parameter to AGGLVL8.
PDCCH BLER Target	NRDUCellPdcch.PdcchBlerTarget	None	Set this parameter as required.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Aggregation Level Adaptation Policy	NRDUCellPdcch.<i>PdcchAggLvlAdaptPol</i>	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PUSCH_DTX_AGG_LVL_ADAPT_SW	This option is selected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Maximum Aggregation Level	NRDUCellPdcch.<i>Algo.PdcchMaxAggLevel</i>	None	Use the recommended value. After thoroughly evaluating the network impacts, set this parameter to AGG_LVL_8 or AGG_LVL_4 under an extremely heavy load (for example, when there are more than 600 UEs accessing a cell per second).
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	UL_DL_CCE_RATIO_ADAPT_SW	Use the recommended value.
PDCCH Algorithm Extension Switch	NRDUCellPdcch.<i>PdcchAlgoExtSwitch</i>	UE_PDCCH_SYM_NUM_ADAPT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Use the recommended value.
UE PDCCH Adaptation Max Symbol Number	NRDUCellPdcch.<i>UePdcchAdaptMaxSymNum</i>	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	AGG_LVL_INI T_SELECT_OP T_SW	Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_MUL TI_BWP1_AL GO_SW	This option is selected by default, which is also the recommended setting.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	PDCCH_RES_ ALLOC_ENH_ SW	Use the recommended value.
IBLER-based PDCCH SINR Offset	NRDUCellPdcch.<i>IblerPdcchSinrOffset</i>	None	Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PUSCH_NAC K_DTX_AGG_ LVL_OPT_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_AGG_ LVL_OPT_S W	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	MULTI_SYM_ CORESET_S W	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.<i>UlpuschAlgoSwitch</i>	PUSCH_ACK_ DTX_DETECT_ _SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PUSCH_HAR Q_DTX_AGG_ LVL_OPT_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_ANT_DIFF_OPT_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	UL_DL_CCE_SHARING_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_BLIND_DETECT_CENT_OPT_SW	You are advised to select this option.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_CCE_CENTRAL_ALLOC_SW	You are advised to select this option.
Non-DTX SINR Increase Step	NRDUCellPdcch.<i>NonDtxSinrIncreaseStep</i>	None	Use the recommended value.
UL CCE Ratio Upper Limit	NRDUCellPdcch.<i>UlCceRatioUpperLimit</i>	None	Use the recommended value.
Reference Resource Type for PDCCH Adjust	NRDUCellPdcch.<i>PDCCHAdjRefResType</i>	UL_PREALLOCATION_CCE_RES	You are advised to select this option.
Reference Resource Type for PDCCH Adjust	NRDUCellPdcch.<i>PDCCHAdjRefResType</i>	APERIODIC_CSI_CCE_RES	Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i>	PDCCH_COMM_SLOT_RATIO_ADJ_SW	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink CCE Percentage Maximum Adjustment Step	NRDUCellPdcch.UlCcePctMaxAdjStep	None	Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	PDCCH_MULTIBWP1_ALGO_ENH_SW	This option is selected by default, which is also the recommended setting.
Process Switch	NRCCellAlgoSwitch.ProcessSwitch	RRC_REESTABLISH_TO_SETUP_SW	Use the recommended value.
PDCCH Algorithm Switch	NRDUCellPdcch.PdcchAlgoSwitch	UL_DL_COMM_CCE_SHARING_SW	Use the recommended value.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	UL_COMM_LOW_PRI_CCE_UNRSV_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	MULTI_BWP1_TOP_UE_CCE_ALLOC_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	MULTI_SEARCH_SPACE_SW	This option is deselected by default. You are advised to change the setting under the guidance of Huawei engineers.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	PDCCH_BLIND_DET_ASSIGN_OPT_SW	You are advised to select this option when configuration of multiple PDCCH search spaces (controlled by the MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i> parameter) is in effect. This can balance blind detection times across different PDCCH CCE aggregation levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times.
PDCCH Algorithm Extension Switch	NRDUCellPdcch.<i>PdcchAlgoExtSwitch</i>	MMW_PDCC_H_COVERAGE_ENH_SW	You are advised to select this option in weak-coverage and light-load scenarios.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	MULTI_BWP1_RATIO_STAT_OPT_SW	You are advised to select this option in cells serving a small number of UEs with limited service types (for example, only uplink or downlink services).
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	EQUIV_CCE_RATIO_SYM_ADPT_SW	You are advised to select this option when CCE resources are insufficient.
PDCCH Equivalent CCE Ratio Threshold	NRDUCellPdcch.<i>Algo.PdcchEquivalentCceRatioThld</i>	None	Use the recommended value.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	AGG_LEVEL_FAST_DECREASE_SW	You are advised to select this option when CCE resources are insufficient.
Aggregate Level Fast Decrease Power Threshold	NRDUCellPdcch.<i>Algo.AggLvlFastDcrPwrThld</i>	None	Use the recommended value.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	DTX_FILTER_SW	You are advised to select this option when CCE resources are insufficient.

Parameter Name	Parameter ID	Option	Setting Notes
CORESET0 Offset Policy	NRDUCellPdcchAlgo.Coreset0OffsetPol	None	Use the recommended value.
PDCCH Algorithm Extension Switch	NRDUCellPdcchPdcchAlgoExtSwitch	PDCCH_SYM_ADAPT_OPT_SW	You are advised to select this option in extreme CCE allocation scenarios under light load.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following provides activation command examples for the sub-functions of PDCCH channel management.

Low-frequency TDD command examples

```
//Configuring a static number of PDCCH symbols for a cell or enabling PDCCH symbol number adaptation as required
//Setting the number of PDCCH symbols to two for a cell (required only when PDCCH symbol number adaptation is disabled)
MOD NRDUCCELLPDCCH: NrDuCellId=0, OccupiedSymbolNum=2SYM;
//Enabling PDCCH symbol number adaptation
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-1;
//Enabling equivalent-CCE-usage-based PDCCH symbol adaptation and setting the equivalent PDCCH CCE usage threshold
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=EQUIV_CCE_RATIO_SYM_ADPT_SW-1;
MOD NRDUCCELLPDCCHALGO: NrDuCellId=0, PdcchEquivCceRatioThld=65;
//Setting the maximum number of symbols allowed for PDCCH symbol number adaptation to two when PDCCH symbol number adaptation is enabled
MOD NRDUCCELLPDCCH: NrDuCellId=0, UePdcchAdaptMaxSymNum=2SYM;
//Enabling PDCCH symbol number adaptation for multiple BWP1s
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-1;
//Configuring a static proportion of PDCCH CCEs for uplink scheduling information for a cell or enabling PDCCH uplink-to-downlink CCE ratio adaptation as required
//Setting the proportion of PDCCH CCEs for uplink scheduling information to 50% for a cell (required only when PDCCH uplink-to-downlink CCE ratio adaptation is disabled)
MOD NRDUCCELLPDCCH: NrDuCellId=0, UIMaxCcePct=50;
//Enabling PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-1;
//Enabling the multi-BWP1 PDCCH transmission algorithm
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-1;
//Enabling the enhanced multi-BWP1 PDCCH transmission algorithm
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_ENH_SW-1;
```

```
//Enabling multi-BWP1 ratio statistics optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=MULTI_BWP1_RATIO_STAT_OPT_SW-1;
//Enabling PDCCH uplink and downlink CCE sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
//Setting the period for reserving PDCCH CCEs for uplink scheduling information in the slot for aperiodic CSI
report scheduling to 5 ms
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchULCceRsvPeriod=MS5;
//Setting the maximum aggregation level for the common PDCCH to 4
MOD NRDUCELLPDCCH: NrDuCellId=0, MaxCommonPdcchAggLvl=AGGLVL4;
//Setting the target PDCCH BLER directly for the basic function of dynamic CCE aggregation level selection
for the UE-specific PDCCH (as this function is not under switch control)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchBlrTarget=3;
//Enabling initial CCE aggregation level selection optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-1;
//Enabling initial CCE aggregation level selection optimization for multiple BWPs
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-1;
//Setting the PDCCH aggregation level adaptation policy to separate adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAggLvlAdaptPol=SEPARATE_ADAPT;
//Setting the BLER-based PDCCH SINR offset to -4 dB
MOD NRDUCELLPDCCH: NrDuCellId=0, IblcrPdcchSinrOffset=-4;
//Enabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-1;
//Enabling aggregation level optimization based on PUSCH NACK/DTX
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_NACK_DTX_AGG_LVL_OPT_SW-1;
//Enabling PDCCH aggregation level optimization in the case of consecutive DTXs
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_AGG_LVL_OPT_SW-1;
//Enabling PDCCH resource allocation enhancement
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_RES_ALLOC_ENH_SW-1;
//Setting the maximum PDCCH aggregation level to 16
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchMaxAggLevel=AGG_LVL_16;
//Setting the PDCCH frequency-domain resource range for a cell to a full bandwidth
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedRbNum=0;
//Enabling the PDCCH multi-symbol resource pool
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SYM_CORESET_SW-1;
//Enabling configuration of multiple PDCCH search spaces
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SEARCH_SPACE_SW-1;
//((Recommended when configuration of multiple PDCCH search spaces is in effect) Enabling PDCCH blind
detection assignment optimization, which aims to balance blind detection times across different PDCCH
aggregation levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind
detection times
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-1;
//Enabling outer-loop robustness improvement for PDCCH aggregation level adaptation
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=PUSCH_ACK_DTX_DETECT_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_HARQ_DTX_AGG_LVL_OPT_SW-1;
//Configuring the upper limit of the PDCCH uplink CCE proportion
MOD NRDUCELLPDCCH: NrDuCellId=0, ULCceRatioUpperLimit=100;
//Configuring enhanced PDCCH resource adaptation for low-priority scheduling to indicate that PDCCH
scheduling resource adaptation does not depend on uplink preallocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PDCCHAdjRefResType=UL_PREALLOCATION_CCE_RES-0;
//Configuring enhanced PDCCH resource adaptation for low-priority scheduling to indicate that PDCCH
scheduling resource adaptation does not depend on aperiodic CSI scheduling
MOD NRDUCELLPDCCH: NrDuCellId=0, PDCCHAdjRefResType=APERIODIC_CSI_CCE_RES-0;
//Setting the NonDtxSinrIncreaseStep parameter for PDCCH aggregation level optimization in the case of
consecutive non-DTXs
MOD NRDUCELLPDCCH: NrDuCellId=0, NonDtxSinrIncreaseStep=0;
//Enabling PDCCH blind detection number optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_BLIND_DETECT_CNT_OPT_SW-1;
//Enabling centralized PDCCH CCE resource allocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_CCE_CENTRAL_ALLOC_SW-1;
//Enabling PDCCH CCE ratio adjustment in common slots
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_COMM_SLOT_RATIO_ADJ_SW-1;
//Configuring the maximum adjustment step for the PDCCH uplink CCE proportion
MOD NRDUCELLPDCCH: NrDuCellId=0, ULCcePctMaxAdjStep=15;
//Enabling PDCCH uplink and downlink common CCE sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_COMM_CCE_SHARING_SW-1;
//Enabling transition from RRC reestablishment to RRC setup
MOD NRCELLALGOSWITCH: NrCellId=0, ProcessSwitch=RRC_REESTAB_TO_SETUP_SW-1;
//Enabling CCE resource optimization for low-priority scheduling based on uplink common DCI
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=UL_COMM_LOW_PRI_CCE_UNRSV_SW-1;
```

```
//Enabling multi-BWP1 CCE allocation for top-priority UEs
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_BWP1_TOP_UE_CCE_ALLOC_SW-1;
//Enabling fast aggregation level reduction and setting the power threshold for this function
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=AGG_LEVEL_FAST_DECREASE_SW-1;
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, AggLvlFastDcrPwrThld=1;
//Enabling optimized PDCCH symbol adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=PDCCH_SYM_ADAPT_OPT_SW-1;
//Enabling DTX filtering
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=DTX_FILTER_SW-1;
//Setting Coreset0OffsetPol to EDGE_FIRST
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, Coreset0OffsetPol=EDGE_FIRST;
```

High-frequency TDD command examples

```
//Configuring a static number of PDCCH symbols for a cell or enabling PDCCH symbol number adaptation
as required
//Setting the number of PDCCH symbols to two symbols in some slots and one symbol in others for a cell
(required only when PDCCH symbol number adaptation is disabled)
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedSymbolNum=PART_SLOT_2SYM;
//Enabling PDCCH symbol number adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-1;
//Configuring a static proportion of PDCCH CCEs for uplink scheduling information for a cell or enabling
PDCCH uplink-to-downlink CCE ratio adaptation as required
//Changing the number of PDCCH symbols to one for a cell (when the number of PDCCH symbols is set to
one in some slots and two in others or set to two, neither static nor adaptive configuration of the
proportion of PDCCH CCEs for uplink scheduling information takes effect for the cell)
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedSymbolNum=1SYM;
//Setting the proportion of PDCCH CCEs for uplink scheduling information to 50% for a cell (required only
when PDCCH uplink-to-downlink CCE ratio adaptation is disabled)
MOD NRDUCELLPDCCH: NrDuCellId=0, UIMaxCcePct=50;
//Enabling PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-1;
//Setting the maximum aggregation level for the common PDCCH to 4
MOD NRDUCELLPDCCH: NrDuCellId=0, MaxCommonPdcchAggLvl=AGGLVL4;
//Setting the target PDCCH BLER directly for the basic function of dynamic CCE aggregation level selection
for the UE-specific PDCCH (as this function is not under switch control)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchBlerTarget=3;
//Enabling initial CCE aggregation level selection optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-1;
//Setting the BLER-based PDCCH SINR offset to -4 dB
MOD NRDUCELLPDCCH: NrDuCellId=0, IblerPdcchSinrOffset=-4;
//Enabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-1;
//Setting the PDCCH frequency-domain resource range for a cell to a full bandwidth
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedRbNum=0;
//Enabling PDCCH antenna difference optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_ANT_DIFF_OPT_SW-1;
//Configuring enhanced PDCCH resource adaptation for low-priority scheduling to indicate that PDCCH
scheduling resource adaptation does not depend on uplink preallocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PDCCHAdjRefResType=UL_PREALLOCATION_CCE_RES-0;
//Configuring enhanced PDCCH resource adaptation for low-priority scheduling to indicate that PDCCH
scheduling resource adaptation does not depend on aperiodic CSI scheduling
MOD NRDUCELLPDCCH: NrDuCellId=0, PDCCHAdjRefResType=APERIODIC_CSI_CCE_RES-0;
//Setting the NonDtxSinrIncreaseStep parameter for PDCCH aggregation level optimization in the case of
consecutive non-DTXs
MOD NRDUCELLPDCCH: NrDuCellId=0, NonDtxSinrIncreaseStep=0;
//Enabling mmWave PDCCH coverage enhancement (in weak-coverage and light-load scenarios)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=MMW_PDCCH_COVERAGE_ENH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples for the sub-functions. You can determine whether to restore the settings of other parameters based on actual network conditions.

Low-frequency TDD command examples

```
//Setting Coreset0OffsetPol to DEFAULT
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, Coreset0OffsetPol=DEFAULT;
```

```
//Disabling optimized PDCCH symbol adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=PDCCH_SYM_ADAPT_OPT_SW-0;
//Disabling equivalent-CCE-usage-based PDCCH symbol adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=EQUIV_CCE_RATIO_SYM_ADPT_SW-0;
//Disabling PDCCH symbol number adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-0;
//Disabling PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-0;
//Disabling multi-BWP1 ratio statistics optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=MULTI_BWP1_RATIO_STAT_OPT_SW-0;
//Disabling the enhanced multi-BWP1 PDCCH transmission algorithm
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_ENH_SW-0;
//Disabling the multi-BWP1 PDCCH transmission algorithm
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-0;
//Disabling PDCCH uplink and downlink CCE sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-0;
//Disabling initial CCE aggregation level selection optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-0;
//Disabling initial CCE aggregation level selection optimization for multiple BWP1s
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-0;
//Disabling aggregation level optimization based on PUSCH NACK/DTX
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_NACK_DTX_AGG_LVL_OPT_SW-0;
//Disabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-0;
//Disabling PDCCH aggregation level optimization in the case of consecutive DTXs
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_AGG_LVL_OPT_SW-0;
//Disabling PDCCH resource allocation enhancement
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_RES_ALLOC_ENH_SW-0;
//Disabling PDCCH blind detection assignment optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-0;
//Disabling the PDCCH multi-symbol resource pool
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SYM_CORESET_SW-0;
//Disabling configuration of multiple PDCCH search spaces
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SEARCH_SPACE_SW-0;
//Disabling outer-loop robustness improvement for PDCCH aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_HARQ_DTX_AGG_LVL_OPT_SW-0;
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=PUSCH_ACK_DTX_DETECT_SW-0;
//Disabling PDCCH blind detection number optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_BLIND_DETECT_CNT_OPT_SW-0;
//Disabling centralized PDCCH CCE resource allocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_CCE_CENTRAL_ALLOC_SW-0;
//Disabling PDCCH CCE ratio adjustment in common slots
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_COMM_SLOT_RATIO_ADJ_SW-0;
//Disabling PDCCH uplink and downlink common CCE sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_COMM_CCE_SHARING_SW-0;
//Disabling transition from RRC reestablishment to RRC setup
MOD NRCELLALGOSWITCH: NrCellId=0, ProcessSwitch=RRC_REESTAB_TO_SETUP_SW-0;
//Disabling CCE resource optimization for low-priority scheduling based on uplink common DCI
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=UL_COMM_LOW_PRI_CCE_UNRSV_SW-0;
//Disabling multi-BWP1 CCE allocation for top-priority UEs
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_BWP1_TOP_UE_CCE_ALLOC_SW-0;
//Disabling fast aggregation level reduction
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=AGG_LEVEL_FAST_DECREASE_SW-0;
//Disabling DTX filtering
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=DTX_FILTER_SW-0;
```

High-frequency TDD command examples

```
//Disabling PDCCH symbol number adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-0;
//Disabling PDCCH uplink-to-downlink CCE ratio adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_RATIO_ADAPT_SW-0;
//Disabling initial CCE aggregation level selection optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-0;
//Disabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-0;
//Disabling PDCCH antenna difference optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_ANT_DIFF_OPT_SW-0;
//Disabling mmWave PDCCH coverage enhancement
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=MMW_PDCCH_COVERAGE_ENH_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the following performance indicators to check whether each sub-function of PDCCH channel management has taken effect.

Table 5-4 Indicators related to the sub-functions involved in PDCCH channel management

Indicator	Formula
Average uplink cell throughput	Cell Uplink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average uplink CCE aggregation level	$(1 \times N.CCE.UL.AggLvl1Num + 2 \times N.CCE.UL.AggLvl2Num + 4 \times N.CCE.UL.AggLvl4Num + 8 \times N.CCE.UL.AggLvl8Num + 16 \times N.CCE.UL.AggLvl16Num) / (N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)$
Average downlink CCE aggregation level	$(1 \times N.CCE.DL.AggLvl1Num + 2 \times N.CCE.DL.AggLvl2Num + 4 \times N.CCE.DL.AggLvl4Num + 8 \times N.CCE.DL.AggLvl8Num + 16 \times N.CCE.DL.AggLvl16Num) / (N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)$
Uplink PRB usage	$N.PR.B.UL.Used.Avg / N.PR.B.UL.Avail.Avg$
Downlink PRB usage	$N.PR.B.DL.Used.Avg / N.PR.B.DL.Avail.Avg$
Number of available CCEs	$N.CCE.Avail.Avg$

Indicator	Formula
CCE usage	$N.CCE.Used.Avg/N.CCE.Avail.Avg$
Uplink CCE allocation success rate	$(N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)/N.CCE.UL.AllocReq.Num$
Downlink CCE allocation success rate	$(N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)/N.CCE.DL.AllocReq.Num$
Number of UEs scheduled in the uplink	$N.User.Schedule.UL.Sum$
Number of UEs scheduled in the downlink	$N.User.Schedule.DL.Sum$
Uplink cell traffic volume	$N.ThpVol.UL.Cell$
Uplink transmission duration in the cell	$N.ThpTime.UL.Cell$
Equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted	$N.CCE.UL.Used.Equivalent/N.CCE.UL.Avail.Equivalent$
Difference between the number of Msg3 receptions and the number of Msg4 responses	$N.RA.Contention.Msg3 - N.RA.Contention.Resolution.Succ$
RRC reestablishment rate	$N.RRC.ReEst.Att/(N.RRC.SetupReq.Msg + N.RRC.ReEst.Att)$
Contention-based access success rate	$N.RA.Contention.Resolution.Succ/N.RA.Contention.Att \times 100\%$

5.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

6 PUCCH Channel Management

6.1 Principles

6.1.1 Overview

The PUCCH carries uplink control information (UCI), which contains the following:

- Acknowledgement (ACK) and negative acknowledgement (NACK) feedback
Hybrid automatic repeat request (HARQ) ACK and NACK feedback for PDSCH demodulation
- Channel state information (CSI)
CSI-RS measurement results, such as CQI, precoding matrix indication (PMI), and rank indication (RI)

 NOTE

CSI-RS refers to CSI-RS for channel measurement (CSI-RS for CM), which is used for downlink channel state measurement. For more information on CSI-RS for CM, see [7 CSI-RS Resource Management](#).

- Uplink scheduling request (SR)
Resource request for PUSCH scheduling

UCI formats include format 0, format 1, format 2, format 3, and format 4. [Table 6-1](#) lists the characteristics of different PUCCH formats. For more information, see section 6.3 "Uplink control information" in 3GPP TS 38.212 (Release 15).

Table 6-1 Different PUCCH formats

PUCCH Format	Format Type	Number of Occupied Symbols	Number of Occupied PRBs per UE	Transmitted Data Volume	Application Scenario
Format 0	Short format	1 or 2	1	Small amount of data (≤ 2 bits)	Ultra-low latency
Format 1	Long format	4 to 14	1	Small amount of data (≤ 2 bits)	Coverage improvement
Format 2	Short format	1 or 2	1 to 16	Large amount of data (> 2 bits)	Ultra-low latency
Format 3	Long format	4 to 14	1 to 6, 8 to 10, 12, 15, or 16	Large amount of data (> 2 bits)	Coverage improvement
Format 4	Long format	4 to 14	1	Medium amount of data (> 2 bits)	Coverage improvement and UE capacity increase

PUCCH channel management works as follows:

1. The gNodeB performs cell-level PUCCH configuration. For more information, see [6.1.2 Cell-Level PUCCH Resource Configuration](#) and [6.1.3 Cell-Level PUCCH Resource Configuration Enhancement \(Low-Frequency TDD/High-Frequency TDD\)](#).
2. The gNodeB allocates resources to UEs based on cell-level PUCCH configurations. For more information, see [6.1.4 PUCCH Resource Configuration Modes](#).
3. Based on the resources allocated by the gNodeB, UEs select a proper resource unit for UCI transmission over the PUCCH. For more information, see [6.1.5 Resource Unit Selection by UEs](#).

For other PUCCH-related functions, see [6.1.6 UE Compatibility Processing](#), [6.1.7 PUCCH Reception Enhancement \(Low-Frequency TDD\)](#), and [6.1.8 PUCCH Modulation Scheme](#).

6.1.2 Cell-Level PUCCH Resource Configuration

PUCCH time-frequency resources are configured on a per cell basis. Cell-level PUCCH time-frequency formats include short formats and long formats, depending on the setting of the **NRDUCellPucch.StructureType** parameter. In the current version, the formats supported by each RAT are as follows:

In low-frequency TDD, both short and long formats are supported.

In high-frequency TDD, only short formats are supported.

6.1.2.1 Short Format (Low-Frequency TDD/High-Frequency TDD)

Short PUCCH formats include format 0 and format 2, with their time-frequency formats illustrated in Figure 6-1 and Figure 6-2, respectively.

Figure 6-1 Per-RB time-frequency format in format 0

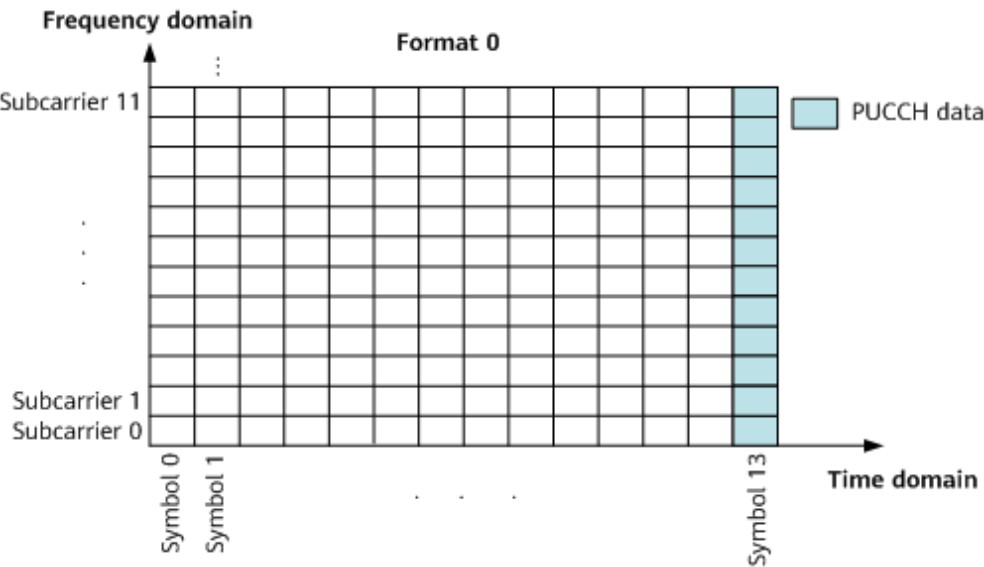
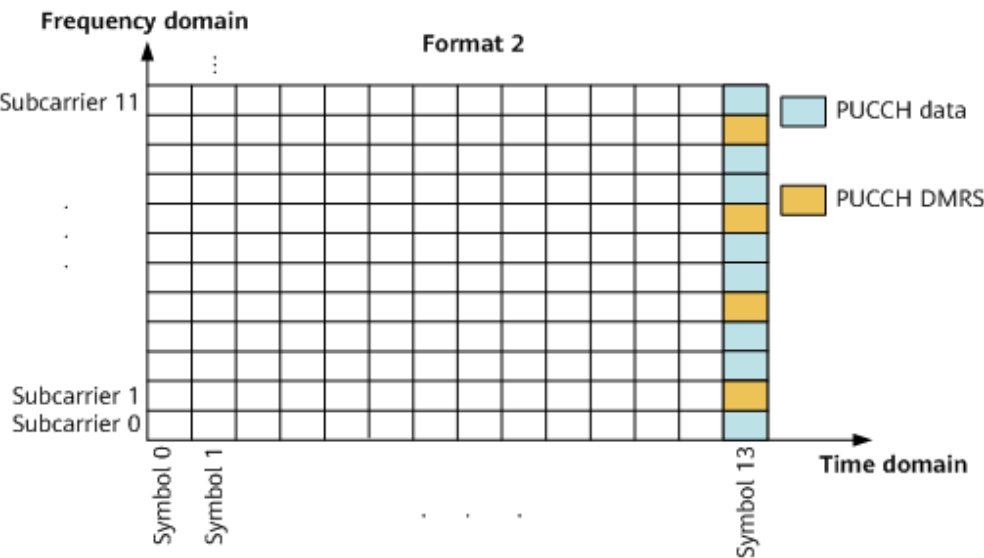


Figure 6-2 Per-RB time-frequency format in format 2



When a short format is used in low-frequency TDD, the time-domain resources are specified by the **NRDUCellPucch.ShortPucchSymbolNum** parameter, which can only be set to **SYMBOL1** or **SYMBOL2**, and a full bandwidth is used in the frequency domain.

As only short formats are supported in high frequency bands, static configuration and adaptive configuration are supported in the time domain, and a full bandwidth is used in the frequency domain.

- Static configuration: If the **PUCCH_RES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is deselected, the time-domain range is specified by the **NRDUCellPucch.ShortPucchSymbolNum** parameter.
- Adaptive configuration: PUCCH symbol number adaptation takes effect for a cell only when the **PUCCH_RES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is selected, the **NRDUCellPucch.ShortPucchSymbolNum** parameter is not set to **SYMBOL1** or **SYMBOL2**, and the sector corresponding to the cell serves at least 20 RRC_CONNECTED UEs.

After this function takes effect, the total number of symbols configured for the PUCCH remains unchanged. However, the number of symbols available for the PUCCH is adaptively adjusted based on the requirements of PDSCH scheduling for HARQ feedback and the requirements of PUSCH scheduling for symbols. When a small amount of HARQ feedback is required for PDSCH scheduling but a large number of symbols are required for PUSCH scheduling, the gNodeB reduces the number of symbols available for the PUCCH and allocates some symbols occupied by the PUCCH to the PUSCH, increasing the uplink cell throughput.

6.1.2.2 Long Format (Low-Frequency TDD)

Long PUCCH formats include format 1, format 3, and format 4, with their time-frequency formats illustrated in [Figure 6-3](#) and [Figure 6-4](#), separately.

Figure 6-3 Per-RB time-frequency format in format 1

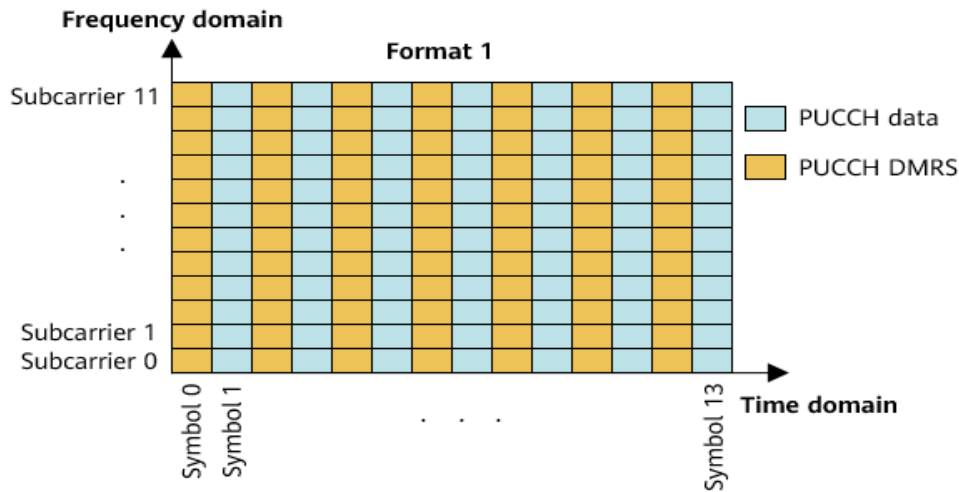
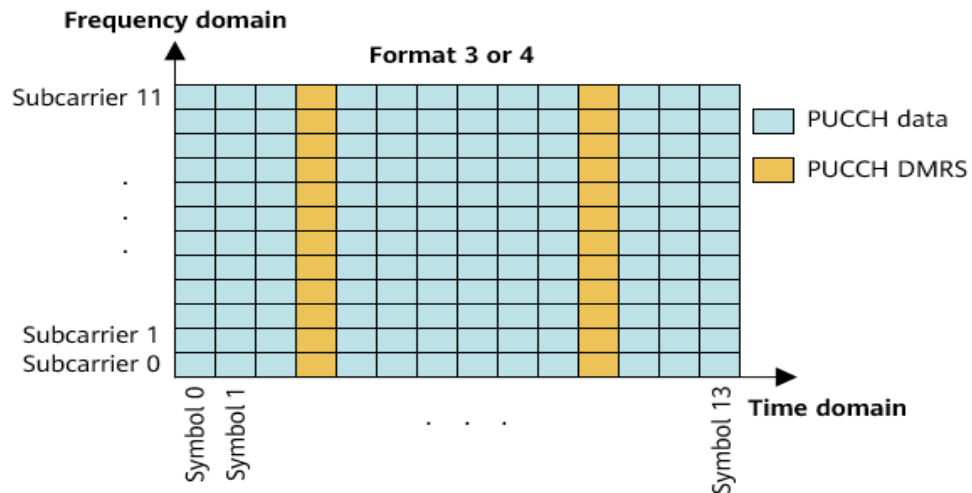


Figure 6-4 Per-RB time-frequency format in format 3 or format 4



Specifically, format 1, format 3, and format 4 always use 14 symbols in the time domain, and support adaptive adjustment and static configuration of PUCCH RBs in the frequency domain. PUCCH RB adaptation is controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter.

- When PUCCH RB adaptation is enabled, the number of RBs for the PUCCH in the frequency domain is adaptively adjusted based on the number of UEs and load level.
- When PUCCH RB adaptation is disabled, the PUCCH resources in the frequency domain are statically configured using the following parameters:
 - Number of RBs for ACK and NACK feedback
 - **NRDUCellPucch.Format1RbNum**
 - **NRDUCellPucch.Format3RbNum**
 - **NRDUCellPucch.Format4RbNum**
 - Number of dedicated RBs for CSI reporting
 - **NRDUCellPucch.CsiDedicatedRbNum**
 - **NRDUCellPucch.Format4CsiDedicatedRbNum**

The total number of RBs configured for the PUCCH is limited in specific scenarios. During configuration, the number of RBs in each format must be properly planned as follows:

- For a 20 MHz NR TDD cell, no more than 26 RBs can be configured for the PUCCH.
- If a supplementary uplink (SUL) cell with LTE and NR spectrum sharing disabled provides a bandwidth of 10 MHz, 15 MHz, or 20 MHz, no more than 30 RBs can be configured for the PUCCH.

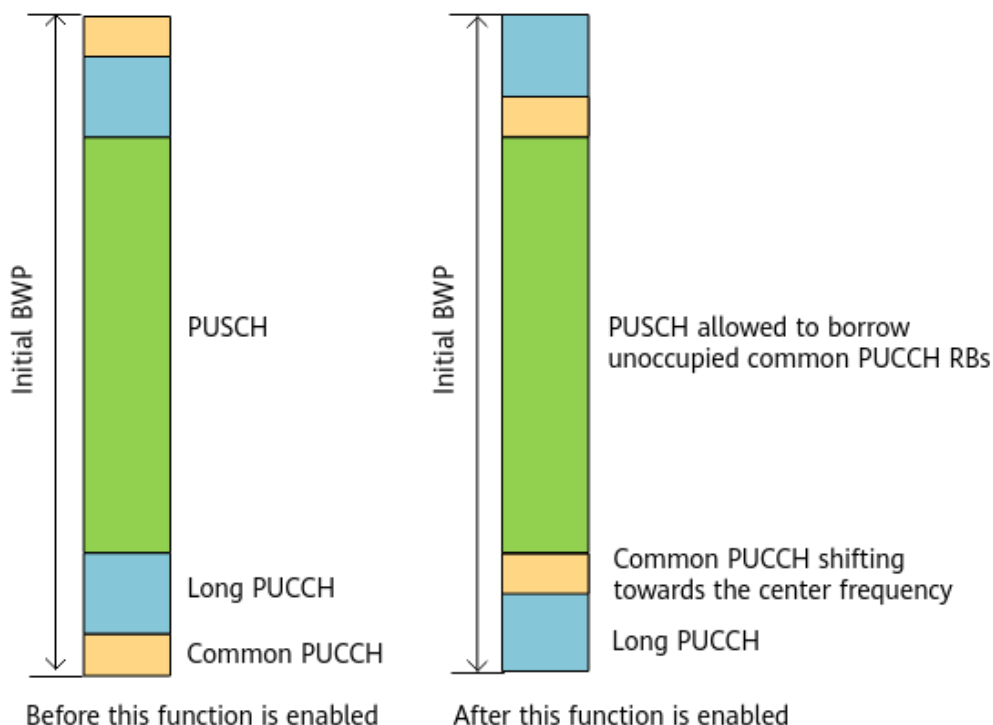
6.1.3 Cell-Level PUCCH Resource Configuration Enhancement (Low-Frequency TDD/High-Frequency TDD)

6.1.3.1 Inward Frequency Shifting of the Common PUCCH (Low-Frequency TDD)

The common PUCCH is indicated by *pucch-ResourceCommon*. According to Table 9.2.1-1: PUCCH resource sets before dedicated PUCCH resource configuration in 3GPP TS 38.213 V16.3.1, 16 resource sets (indicated by indexes 0 to 15) are defined for the common PUCCH. Currently, Huawei base stations use the resource set with index 12, indicating that the common PUCCH always occupies four RBs in the frequency domain.

In light-load scenarios, the common PUCCH may have unoccupied RBs due to low RB usage. To increase the RB usage, inward frequency shifting of the common PUCCH is introduced in low-frequency TDD. As shown in [Figure 6-5](#), with this function, the position of the common PUCCH is shifted towards the center frequency of the initial BWP, allowing PUSCH scheduling to borrow PUCCH RBs. This function is controlled by the **COMMON_PUCCH_POS_INWARD_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. It is recommended that this function be enabled for both the local and neighboring cells. Otherwise, the frequency-domain resources of the common PUCCHs between cells are not aligned, increasing the interference to these channels.

Figure 6-5 Inward frequency shifting of the common PUCCH



6.1.3.2 Quadruple Code Division for Format-4 Resource Allocation (Low-Frequency TDD)

Quadruple code division for format-4 resource allocation is supported in low frequency bands, and is controlled by the **FORMAT4_OCC_LENGTH_N4_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter.

- When this option is selected, quadruple code division for format-4 resource allocation is enabled. In this case, quadruple code division is used for format-4 resource allocation, and four code channels are configured for each RB. After this function is enabled, more RRC_CONNECTED UEs and more UEs to be scheduled in the downlink are allowed, but interference between UEs may increase.
- When this option is deselected, quadruple code division for format-4 resource allocation is disabled. In this case, double code division is used for format-4 resource allocation, and two code channels are configured for each RB.

Quadruple code division for format-4 resource allocation applies only to low-frequency TDD cells/SUL cells with 30 kHz SCS.

6.1.3.3 Format-1 SR Code Channel Allocation Optimization (Low-Frequency TDD)

Format-1 SR code channel allocation optimization is supported in low-frequency TDD, and is controlled by the **FORMAT1_SR_CHN_ALLOC_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter.

- When the **FORMAT1_SR_CHN_ALLOC_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is selected, format-1 SR code channel allocation optimization is enabled. In this case, the following apply:
 - When the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is deselected, the number of format-1 RBs is specified by the **NRDUCellPucch.Format1RbNum** parameter, in which case 15 SR code channels are configured for each RB.
 - When the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is selected, the number of format-1 RBs is adaptively configured based on the number of UEs and load level. When a cell serves a large number of UEs or is under heavy load, the gNodeB configures additional RBs for format 1. In this case, 24 SR code channels are configured for each additional RB and 10 SR code channels are configured for each original RB. When a cell serves a small number of UEs or is under light load, the same number of RBs is always configured for format 1, and 10 SR code channels are configured for each RB.

After format-1 SR code channel allocation optimization is enabled, the maximum number of UEs that can access a cell increases. As the number of UEs in the cell and the number of UEs scheduled in the downlink increase, the probability of HARQ resource conflicts increases, and the number of downlink scheduling times may decrease.

- When the **FORMAT1_SR_CHN_ALLOC_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is deselected, format-1 SR code channel allocation optimization is disabled. In this case, the following apply:

- When the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is deselected, the number of format-1 RBs is specified by the **NRDUCellPucch.Format1RbNum** parameter, in which case 10 SR code channels are configured for each RB.
- When the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is selected, the same number of RBs is always configured for format 1, and 10 SR code channels are configured for each RB.

6.1.3.4 Format-1 SR Code Channel Allocation in RB Saving Mode (Low-Frequency TDD)

Format-1 SR code channel allocation in RB saving mode is supported in low-frequency TDD scenarios, and is controlled by the **SR_CHN_RB_SAVING_MODE_ALLOC_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter.

- When the **SR_CHN_RB_SAVING_MODE_ALLOC_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is selected, SR code channels are allocated in RB saving mode. During SR code channel allocation for UEs, only after all available SR code channels on an RB among format-1 resources are allocated can those on another RB be allocated. Both odd- and even-numbered SR code channels are available for low-frequency TDD cells/SUL cells. When format-1 SR code channel allocation in RB saving mode is enabled in a cell serving a small number of UEs, the number of UEs for which code division multiplexing is performed on each RB increases, and the number of RBs consumed by the PUCCH in the cell decreases. As such, more UEs are allowed to access the network when a single baseband chip serves multiple cells.
- When the **SR_CHN_RB_SAVING_MODE_ALLOC_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is deselected, SR code channels are not allocated in RB saving mode. For low-frequency TDD cells/SUL cells, the base station allocates odd-numbered SR code channels on all RBs of format-1 resources only after the even-numbered SR code channels on these RBs are used up.

This function applies only to low-frequency TDD cells/SUL cells.

6.1.3.5 Format-1 HARQ-ACK Resource Scheduling in RB Saving Mode (Low-Frequency TDD)

Format-1 HARQ-ACK resource scheduling in RB saving mode is supported in low-frequency TDD, and is controlled by the **HARQ_ACK_RB_SAVING_SCH_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter.

- When the **HARQ_ACK_RB_SAVING_SCH_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is selected, HARQ-ACK resources are scheduled in RB saving mode. During HARQ-ACK resource allocation for UEs, only the RBs occupied by UEs sending SR are available for scheduling.

When format-1 HARQ-ACK resource scheduling in RB saving mode is enabled in a cell serving a small number of UEs, the number of UEs for which code division multiplexing is performed on each RB increases, and the number of

RBs consumed by the PUCCH in the cell decreases. As such, more UEs are allowed to access the network when a single baseband chip serves multiple cells. However, in a light- or medium-load cell serving a majority of active UEs in the downlink, the probability of HARQ-ACK resource conflicts increases. As a result, the average number of UEs scheduled in the downlink in the cell, average downlink cell throughput, and downlink user-perceived throughput may decrease. The average transmission delay of the first packet that enters the empty buffer, average RLC processing delay of downlink data packets, and DU service transmission delay may be prolonged.

- When the **HARQ_ACK_RB_SAVING_SCH_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is deselected, HARQ-ACK resources are not scheduled in RB saving mode. During HARQ-ACK resource allocation for UEs, all RBs allocated to UEs are available for scheduling.

6.1.3.6 Frequency-Domain Interference Optimization for PUCCH Format 1 (Low-Frequency TDD)

- Interference optimization for format-1 SR allocation
Interference optimization for format-1 SR allocation is supported in low-frequency cells, and is controlled by the **SR_ALLOC_INTRF_OPT_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter.
 - When the **SR_ALLOC_INTRF_OPT_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is selected, the frequency-domain resources allocated to format-1 SRs for different cells are staggered based on the odd-even PCI principle. This reduces inter-cell interference to format-1 SRs and improves the format-1 SR demodulation performance.
 - When the **SR_ALLOC_INTRF_OPT_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is deselected, the frequency-domain resources allocated to format-1 SRs for different cells are aligned.
- Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3
Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 is supported in low-frequency TDD cells, and is controlled by the **FORMAT1_FREQ_INTRF_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter.
 - If the **FORMAT1_FREQ_INTRF_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is selected, the frequency-domain resources in PUCCH format 1 for different cells are staggered based on the PCI modulo 3 value. This reduces inter-cell interference to format-1 resources and improves the format-1 demodulation performance.
 - If the **FORMAT1_FREQ_INTRF_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter is deselected, the frequency-domain resources in PUCCH format 1 for different cells are aligned.

6.1.3.7 PUCCH Resource Reduction for Heavy-Traffic UEs in Single-Beam Scenarios (High-Frequency TDD)

In high-frequency TDD, for NR cells with single-beam coverage (**NRDUCellTrpMmwaveBeam.CoverageScenario** set to **SCENARIO_103**) and **NRDUCell.SlotAssignment** set to **4_1_DDDSU**, PUCCH resource reduction for heavy-traffic UEs is supported. This function is controlled by the **NRDUCellPucch.PucchResrcRednMthd** parameter. The parameter settings must be consistent within a sector. Otherwise, this function may not take effect in some cells. In addition, the number of RRC_CONNECTED UEs in a sector must be less than or equal to four times the total number of AAUs serving the sector.

- When the **NRDUCellPucch.PucchResrcRednMthd** parameter is set to **MASS_FLOW** in a cell serving heavy-traffic UEs, PUCCH resource reduction for heavy-traffic UEs takes effect, reducing the PUCCH resource consumption in the cell.
- When the **NRDUCellPucch.PucchResrcRednMthd** parameter is set to **OFF**, PUCCH resource reduction is disabled.

6.1.3.8 PUCCH Format-1 RB Resource Control (Low-Frequency TDD)

PUCCH format-1 RB resource control is controlled by the **FORMAT1_RB_RES_CTRL_SW** option of the **gNodeBAlgo.PucchSwitch** parameter. When the **FORMAT1_RB_RES_CTRL_SW** option of the **gNodeBAlgo.PucchSwitch** parameter is selected, dynamic allocation control applies to format-1 RB resources on a baseband processing unit. When this option is deselected, this function does not take effect. PUCCH format-1 RB resource control applies only to low-frequency TDD cells and SUL cells.

6.1.3.9 Inter-TRP RB Resource Sharing (Low-Frequency TDD)

In low-frequency TDD, inter-TRP RB resource sharing is introduced and is controlled by the **INTER_TRP_RB_SHARE_POLICY_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. If this option is selected for a cell with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**, a single-TRP mechanism is used for PUCCH RB resource sharing. If this option is selected for a cell with **NRDUCell.NrDuCellNetworkingMode** set to a value other than **NORMAL_CELL**, a multi-TRP mechanism is used for PUCCH RB resource sharing.

6.1.3.10 PUCCH RB Adaptation Optimization (Low-Frequency TDD)

In low-frequency TDD, when PUCCH RB adaptation is enabled (by selecting the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter), the number of RBs for the PUCCH in the frequency domain is adaptively adjusted based on the number of UEs and load level. However, if the cell serves only a small number of UEs, reserving a large number of PUCCH RBs is a waste of resources. To further optimize the use of PUCCH RBs and increase RB usage, PUCCH RB adaptation optimization is introduced. This function is controlled by the **PUCCH_RBRES_ADAPT_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. Selecting this option reduces the number of RBs used for format-1 periodic CSI reporting when PUCCH RB adaptation is enabled.

6.1.3.11 Refined Control of Format-1 Resources

In low frequency bands, to optimize format-1 RB allocation on a BBP by freeing up some RBs previously occupied by format 1, refined control of format-1 resources can be enabled by selecting the **F1_RES_REFINED_CTRL_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. This function enables refined management of format-1 resources based on the number of activated cells and the number of configured format-1 RBs on the BBP.

6.1.3.12 Format-3 and Format-4 ACK RB Resource Adaptation (Low-Frequency TDD)

In low-frequency TDD, format-3 and format-4 ACK RB resource adaptation (controlled by the **F3_AND_F4_ACK_RB_RES_ADAPT_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter) is introduced to reduce the number of RBs used for format-3 and format-4 ACKs after PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter) is enabled. After format-3 and format-4 ACK RB resource adaptation is enabled, its effectiveness depends on the parameter settings listed in [Table 6-2](#). After this function takes effect, the RBs allocated for format-3 and format-4 ACKs are adaptively adjusted based on the resource usage. Format 3 can use four, six, or eight RBs, while format 4 can use two, four, or six RBs.

Table 6-2 Thresholds for enabling format-3 and format-4 ACK RB resource adaptation

Threshold	Parameter	Effective Conditions
Resource usage threshold for format-3 and format-4 RB adaptation	NRDUCellPucch.F3AndF4HarqResUseRatio	Format-3 and format-4 ACK RB resource adaptation takes effect when the resource usage exceeds the NRDUCellPucch.F3AndF4HarqResUseRatio parameter value and the resource allocation failure rate exceeds the NRDUCellPucch.F3AndF4HarqResFailRatio parameter value.
Resource allocation failure rate threshold for format-3 and format-4 RB adaptation	NRDUCellPucch.F3AndF4HarqResFailRatio	

6.1.3.13 Disabling Frequency Hopping for Format-4 Resources (Low-Frequency TDD)

According to section 6.3.2.6 "PUCCH formats 3 and 4" in 3GPP TS 38.211 V18.5.0, format 4 supports code division multiplexing (CDM), while format 3 applies stricter CDM constraints. CDM enables more UEs to share a single RB, thereby reducing RB consumption. In low-frequency TDD, disabling frequency hopping for format-4 resources is introduced to allow more UEs to use format 4 when PUCCH RB consumption is high. This function is controlled by the **F4_NO_FREQ_HOPPING_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch**

parameter. If this option is selected, frequency hopping for format-4 resources is disabled, allowing adaptive selection between format-4 and format-3 ACK RBs based on the PUCCH SINR of UEs. The SINR threshold for switching between format-4 and format-3 ACK RBs is specified by the **NRDUCellPucch.F3AndF4HarqSinrThld** parameter. A larger value of this parameter increases the probability of UEs using format-3 ACK RBs, while a smaller value increases the probability of UEs using format-4 ACK RBs.

6.1.3.14 HARQ Resource Increase

In low frequency bands, to increase the number of format-3 and format-4 RBs in a cell serving UEs that are transmitting a high volume of data, HARQ resource increase can be enabled by selecting the **HARQ_SPEC_INCREASE_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. This function allows the cell to use more format-3 and format-4 resources for HARQ feedback.

6.1.3.15 Maximized Search Within HARQ Resource Set 1

In low frequency bands, to increase the probability of successful format-3 and format-4 RB allocation for ACKs in a cell with limited RB resources, maximized search within HARQ resource set 1 can be enabled by selecting the **HARQ_RES_SET1_MAX_SRCH_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. With this function, the base station searches for resources in HARQ resource set 1 for multiple times for UEs requesting format-3 and format-4 resources for HARQ feedback, increasing the request success rate.

6.1.3.16 PUCCH Symbol Length Number Increase

To minimize signal interference, PUCCH signals can be punctured through resource mapping. For details about puncturing, see section 6.3.2.4.1 "Sequence modulation" in 3GPP TS 38.211 V18.6.0. As more features and functions evolve, the base station is expected to support additional puncturing schemes. To enable this, the PUCCH symbol length number increase function is introduced. This function is controlled by the **PUCCH_SYM_TYPE_NUM_INCR_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. It increases the number of configurable PUCCH puncturing schemes in each TTI, allowing support for more PUCCH symbol lengths per TTI.

6.1.4 PUCCH Resource Configuration Modes

According to 3GPP specifications, a gNodeB can configure PUCCH resources for UEs through RRC signaling messages, semi-persistently (activation and deactivation through the MAC CE), or adaptively (through PDCCH DCI). MAC CE is short for Media Access Control control element.

3GPP defines the following restrictions on PUCCH resource configuration:

- PUCCH resources for SR transmission can be configured only through RRC signaling messages.
- PUCCH resources for periodic CSI transmission can be configured either through RRC signaling messages or semi-persistently.
- PUCCH resources for ACK and NACK transmission can be configured either through RRC signaling messages or adaptively.

The current version is subject to the following restrictions on PUCCH resource configuration:

- **For SR Transmission:** PUCCH resources can only be configured statically.
- **For CSI Reporting:** PUCCH resources can only be configured statically.
- **For ACK/NACK Transmission:** PUCCH resources can only be configured adaptively.

 NOTE

For more information on SR resource configuration, see the description of `SchedulingRequestResourceConfig` in section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V16.0.0.

For SR Transmission

PUCCH resources can only be configured statically. In static PUCCH resource allocation for SR transmission, the gNodeB allocates a set of resources to a UE through a higher-layer RRC message, including configurations such as the time-frequency position, index ID, start symbol, and number of symbols.

The SR period can be configured either statically or adaptively.

- Static SR period configuration: When the **SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is deselected, the SR period is always specified by the **NRDUCellPucch.SrPeriod** parameter.
- SR period adaptation: When the **SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter is selected, the gNodeB adaptively adjusts the SR period based on the number of UEs in the cell and the number of UEs served by the BBP. If a cell serves a small number of UEs, a short period is configured for newly admitted UEs. However, as the number of UEs served by the cell increases, the gNodeB configures a longer SR period for some of the UEs originally configured with short periods. This ensures that the number of UEs in a cell can reach the allowed maximum number. When the BBP serving a cell serves a large number of UEs, only a long period can be configured for newly admitted UEs even if the cell serves a small number of UEs.
 - Enhanced SR period adaptation: When SR period adaptation is enabled, enhanced SR period adaptation is introduced to allow the base station to configure a short period for newly admitted UEs in a cell serving a small number of UEs. This function is controlled by the **ENHANCED_SR_PERIOD_ADAPT_SW** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter.
 - When this option is selected for a cell serving a small number of UEs, the gNodeB adaptively adjusts the SR period based only on the number of UEs in the cell, regardless of the number of UEs served by the BBP serving the cell. When this option is selected for a cell serving a large number of UEs, the gNodeB adaptively adjusts the SR period based on the number of UEs in the cell and the number of UEs served by the BBP.

After enhanced SR period adaptation is enabled, the access delay of newly admitted UEs is shortened. However, for low-frequency TDD cells with a bandwidth of 60 MHz or higher, fewer PUSCH RBs will be

available as more PUCCH RBs are required, thereby decreasing the average uplink cell throughput.

- When this option is deselected, the gNodeB adaptively adjusts the SR period based on the number of UEs in the cell and the number of UEs served by the BBP.
- Minimum SR period adjustment for adaptation: When SR period adaptation is enabled, the minimum SR period for adaptation can be further adjusted using the **SR_PERIOD_OPT_SWITCH** option of the **NRDUCellPucch.SrResoureAlgoSwitch** parameter.
 - When this option is selected, the minimum SR period for adaptation decreases, reducing the delay for UEs.
 - When this option is deselected, the minimum SR period for adaptation remains unchanged.
- RLC timer adaptation along with the SR period: When SR period adaptation is enabled, the number of RLC retransmissions is likely to reach the maximum value before this happens with SR retransmissions, leading to service drops. To prevent this, RLC timer adaptation along with the SR period is introduced. This function is controlled by the **RLC_TIMER_ADAPT_WITH_SR_PRD_SW** option of the **NRDUCellServExp.RlcSwitch** parameter.
 - When this option is selected, the RLC timer length that takes effect is adaptively changed with the SR period.
 - When this option is deselected, configure gNodeB RLC parameters by referring to [Table 6-3](#).

Table 6-3 Recommended gNodeB RLC parameter settings when SR period adaptation is enabled

Parameter Name	Parameter ID	Recommended Value
Polling PDU Retransmission Timer for UE	gNBRlcParamGroup.UePollingPduRetransTimer	MS100(100)
Polling PDU Retransmission Timer for gNodeB	gNBRlcParamGroup.gNBPollingPduRetransTimer	MS100(100)
AM Status Report Prohibit Timer for UE	gNBRlcParamGroup.UeAmStatusRptProhibitTmr	MS100(100)
AM Status Report Prohibit Timer for gNodeB	gNBRlcParamGroup.gNBAmStatusRptProhibitTmr	MS100(100)

If a UE does not receive the DCI carrying uplink grants delivered by the gNodeB after sending an SR to the gNodeB, it will repeatedly send the SR at regular

intervals. To prevent UEs from frequently sending SRs, the sr-ProhibitTimer (SR prohibit timer) is introduced, as stipulated in section 5.4.4 "Scheduling Request" in 3GPP TS 38.321 V16.2.1. The SR prohibit timer starts immediately after a UE sends an SR to the gNodeB.

- If the UE receives the DCI carrying uplink grants delivered by the gNodeB, the SR prohibit timer stops and PUSCH transmission starts.
- If the UE does not receive the DCI carrying uplink grants delivered by the gNodeB, it will resend the SR to the gNodeB after the SR prohibit timer expires.

For Huawei gNodeBs, the SR prohibit timer can be configured using the **NRDUCellPucch.SrProhibitTimer** parameter.

- If this parameter is set to **MS0**, the SR prohibit timer does not take effect for UEs in the cell.
- Any other value of this parameter indicates the length of the SR prohibit timer.
 - If this parameter is set to a value greater than the SR period and the gNodeB does not promptly respond to the SR sent by a UE, the UE will miss multiple opportunities to send PUSCH data, so data will take a long time to buffer.
 - If this parameter is set to a value less than the SR period and the gNodeB does not respond to the SR sent by a UE, the UE will repeatedly send the SR based on the configured SR period, which may cause UE resynchronization.

Modification of this parameter takes effect only for UEs that access the network after the modification.

For CSI Reporting

PUCCH resources can only be configured statically. In static PUCCH resource allocation for CSI reporting, the gNodeB allocates a set of resources to a UE through a higher-layer RRC message, including configurations such as the time-frequency position, index ID, start symbol, and number of symbols.

The CSI reporting period for a low-frequency cell can be configured either statically or adaptively.

- Static configuration: When CSI reporting period adaptation (controlled by the **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.CsiResoureAlgoSwitch** parameter) is disabled, the CSI reporting period is always specified by the **NRDUCellPucch.CsiReportPeriod** parameter.

For TDD cells, the CSI reporting period is equal to the larger of either this parameter value or the **NRDUCellCsirs.CsiPeriod** parameter value.

For SUL cells, the CSI reporting period is equal to the larger of either this parameter value or half the **NRDUCellCsirs.CsiPeriod** parameter value.

- Adaptive configuration: When CSI reporting period adaptation (controlled by the **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.CsiResoureAlgoSwitch** parameter) is enabled, the gNodeB adaptively adjusts the CSI reporting period based on the number of UEs. If a

cell serves a small number of UEs, a short period is configured for newly admitted UEs. However, as the number of UEs served by the cell increases, the gNodeB configures a longer CSI reporting period for some of the UEs originally configured with short periods. This ensures that the number of UEs periodically transmitting CSI in a cell can reach the allowed maximum number. The maximum CSI reporting period for UEs is 160 ms.

In high-frequency cells, CSI can only be reported aperiodically, which is performed on the PUSCH. For details about the PUSCH, see *Uplink Scheduling*.

For ACK/NACK Transmission

PUCCH resources can be configured adaptively. In adaptive PUCCH resource allocation for ACK and NACK transmission:

1. The gNodeB allocates multiple (a maximum of four is defined in 3GPP specifications and two are supported in the current version) sets of resources to a UE through a higher-layer RRC message. Each resource set includes up to eight PUCCH resources, with each providing configurations such as the time-frequency position, index ID, start symbol, and number of symbols.
2. Based on PDSCH data scheduling, the gNodeB allocates one set of idle PUCCH resources to a UE to transmit HARQ feedback, and signals the resource set index through the DCI.

Scheduling optimization for HARQ-ACK resource set 0 is introduced to optimize the allocation and scheduling of resources in HARQ-ACK resource set 0, thereby improving the resource usage of HARQ-ACK resource set 0. This function is controlled by the **HARQ_ACK_RES_SET0_SCH_OPT_SW** option of the **NRDUCellPucch.PucchPerformanceSw** parameter. It is recommended that this function be enabled when a large number of UEs are scheduled in the downlink. Once enabled, this function allows more resources in HARQ-ACK resource set 0 to be scheduled, increasing the number of UEs that can be scheduled in the downlink in the cell as well as the average downlink cell throughput. However, as the number of UEs scheduled in the downlink increases, interference to neighboring cells may increase. In other scenarios, this function has no negative or positive effects.

NOTE

For more information on the resource set definition, see the description of PUCCH-Config in section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V16.0.0.

6.1.5 Resource Unit Selection by UEs

A UE selects appropriate time-frequency resources to transmit UCI based on the time-frequency resource range indicated in the resource set. The UE selects a resource unit based on the PUCCH time-frequency format.

- 3GPP specifications define the resource unit in format 0 (short format) as one RB.
- 3GPP specifications define the resource unit in format 2 (short format) as 1–16 RBs.
- 3GPP specifications define the resource unit in format 1 (long format) as one RB.

- 3GPP specifications define the resource unit in format 3 (long format) as 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, or 16 RBs. However, in the current version, the resource unit is only one RB in format 3 (long format).
- 3GPP specifications define the resource unit in format 4 (long format) as one RB.

6.1.6 UE Compatibility Processing

The format-4 capability (*pucch-F4-WithFH*) reported by a UE may be inconsistent with the actual capability of the UE. In this case, if the gNodeB configures format-4 resources for the UE, the UE will fail to send information on the PUCCH. To solve this problem, format-4 PUCCH capability adaptation is introduced. This function is controlled by the **PUCCH_F4_WITHFH_SW** option of the **NRDUCellPucch.UePucchCapabilityAlgoSw** parameter.

- If this option is selected, PUCCH format 4 does not take effect for UEs that report the format-4 capability over the air interface but do not actually support it. This means that format-4 PUCCH resources are not configured for such UEs. After this function is enabled, the downlink user-perceived throughput increases for such UEs. However, fewer UEs can be allocated periodic CSI resources in the cell. As the number of UEs served by the cell increases, the downlink cell throughput will decrease.
- If this option is deselected, PUCCH format 4 takes effect for UEs that report the format-4 capability over the air interface.

6.1.7 PUCCH Reception Enhancement (Low-Frequency TDD)

The gNodeB receives PUCCH signals through maximum ratio combining (MRC) or interference rejection combining (IRC). Both methods weight and combine the signals on receive antennas, but differ in weights.

- MRC weights and combines the signals on receive antennas based on the maximum ratio to maximize the signal-to-noise ratio (SNR).
- IRC weights and combines the signals on receive antennas based on interference rejection to maximize the SINR.

In this version, PUCCH MRC/IRC adaptation is supported in low frequency bands. This function is controlled by the **PUCCH_MRC_IRC_SW** option of the **NRDUCellPucch.PucchReceiveEnhSwitch** parameter. It takes effect only for long PUCCH formats.

- If this option is selected, the gNodeB adaptively selects MRC or IRC based on the interference and noise received in the uplink.
- If this option is deselected, the gNodeB always uses MRC.

6.1.8 PUCCH Modulation Scheme

In the current version, formats 0 and 2 are used as short PUCCH formats. No modulation scheme is used in format 0, and the modulation scheme quadrature phase shift keying (QPSK) is used in format 2.

As for long PUCCH formats, formats 1, 3, and 4 are used. The modulation scheme binary phase shift keying (BPSK) is used in format 1 with one bit, and the modulation scheme QPSK is used in format 1 with two bits as well as format 3 and format 4.

6.2 Network Analysis

6.2.1 Benefits

PUCCH channel management improves network performance by properly utilizing PUCCH resources.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).
[Table 6-4](#) describes the network impacts of each sub-function involved in PUCCH channel management. For the definitions of related indicators, see [6.4.2 Activation Verification](#).

Table 6-4 Network impacts of the sub-functions involved in PUCCH channel management

Sub-Function Name	Network Impacts
PUCCH symbol number adaptation	In high frequency bands, after this function takes effect, the average uplink cell throughput increases. As a result, the number of downlink RLC status reports or TCP response packets increases, which may decrease the downlink cell throughput and user-perceived rate.
PUCCH RB adaptation	In low frequency bands, after this function is enabled, PUCCH frequency-domain configurations will change, which may cause the random access success rate and reported CQI values to fluctuate.
Inward frequency shifting of the common PUCCH	In low-frequency TDD, after this function takes effect, the uplink user-perceived rates of large-packet UEs increase in light-load scenarios. However, in medium- and heavy-load scenarios, this function leads to increased UE access delay and fluctuations in the contention-based access success rate, downlink IBLER, and downlink RBLER. If the configurations for the local and neighboring cells are inconsistent, the frequency-domain resources of the common PUCCHs between cells are not aligned. As a result, the interference to these channels increases, affecting the average uplink and downlink UE throughputs.

Sub-Function Name	Network Impacts
Quadruple code division for format-4 resource allocation	In low frequency bands, after this function is enabled, quadruple code division is used for format-4 resource allocation. This allows more RRC_CONNECTED UEs and more UEs to be scheduled in the downlink, but may increase interference between UEs. It is recommended that this function be enabled only in scenarios with heavy CA loads. In other scenarios, enabling this function may decrease the average uplink and downlink cell throughputs, increase the service drop rate, and worsen the cell edge user experience. After this function takes effect in high-speed scenarios, the following changes occur compared to double code division: • Reduced format-4 RB consumption and more available PUSCH RBs may increase the average number of available PRBs for the PUSCH and cause fluctuations in the uplink PRB usage of UEs, total number of PRBs that are paired for uplink MIMO, and contention-based access success rate. • Potentially increased inter-UE interference on CSI and ACK may raise the N.U.L.NI.Avg counter value and cause fluctuations in the number of times the measured PUCCH SINR values fall into different ranges, average CQI, uplink and downlink IBLERs and RBLERs, and uplink and downlink MCS indexes.
Format-1 SR code channel allocation optimization	In low frequency bands, after this function is enabled, the maximum number of UEs that can access a cell increases. As the number of UEs in the cell and the number of UEs scheduled in the downlink increase, the probability of HARQ resource conflicts increases, and the number of downlink scheduling times may decrease.
Format-1 SR code channel allocation in RB saving mode	After this function is enabled in a cell serving a small number of UEs, the number of UEs for which code division multiplexing is performed on each RB increases, and the number of RBs consumed by the PUCCH in the cell decreases. As such, more UEs are allowed to access the network when a single baseband chip serves multiple cells.

Sub-Function Name	Network Impacts
Format-1 HARQ-ACK resource scheduling in RB saving mode	After this function is enabled in a cell serving a small number of UEs, the number of UEs for which code division multiplexing is performed on each RB increases, and the number of RBs consumed by the PUCCH in the cell decreases. As such, more UEs are allowed to access the network when a single baseband chip serves multiple cells. However, in a light- or medium-load cell serving a majority of active UEs in the downlink, the probability of HARQ-ACK resource conflicts increases. As a result, the average number of UEs scheduled in the downlink in the cell, average downlink cell throughput, and downlink user-perceived throughput may decrease. The average transmission delay of the first packet that enters the empty buffer, average RLC processing delay of downlink data packets, and DU service transmission delay may be prolonged. In low-frequency TDD, after this function is enabled, it may not take effect when the BBP serves a small number of cells.
Frequency-domain interference optimization for PUCCH format 1	Interference optimization for format-1 SR allocation and frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 are recommended in areas with strong inter-cell format-1 interference. They can reduce inter-cell interference to format-1 resources and improve the format-1 demodulation performance. Before enabling these functions, you are advised to properly plan PCIs based on their mod 3 values for the cells to be enabled with either of these functions. Otherwise, interference cannot be reduced.
PUCCH resource reduction for heavy-traffic UEs in single-beam scenarios	In high frequency bands, after this function takes effect, the average uplink cell throughput and average uplink UE throughput increase. If RRC reconfiguration is performed for a UE, temporary interference may occur during RRC reconfiguration, causing HARQ demodulation errors. This may affect the average downlink cell throughput and average downlink UE throughput.
PUCCH format-1 RB resource control	In low-frequency TDD, when this function is enabled, dynamic allocation control applies to format-1 RB resources on a baseband processing unit, allowing more UEs to access a cell with a large number of RRC_CONNECTED UEs (depending on live network conditions) or more than two RedCap BWPs, which may decrease the average uplink and downlink UE throughputs.

Sub-Function Name	Network Impacts
Inter-TRP RB resource sharing	In low-frequency TDD, when this function is enabled, the average downlink UE throughput fluctuates depending on the cell networking mode (specified by the NRDUCell.NrDuCellNetworkingMode parameter) and UE distribution in TRP coverage areas. Assume that NRDUCell.NrDuCellNetworkingMode is set to a value other than NORMAL_CELL for a cell. The average downlink UE throughput may increase when UEs are sparsely distributed in the coverage areas of multiple TRPs for the cell and may decrease when UEs are densely distributed in the coverage areas of some TRPs for the cell.
PUCCH RB adaptation optimization	In low-frequency TDD, after this function takes effect, PUCCH RB consumption is reduced. In light-load scenarios, the uplink peak rate of UEs increases and the reported CQI values may slightly fluctuate. In medium- and heavy-load scenarios, the uplink and downlink user-perceived rates decrease, the IBLERs for uplink and downlink scheduling fluctuate, the average CCE aggregation level increases, and the CCE usage and CCE allocation success rate fluctuate. In addition, the values of the following indicators may also fluctuate: contention-based access success rate, duration in which BWP2 is in effect in the cell, proportion of DRX sleep time in the cell, uplink and downlink RBLERs, and uplink and downlink MCS indexes.
Refined control of format-1 resources	After this function takes effect, some RBs previously occupied by format 1 are freed up, reducing the RBs used to transmit HARQ-ACKs with two or fewer bits. This allows the cell to serve more UEs transmitting HARQ-ACKs with more than two bits, increasing the number of successful PUCCH HARQ-ACK resource requests (N.PUCCH.ACK.AllocReq.Succ) and the average downlink cell and UE throughputs.

Sub-Function Name	Network Impacts
Format-3 and format-4 ACK RB resource adaptation	After this function takes effect, PUCCH RB consumption is reduced. In addition, the proportions of RBs used for format-3 and format-4 ACKs change, causing the PUCCH SINR to fluctuate and potential fluctuations in other indicators such as contention-based access success rate, uplink and downlink MCS indexes, and uplink and downlink RBLERs. In light-load scenarios, the uplink user-perceived rate increases. In medium- and heavy-load scenarios, the average uplink and downlink UE throughputs as well as the uplink and downlink IBLERs may fluctuate, and the average uplink and downlink CCE aggregation levels may increase. A smaller value of the NRDUCellPucch.F3AndF4HarqResUseRatio parameter results in a higher probability of triggering format-3 and format-4 ACK RB extension in the case of a low format-3 and format-4 ACK resource usage. A larger value of this parameter results in possible delayed format-3 and format-4 ACK RB extension. A smaller value of the NRDUCellPucch.F3AndF4HarqResFailRatio parameter results in a higher probability of triggering format-3 and format-4 ACK RB extension in the case of a low format-3 and format-4 ACK resource allocation failure rate. A larger value of this parameter results in possible delayed format-3 and format-4 ACK RB extension.
Disabling frequency hopping for format-4 resources	In low-frequency TDD, after this function takes effect, the average downlink UE throughput increases in extremely heavy-load scenarios or scenarios with good coverage and may decrease in other scenarios. In addition, if the NRDUCellPucch.F3AndF4HarqSinnThld parameter is set to an excessively large or small value, the average downlink cell throughput and average downlink UE throughput may decrease.
HARQ resource increase	After this function takes effect, additional format-3 and format-4 resources are available for HARQ feedback, increasing the RBs used to transmit HARQ-ACKs with more than two bits. This allows the cell to serve more UEs transmitting such HARQ-ACKs, increasing the number of successful PUCCH HARQ-ACK resource requests (N.PUCCH.ACK.AllocReq.Succ) and the average downlink cell and UE throughputs.
Maximized search within HARQ resource set 1	After this function takes effect, the probability of successful format-3 and format-4 RB allocation for ACKs increases, resulting in more successful PUCCH HARQ-ACK resource requests (N.PUCCH.ACK.AllocReq.Succ) and more RBs used to transmit HARQ-ACKs with more than two bits. This allows the cell to serve more UEs transmitting such HARQ-ACKs, increasing the average downlink cell and UE throughputs.

Sub-Function Name	Network Impacts
PUCCH symbol length number increase	After this function takes effect, more PUCCH puncturing schemes are allowed in each TTI, preventing some UEs from ACK feedback or CSI reporting failures and causing the following impacts on indicators. • The N.RLC.DL.Cell5QI.Delay, N.MAC.DL.Cell5QI.Delay, and N.DL.PDSCH.Tti.Num counter values and the uplink and downlink IBLERs and RBLERs remain unchanged or slightly decrease. • The number of times the measured PUCCH SINR values fall into different ranges fluctuates. • The average uplink and downlink cell throughputs and average uplink and downlink UE throughputs remain unchanged or slightly increase.
Scheduling optimization for HARQ-ACK resource set 0	After this function is enabled, more resources in HARQ-ACK resource set 0 can be scheduled, increasing the number of UEs that can be scheduled in the downlink in the cell as well as the average downlink cell throughput. However, as the number of UEs scheduled in the downlink increases, interference to neighboring cells may increase. In other scenarios, this function has no negative or positive effects.
Format-4 PUCCH capability adaptation	After this function is enabled, the downlink user-perceived throughput increases for UEs that report the format-4 PUCCH capability over the air interface but do not actually support it. However, fewer UEs can be allocated periodic CSI resources in the cell. As the number of UEs served by the cell increases, the downlink cell throughput will decrease.
PUCCH MRC/IRC adaptation	In low frequency bands, it is recommended that this function be enabled when there is strong interference. After this function is enabled, if there is uplink interference, the PUCCH coverage performance improves, and the average downlink throughput of UEs that face severe interference may increase. You can observe the N.UL.NI.Avg counter and create a cumulative distribution function (CDF) plot. If the value on the horizontal axis that corresponds to the value 50% on the vertical axis is greater than -111 dBm (for low-frequency TDD), there is strong interference. In this case, you are advised to enable PUCCH MRC/IRC adaptation.

6.3 Requirements

6.3.1 Licenses

This basic function is not under license control.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	PUCCH RB adaptation optimization (controlled by the PUCCH_RBRES_ADAPT_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter) can be enabled only when PUCCH RB adaptation is enabled.
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	Format-3 and format-4 ACK RB resource adaptation (controlled by the F3_AND_F4_ACK_RB_RES_ADAPT_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter) can be enabled only when PUCCH RB adaptation is enabled.
Low-frequency TDD	Long PUCCH format	NRDUCellPucch.StructureType	<i>Channel Management</i>	PUCCH MRC/IRC adaptation (controlled by the PUCCH_MRC_IRC_SW option of the NRDUCellPucch.PucchReceiveEnhSwitch parameter) can be enabled only when long PUCCH formats (with NRDUCellPucch.StructureType set to LONG_STRUCTURE) are used in the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Long PUCCH format	NRDUCellPucch.StructureType	<i>Channel Management</i>	Frequency-domain interference optimization for PUCCH format 1 (controlled by the FORMAT1_FREQ_INTRF_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter and the SR_ALLOC_INTRF_OPT_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) can be enabled only when long PUCCH formats (with NRDUCellPucch.StructureType set to LONG_STRUCTURE) are used in the cell.
Low-frequency TDD	Long PUCCH format	NRDUCellPucch.StructureType	<i>Channel Management</i>	Quadruple code division for format-4 resource allocation (controlled by the FORMAT4_OCC_LENGTH_N4_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) can be enabled only when long PUCCH formats (with NRDUCellPucch.StructureType set to LONG_STRUCTURE) are used in the cell.
Low-frequency TDD	Full-bandwidth initial BWP configuration	INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicySwitch parameter	<i>5G Networking and Signaling</i>	Inward frequency shifting of the common PUCCH (controlled by the COMMON_PUCCH_POS_INWARD_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter) can be enabled only when full-bandwidth initial BWP configuration is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PRACH frequency-domain start position	NRDUCell.PrachFreqStartPosition	<i>Random Access Control</i>	Inward frequency shifting of the common PUCCH (controlled by the COMMON_PUCCH_POS_INWARD_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter) can be enabled only when the parameter specifying the PRACH frequency-domain start position is set to 65535 or a value greater than or equal to 6 .
High-frequency TDD	None	None	None	None

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink Msg4 payload reduction	MSG4_PAYLOAD_REDUCE_SW option of the NRDUCellDisch.DLSchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Downlink Msg4 payload reduction must be disabled when NRDUCellPucch.PucchResrcRednMthd is set to MASS_FLOW and NRDUCellTrpMmwav-Beam.CoverageScenario is set to SCENARIO_103 .
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellPucch.StructureType parameter must be set to LONG_STRUCTURE .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	SRS resource management	NRDUCellSrs.SrsResStartPosition	None	PUCCH resource reduction must be disabled when NRDUCellPucch.ShortPucchSymbolNum is set to SYMBOL2 and NRDUCellSrs.SrsResStartPosition is set to 1.
High-frequency TDD	Short PUCCH Symbol Number	NRDUCellPucch.ShortPucchSymbolNum	<i>Channel Management</i>	
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	In low-frequency TDD, High-speed Railway Superior Experience cannot work with the following PUCCH sub-functions: - SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) - Enhanced SR period adaptation (controlled by the ENHANCED_SR_PERIOD_ADAPT_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) - CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	When SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) is disabled and the NRDUCellPucch.SrPeriod parameter is set to a value less than SLOT20 , SRS carrier switching cannot be enabled.
High-frequency TDD Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SWITCH option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When inter-FR CA is enabled, the NRDUCellPucch.StructureType parameter (specifying the cell-level PUCCH time-frequency format) cannot be set to SHORT_STRUCTURE .
Low-frequency TDD	QCI-specific SR period configuration	SR_PERIOD_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	When the NRDUCell.DuplexMode parameter is set to CELL_SUL or CELL_SDL , SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) cannot be enabled together with QCI-specific SR period configuration.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	When the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL for an NR DU cell, format-1 HARQ-ACK resource scheduling in RB saving mode (controlled by the HARQ_ACK_RB_SAVING_SWITCH_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	When the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE for an NR DU cell, format-1 HARQ-ACK resource scheduling in RB saving mode (controlled by the HARQ_ACK_RB_SAVING_SCH_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) cannot be enabled.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Proactive avoidance of remote interference cannot work with short PUCCH formats (with NRDUCellPucch.StructureType set to SHORT_STRUCTURE).
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	When an NR DU cell is a fusion cell, format-1 HARQ-ACK resource scheduling in RB saving mode (controlled by the HARQ_ACK_RB_SAVING_SCH_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	After the RedCap Introduction function is enabled, the sum of the values of parameters NRDUCellPucch.Format1RbNum, NRDUCellPucch.CsiDedicatedRbNum, NRDUCellPucch.Format3RbNum, NRDUCellPucch.Format4RbNum, and NRDUCellPucch.Format4CsiDedicatedRbNum cannot exceed 32 for an NR TDD cell that meets all of the following conditions: • The PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgorithmSwitch parameter is deselected. • The PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommonChnIntrfAvoidSwitch parameter is selected. • The UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected, and other options of this parameter are deselected. • The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_MANUAL. • The NRDUCellCoreset.CommonCtrlResRbNum

RAT	Function Name	Function Switch	Reference	Description
				parameter is set to RB48 or RB96. - The NRDUCellBwp.UlMinCarrierBw parameter is set to NOT_CONFIG or a value greater than 20M. - The NRDUCell.FrequencyBand parameter is set to N79. - The SSB center frequency position is greater than RB23.
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 (controlled by the FORMAT1_FREQ_INTRF_OPTION_SW option of the NRDUCellPucch.PucchAlgorithmSwitch parameter) cannot be enabled when both of the following conditions are met: - UE bandwidth adaptation (controlled by the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter) is disabled. - The NRDUCell.UlBandwidth parameter is set to CELL_BW_30M or CELL_BW_70M.
Low-frequency TDD	Uplink bandwidth	NRDUCell.UlBandwidth	None	
High-frequency TDD	SRS bandwidth adaptation	SRS_BANDWIDTH_ADAPTIVE_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	When SRS bandwidth adaptation is enabled, PUCCH resource reduction for heavy-traffic UEs (with NRDUCellPucch.PucchResourceRednMthd set to MASS_FLOW) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Increased downlink symbol proportion in self-contained slots	S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDischRes.DischResAlgoSw parameter	None	PUCCH resource reduction must be disabled when the S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDischRes.DischResAlgoSw parameter is selected.
Low-frequency TDD	Policy for PUCCH resource sharing with the PUSCH	NRDUCellPusch.PuschSharePucchResPol	None	- When frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 is enabled, the NRDUCellPusch.PuschSharePucchResPol parameter cannot be set to FULLY_SHARED_TIGHT_CONTROL. - When frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 is enabled and common PUCCH RB resource optimization (controlled by the COMM_PUCCH_RB_RES_OPT_SW option of the NRDUCellPusch.PucchAlgoSwitch parameter) is disabled, the NRDUCellPusch.PuschSharePucchResPol parameter cannot be set to PARTIALLY_SHARED.
Low-frequency TDD	SRS transmission using common PUCCH resources	COMMON_PUCCH_AVOID_SRS_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	None	When SRS transmission using common PUCCH resources is enabled, inward frequency shifting of the common PUCCH (controlled by the COMMON_PUCCH_POS_INWARD_SW option of the NRDUCellPusch.PucchAlgoExtSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Common PUCCH RB resource optimization	COMM_PUCCH_RB_RES_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	None	When common PUCCH RB resource optimization is enabled, inward frequency shifting of the common PUCCH (controlled by the COMMON_PUCCH_POS_INWARD_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink narrowband interference avoidance	UL_NARROWBAND_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 (controlled by the FORMAT1_FREQ_INTRF_OPT_SW option of the NRDUCellPucch.PucchAlgorithmSwitch parameter) cannot be enabled when both of the following conditions are met: - Uplink narrowband interference avoidance (controlled by the UL_NARROWBAND_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. - The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter (specifying the interference identification method) is set to UL_MANUAL. Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 (controlled by the FORMAT1_FREQ_INTRF_OPT_SW option of the NRDUCellPucch.PucchAlgorithmSwitch parameter) cannot be enabled when all of the following conditions are met: - PUCCH interference avoidance (controlled by the PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter) is enabled. - Uplink cell-level interference avoidance
Low-frequency TDD	Interference Identification Method	NRDUCellIntrfIdent.IntrfIdentificationMethod	None	
Low-frequency TDD	PUCCH interference avoidance	PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	None	
Low-frequency TDD	Uplink cell-level interference avoidance	UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	
Low-frequency TDD	Downlink UE-level interference avoidance	DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	None	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink UE-level interference avoidance	UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	(controlled by the UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. - Downlink UE-level interference avoidance (controlled by the DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. - The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter (specifying the interference identification method) is set to UL_DL_MANUAL. - The power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter) is disabled. Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 (controlled by the FORMAT1_FREQ_INTRF_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter) cannot be enabled when all of the following conditions are met: - PUCCH interference avoidance (controlled by the PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoid-Switch parameter) is enabled.

RAT	Function Name	Function Switch	Reference	Description
				- Uplink UE-level interference avoidance (controlled by the UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. - The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter (specifying the interference identification method) is set to UL_MANUAL. - The power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter) is disabled.

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When short PUCCH time-frequency formats (with NRDUCellPucch.StructureType set to SHORT_STRUCTURE) are used in low-frequency cells, intra-band CA does not take effect. In this case, long formats must be used. When short PUCCH time-frequency formats (with NRDUCellPucch.StructureType set to SHORT_STRUCTURE) are used in high-frequency cells, intra-band CA is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_IN TER_BAND_C A_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When short PUCCH time-frequency formats (with NRDUCellPucch.StructureType set to SHORT_STRUCTURE) are used in low-frequency cells, downlink intra-FR inter-band CA does not take effect. In this case, long formats must be used. High-frequency cells currently do not support downlink intra-FR inter-band CA.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	- When SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) is disabled and the NRDUCellPucch.SrPeriod parameter is set to a value less than SLOT20, SRS carrier switching does not take effect. - When SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) is disabled and the NRDUCellPucch.SrPeriod parameter is set to a value greater than or equal to SLOT20, SRS carrier switching takes effect and does not impact the UEs' adaptively configured SR period. - When SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) is enabled, SRS carrier switching takes effect. The SR period adaptively configured for UEs on which SRS carrier switching takes effect is at least 20 slots.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	NRDUCellUePwrSaving.BwpPwrSavingSw	<i>UE Power Saving</i>	- When the power saving BWP function takes effect, cells support only long PUCCH time-frequency formats (with NRDUCellPucch.StructureType set to LONG_STRUCTURE). - The impact relationship between the power saving BWP and the number of format-1 RBs is as follows: When PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) is disabled: - In NSA networking, if NRDUCellPucch.Format1RbNum is set to RB2, BWP2 (narrow bandwidth) cannot be allocated to UEs. - In SA networking, if full-bandwidth initial BWP configuration (controlled by the INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicy-Switch parameter) is enabled and the NRDUCellPucch.Format1RbNum parameter is set to RB2, BWP2 (narrow bandwidth) cannot be allocated to UEs. - The impact relationship between the power saving BWP and the

RAT	Function Name	Function Switch	Reference	Description
				number of format-3 RBs is as follows: When PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) is disabled and the NRDUCellPucch.Format3RbNum parameter is set to RB2 , BWP2 (narrow bandwidth) cannot be allocated to UEs.
Low-frequency TDD	Power saving BWP	NRDUCellUePwrSaving.BwpPwrSavingSw	<i>UE Power Saving</i>	When both PUCCH format-1 RB resource control and power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter) are enabled, more BWP2-capable UEs can access the network. In this case, however, the average uplink and downlink UE throughputs may decrease.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both quadruple code division for format-4 resource allocation (controlled by the FORMAT4_OCC_LENGTH_N4_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) and proactive avoidance of remote interference are enabled in high-speed cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED), the PUCCH resource allocation success rate decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station DL CoMP takes effect, it is recommended that the following functions be enabled: PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) and CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter). Enabling these functions protects UEs not involved in intra-base-station DL CoMP against service drops that occur due to resource preemption by intra-base-station DL CoMP UEs.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	After PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) is enabled, a hyper cell will have more RBs reserved for the PUCCH than a common cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	When UE bandwidth adaptation is enabled in TDD cells, NRDUCellBwp.UlMinCarrierBw is set to 20M , and PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) is disabled, the sum value of the following parameters should not exceed 32: NRDUCellPucch.Format1RbNum , NRDUCellPucch.CsiDedicatedRbNum , NRDUCellPucch.Format3RbNum , NRDUCellPucch.Format4RbNum , and NRDUCellPucch.Format4CsiDedicatedRbNum .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink preallocation	• UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter • UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	Reconfiguring the SR period (specified by the NRDUCellPucch.SrPeriod parameter) to a smaller value or reducing it through SR period adaptation (controlled by the SR_PERIOD_OPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) decreases the number of times basic and smart preallocation are performed. This reduces the average number of PRBs used by UEs involved in uplink preallocation (N.PRB.UL.Prealloc.Used.Avg). If there is a large volume of data configured for preallocation, the number of uplink padding bits decreases, reducing the average number of PRBs used by DRBs on the PUSCH (N.PRB.UL.DrbUsed.Avg) as well as the average uplink cell throughput (Cell Uplink Average Throughput (DU)).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When Co-MM takes effect, you are advised to enable PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) and CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter). Enabling these functions protects non-Co-MM UEs against service drops that occur due to resource preemption by Co-MM UEs.
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When the load is heavy in the uplink and there are a large number of UEs, the proportion of CCE resources used for data scheduling decreases due to an increase in the number of SRs. When packet-length-based scheduling optimization is enabled, it is recommended that SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When the load is heavy in the uplink, the uplink interference increases. As a result, the PUCCH demodulation performance deteriorates. When packet-length-based scheduling optimization is enabled, it is recommended that PUCCH MRC/IRC adaptation (controlled by the PUCCH_MRC_IRC_SW option of the NRDUCellPucch.PucchReceiveEnhSwitch parameter) be enabled.
Low-frequency TDD	Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter) be enabled to increase the proportion of UEs with periodic CSIs and reduce PDCCH resource consumption.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Short PUCCH Symbol Number	NRDUCellPucch.ShortPucchSymbolNum	<i>Channel Management</i>	PUCCH symbol number adaptation takes effect for a cell only when the NRDUCellPucch.ShortPucchSymbolNum parameter is not set to SYMBOL1 or SYMBOL2 , the PUCCH_RES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter is selected, and the sector corresponding to the cell serves at least 20 RRC_CONNECTED UEs.
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	When SR period adaptation or enhanced SR period adaptation is enabled, you are advised to also enable PUCCH RB adaptation and format-1 SR code channel allocation optimization. Otherwise, the maximum number of UEs supported by SR resources, the RRC Setup Success Rate (CU) , and the SgNB Addition Success Rate (CU) may decrease.
Low-frequency TDD	Format-1 SR code channel allocation optimization	FORMAT1_SR_CHN_ALLOC_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	
Low-frequency TDD	Capacity improvement for SR period adaptation	SR_PRD_ADAPT_CAPB_IMPROVE_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter	None	When both PUCCH format-1 RB resource control and capacity improvement for SR period adaptation are enabled for a cell serving a large number of RRC_CONNECTED UEs, the SR period and RTT delay increase and the average uplink and downlink UE throughputs may fluctuate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced SR period adaptation	ENHANCED_SR_PERIOD_ADAPT_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter	<i>Channel Management</i>	When both PUCCH format-1 RB resource control and enhanced SR period adaptation are enabled for a cell serving a large number of RRC_CONNECTED UEs, the SR period and RTT delay increase and the average uplink and downlink UE throughputs may fluctuate.
Low-frequency TDD	Format-1 HARQ-ACK resource scheduling in RB saving mode	HARQ_ACK_RB_SAVING_SCH_SW option of the NRDUCellPucch.SrResourceAlgoSwitch parameter	<i>Channel Management</i>	When both PUCCH format-1 RB resource control and format-1 HARQ-ACK resource scheduling in RB saving mode are enabled, the average uplink and downlink UE throughputs may increase, the CCE aggregation level may decrease, the uplink CCE allocation success rate may increase, and the first packet transmission delay at the RLC layer may decrease in the cell.
Low-frequency TDD	CSI reporting period adaptation	CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter	<i>Channel Management</i>	When both CSI reporting period adaptation and PUCCH RB adaptation optimization (controlled by the PUCCH_RBRES_ADAPT_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter) are enabled, the CSI reporting period is prolonged.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-TRP format-4 resource sharing for HARQ feedback	INTER_TRP_F4_HARQ_SHARE_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter	<i>Hyper Cell Combination</i>	When inter-TRP format-4 resource sharing for HARQ feedback is enabled together with format-3 and format-4 ACK RB resource adaptation (controlled by the F3_AND_F4_ACK_RB_RESOURCE_ADAPT_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter), only format-3 and format-4 ACK RB resource adaptation takes effect.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailBandwidth set to a value other than 20M	<i>RedCap</i>	When both multiple BWP1s for RedCap and CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter) are enabled: For TDD cells, CSI reporting period adaptation takes effect for RedCap UEs when the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter is set to a value less than or equal to 40M; the NRDUCellCsirs.CsiPeriod parameter (specifying the CSI reporting period) is fixed to SLOT320 in other scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value other than 20M	<i>RedCap</i>	When both multiple BWP1s for RedCap and PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) are enabled, PUCCH RB adaptation takes effect only on RedCap UEs using BWP1s at cell bandwidth edges.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	When both RedCap and SR period adaptation (controlled by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResourceAlgoSwitch parameter) are enabled: For TDD cells, the SR period SLOT160 always takes effect.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	When both RedCap and CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter) are enabled: For TDD cells, the CSI reporting period SLOT320 always takes effect.
Low-frequency TDD	QCI-specific SR period configuration	SR_PERIOD_QCI_SW option of the gNBDSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	When both SR period adaptation and QCI-specific SR period configuration are enabled, the gNBDSchParamGroup.SrPeriod parameter can only be set to 20SLOT , 40SLOT , or 80SLOT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RLC timer adaptation along with the SR period	RLC_TIMER_ADAPT_WITH_SR_PRD_SW option of the NRDUCellServExp.RlcSwitch parameter	None	When SR period adaptation is enabled, the number of RLC retransmissions is likely to reach the maximum value before this happens with SR retransmissions, leading to service drops. To prevent this, you are advised to enable RLC timer adaptation along with the SR period (controlled by the RLC_TIMER_ADAPT_WITH_SR_PRD_SW option of the NRDUCellServExp.RlcSwitch parameter). If RLC timer adaptation along with the SR period is disabled, configure gNodeB RLC parameters by referring to Table 6-3 .

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

NOTE

3900 series base stations do not support high-frequency TDD functions.

Boards

- Format-1 SR code channel allocation optimization (low-frequency TDD)
All NR-capable main control boards as well as the UBBPg series and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Format-1 HARQ-ACK resource scheduling in RB saving mode (low-frequency TDD)

All NR-capable main control boards as well as the UBBPg3 series and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- Enhanced SR period adaptation

All NR-capable main control boards as well as the UBBPg series and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- HARQ resource increase

All NR-capable main control boards and baseband processing units (except UBBPfw series) support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- Other basic functions of PUCCH channel management

Except for the RAT, the sub-functions of this function have no other requirements on main control boards and baseband processing units. All main control boards and baseband processing units of the corresponding RAT support these sub-functions. For details about the RATs supported by different main control boards and baseband processing units, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Except for the RAT, the sub-functions of this function have no other requirements on RF modules. All RF modules of the corresponding RAT support these sub-functions. For details about the RATs supported by different RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Cells

Only long PUCCH time-frequency formats (with **NRDUCellPucch.StructureType** set to **LONG_STRUCTURE**) are supported in SUL cells (with **NRDUCell.DuplexMode** set to **CELL_SUL**).

Only short PUCCH time-frequency formats (with **NRDUCellPucch.StructureType** set to **SHORT_STRUCTURE**) are supported in high-frequency cells (with **NRDUCell.FrequencyBand** set to **N257**, **N258**, **N260**, or **N261**).

PUCCH resource reduction for heavy-traffic UEs in single-beam scenarios takes effect only in NR cells with **NRDUCell.SlotAssignment** set to **4_1_DDDSU** and a number of RRC_CONNECTED UEs in a sector less than or equal to four times the total number of AAUs serving the sector.

When the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**:

- The **NRDUCellPucch.ShortPucchSymbolNum** parameter must be set to **SYMBOL1** or **SYMBOL3**.
- The **PUCCH_RES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter must be deselected.

When the **NRDUCell.SlotAssignment** parameter is set to **4_1_DDDSU**, the **NRDUCellPucch.ShortPucchSymbolNum** parameter must be set to **SYMBOL2**, **SYMBOL3**, **SYMBOL4**, or **SYMBOL5**.

Format-1 HARQ-ACK resource scheduling in RB saving mode applies only to low-frequency TDD massive MIMO cells with the **NRDUCell.NrDuCellNetworkingMode** parameter set to **NORMAL_CELL**.

6.3.5 Others

Support for format 4 is optional for UEs. If format 4 is configured, UEs must support this format, which is indicated by the *pucch-F4-WithFH* IE in the UECapabilityInformation message over the Uu interface.

Disabling frequency hopping for format-4 resources requires UEs to have the capability to disable frequency hopping for format-4 resources, meaning they should not report the *pucch-F1-3-4WithoutFH* IE.

- **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter, which controls common PUCCH RB resource optimization
- **PUCCH_COMPRESSION_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter, which controls ultimate PUCCH compression
- **NRDUCellPucch.CommonPucchResComprDegree** parameter, which specifies the compression degree of common PUCCH RBs

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 6-5](#) describes the parameters used for function optimization.

Table 6-5 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Structure Type	NRDUCellPucch.StructureType	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Short PUCCH Symbol Number	NRDUCellPucch.ShortPucchSymbolNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a short time-frequency format is used. Retain the default value.
Format1 RB Number	NRDUCellPucch.Format1RbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a long time-frequency format is used. Retain the default value.
Format3 RB Number	NRDUCellPucch.Format3RbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a long time-frequency format is used. In low-speed TDD cells, if RB adaptation is disabled and mainstream UEs support format 4, it is recommended that this parameter be changed to RB8. You are advised to retain the default value in other scenarios.
Format4 RB Number	NRDUCellPucch.Format4RbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a long time-frequency format is used. In low-speed TDD cells, if RB adaptation is disabled and mainstream UEs support format 4, it is recommended that this parameter be changed to RB6. You are advised to retain the default value in other scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
Format3 CSI- dedicated RB Number	NRDUCellP ucch.CsiDed icatedRbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a long time-frequency format is used. Retain the default value.
Format4 CSI- dedicated RB Number	NRDUCellP ucch.Format4CsiDedica tedRbNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. This parameter must be specified if a long time-frequency format is used. In low-speed TDD cells, if RB adaptation is disabled and mainstream UEs support format 4, the recommended parameter settings (based on the recommended CsiReportPeriod value of SLOT80) are as follows: • If the UBBPfw is used, you are advised to set this parameter to RB2 in scenarios where the number of UEs is less than or equal to 96, and RB4 in scenarios where the number of UEs is greater than 96. • If the UBBPg is used, you are advised to set this parameter to RB2 in scenarios where the number of UEs is less than or equal to 96, RB4 in scenarios where the number of UEs is greater than 96 but less than or equal to 160, RB6 in scenarios where the number of UEs is greater than 160 but less than or equal to 224, and RB8 in scenarios where the number of UEs is greater than 224. You are advised to retain the default value in other scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
SR Period ^a	NRDUCellPucch.SrPeriod	None	The following are recommended SR period settings depending on the number of UEs: For TDD cells established on the UBBPfw: • If the number of UEs is less than or equal to 50, SLOT20 is recommended. • If the number of UEs is greater than 50 but less than or equal to 100, SLOT40 is recommended. • If the number of UEs is greater than 100 but less than or equal to 200, SLOT80 is recommended. • If the number of UEs is greater than 200, SLOT160 is recommended. For low-frequency TDD cells established on the UBBPg: • If the number of UEs is less than or equal to 100, SLOT20 is recommended. • If the number of UEs is greater than 100 but less than or equal to 300, SLOT40 is recommended. • If the number of UEs is greater than 300 but less than or equal to 600, SLOT80 is recommended. • If the number of UEs is greater than 600, SLOT160 is recommended. For TDD high-frequency cells established on the UBBPg: • If the number of UEs is less than or equal to 16, SLOT40 is recommended. • If the number of UEs is greater than 16 but less than or equal to 32, SLOT80 is recommended. • If the number of UEs is greater than 32 but less than or equal to 64, SLOT160 is recommended. • If the number of UEs is greater than 64 but less than or equal to 128, SLOT320 is recommended.

Parameter Name	Parameter ID	Option	Setting Notes
			If the number of UEs is greater than 128, SLOT640 is recommended.
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	SR_PERIOD_ADAPT_SWITCH	The SR_PERIOD_ADAPT_SWITCH option of this parameter specifies whether to enable SR period adaptation. - It is recommended that SR period adaptation be enabled in low-speed cells (for which NRDUCell.HighSpeedFlag is set to LOW_SPEED) to adapt to different UE quantities. - SR period adaptation cannot be enabled in high-speed cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED). When SR period adaptation is enabled, the number of RLC retransmissions is likely to reach the maximum value before this happens with SR retransmissions, leading to service drops. To prevent this, you are advised to enable RLC timer adaptation along with the SR period (controlled by the RLC_TIMER_ADAPT_WITH_SR_PRD_SW option of the NRDUCellServExp.RlcSwitch parameter). If RLC timer adaptation along with the SR period is disabled, configure gNodeB RLC parameters by referring to Table 6-3.
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	ENHANCED_SR_PERIOD_ADAPT_SW	The ENHANCED_SR_PERIOD_ADAPT_SW option of this parameter specifies whether to enable enhanced SR period adaptation. The default setting is recommended. Enhanced SR period adaptation applies only to low-frequency TDD cells set up on UBBPg boards in low-speed scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	SR_PERIOD_OPT_SWITCH	The SR_PERIOD_OPT_SWITCH option of this parameter specifies whether to enable SR period adaptation optimization. The default setting is recommended.
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	SR_CHN_RB_SAVING_MODE_ALLOC_SW	The SR_CHN_RB_SAVING_MODE_ALLOC_SW option of this parameter specifies whether to enable format-1 SR code channel allocation in RB saving mode. It is recommended that this option be selected.
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	HARQ_ACK_RB_SAVING_SCH_SW	The HARQ_ACK_RB_SAVING_SCH_SW option of this parameter specifies whether to enable format-1 HARQ-ACK resource scheduling in RB saving mode. It is recommended that this option be selected in low-frequency TDD massive MIMO cells.
SR Resource Algo Switch	NRDUCellPucch.SrResourceAlgoSwitch	SR_ALLOC_INTRF_OPT_SW	The SR_ALLOC_INTRF_OPT_SW option of this parameter specifies whether to enable interference optimization for format-1 SR allocation. It is recommended that this option be selected in areas with strong inter-cell format-1 interference and retain its default value in other scenarios. Before enabling this function, you are advised to properly plan PCIs based on their mod 3 values for the cells to be enabled with this function. Otherwise, interference cannot be reduced.
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_RES_ADAPTIVE_SWITCH	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The PUCCH_RES_ADAPTIVE_SWITCH option of this parameter specifies whether to enable PUCCH symbol number adaptation. This option applies only to high frequency bands. Set this option to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.AlgoSwitch	PUCCH_RBRES_ADAPTIVE_SWITCH	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The PUCCH_RBRES_ADAPTIVE_SWITCH option of this parameter specifies whether to enable long PUCCH RB adaptation. This option applies only to SUL cells with LTE and NR spectrum sharing disabled/low-frequency TDD cells. Set this option to its recommended value.
PUCCH Algorithm Switch	NRDUCellPucch.AlgoSwitch	FORMAT4_OCC_LENGTH_N4_SWITCH	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The FORMAT4_OCC_LENGTH_N4_SWITCH option of this parameter specifies whether to enable quadruple code division for format-4 resource allocation. This option applies only to low-frequency TDD cells/SUL cells with 30 kHz SCS. You are advised to select this option only in scenarios with heavy CA loads.
PUCCH Algorithm Switch	NRDUCellPucch.AlgoSwitch	FORMAT1_SR_CHN_ALLOC_OPT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The FORMAT1_SR_CHN_ALLOC_OPT_SW option of this parameter specifies whether to enable format-1 SR code channel allocation optimization. Set this option to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.AlgoSwitch	FORMAT1_FREQ_INTRF_OPT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The FORMAT1_FREQ_INTRF_OPT_SW option of this parameter specifies whether to enable frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3. This option applies only to low-frequency TDD cells. It is recommended that this option be selected in areas with strong inter-cell format-1 interference and retain its default value in other scenarios. Before enabling this function, you are advised to properly plan PCIs based on their mod 3 values for the cells to be enabled with this function. Otherwise, interference cannot be reduced.
CSI Report Period	NRDUCellPucch.CsiReportPeriod	None	Use the recommended value.
CSI Resource Algo Switch	NRDUCellPucch.CsiResourceAlgoSwitch	None	Use the recommended value.
CSI Resource Algo Switch	NRDUCellPucch.CsiResourceAlgoSwitch	CSI_REPORT_PERIOD_ADAPT_SWITCH	The CSI_REPORT_PERIOD_ADAPT_SWITCH option of this parameter specifies whether to enable CSI reporting period adaptation. Set this option to its recommended value.
PUCCH Performance Switch	NRDUCellPucch.PerformanceSw	HARQ_ACK_RESOURCE_SET0_SCH_OPT_SW	The HARQ_ACK_RESOURCE_SET0_SCH_OPT_SW option of this parameter specifies whether to enable scheduling optimization for HARQ-ACK resource set 0. Set this option to its recommended value.
SR Prohibit Timer	NRDUCellPucch.SrProhibitTimer	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Receive Enhanced Switch	NRDUCellPucch.PucchReceiveEnhSwitch	PUCCH_MRC_IRC_SW	The PUCCH_MRC_IRC_SW option of this parameter specifies whether to enable PUCCH MRC/IRC adaptation. Set this option to its recommended value.
UE PUCCH Capability Algorithm Switch	NRDUCellPucch.UePucchCapabilityAlgoSw	PUCCH_F4_WITHFH_SW	The PUCCH_F4_WITHFH_SW option of this parameter specifies whether to enable format-4 PUCCH capability adaptation. Set this option to its recommended value.
PUCCH Resource Reduction Method	NRDUCellPucch.PucchResrcRednMthd	None	You are advised to change the value to MASS_FLOW for NR cells with single-beam coverage (NRDUCellTrpMmwav-Beam.CoverageScenario set to SCENARIO_103) and NRDUCell.SlotAssignment set to 4_1_DDDSU . You are advised to retain the default value in other scenarios.
PUCCH Switch	gNodeBAlgo.PucchSwitch	FORMAT1_RB_RES_CTRL_SW	The FORMAT1_RB_RES_CTRL_SW option of this parameter specifies whether to enable PUCCH format-1 RB resource control. Set this option to its recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	INTER_TRP_RB_SHARE_POLICY_SW	The INTER_TRP_RB_SHARE_POLICY_SW option of this parameter specifies whether to enable inter-TRP RB resource sharing. Set this option to its recommended value.
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_RBRES_ADAPT_OPT_SW	The PUCCH_RBRES_ADAPT_OPT_SW parameter specifies whether to enable PUCCH RB adaptation optimization. Set this option to its recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	F3_AND_F4_ACK_RB_RES_ADAPT_SW	The F3_AND_F4_ACK_RB_RES_ADAPT_SW option of this parameter specifies whether to enable format-3 and format-4 ACK RB resource adaptation. Set this option to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Format3 and Format4 HARQ Resource Use Ratio	NRDUCellPucch.F3AndF4HarqResUseRatio	None	Use the recommended value.
Format3 and Format4 HARQ Resource Fail Ratio	NRDUCellPucch.F3AndF4HarqResFailRatio	None	Use the recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	COMMON_PUCCH_POS_INWARD_SW	The COMMON_PUCCH_POS_INWARD_SW option of this parameter specifies whether to enable inward frequency shifting of the common PUCCH. Set this option to its recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	F4_NO_FREQ_HOPPING_SW	The F4_NO_FREQ_HOPPING_SW option of this parameter specifies whether to disable frequency hopping for format-4 resources. Set this option to its recommended value.
Format4 and Format3 HARQ SINR Threshold	NRDUCellPucch.F3AndF4HarqSinrThld	None	Use the recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	F1_RES_REFINED_CTRL_SW	The F1_RES_REFINED_CTRL_SW option of this parameter specifies whether to enable refined control of format-1 resources. Set this option to its recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	HARQ_SPEC_INCREASE_SW	The HARQ_SPEC_INCREASE_SW option of this parameter specifies whether to enable HARQ resource increase. Set this option to its recommended value.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	HARQ_RES_SET1_MAX_SRCH_SW	The HARQ_RES_SET1_MAX_SRCH_SW option of this parameter specifies whether to enable maximized search within HARQ resource set 1. Set this option to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.AlgoSwitch	PUCCH_SYM_TYPE_NUM_INCR_SW	The PUCCH_SYM_TYPE_NUM_INCR_SW option of this parameter specifies whether to enable PUCCH symbol length number increase. Set this option to its recommended value.
a: The actual SR period depends on the SCS and the number of slots. For example, 1 ms is equal to one slot, two slots, four slots, and eight slots when the SCS is 15 kHz, 30 kHz, 60 kHz, and 120 kHz, respectively. For more information, see "SchedulingRequestResourceConfig field descriptions" in 3GPP TS 38.331 V15.5.1.			

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following provides activation command examples for the sub-functions of PUCCH channel management.

Low-frequency TDD command examples

```
//Setting the cell-level PUCCH format to short or long format as required
//Setting the cell-level PUCCH format to short format
MOD NRDUCCELLPUCCH: NrDuCellId=0, StructureType=SHORT_STRUCTURE;
//Setting the number of symbols for the short PUCCH to one when the short PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, ShortPucchSymbolNum=SYMBOL1;
//Setting the cell-level PUCCH format to long format
MOD NRDUCCELLPUCCH: NrDuCellId=0, StructureType=LONG_STRUCTURE;
//Configuring a static number of RBs for the long PUCCH or enabling cell-level long PUCCH RB adaptation
as required when the long PUCCH is used for the cell
//Setting the number of RBs used in format 1 to four when the long PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, Format1RbNum=RB4;
//Setting the number of RBs used in format 3 to sixteen when the long PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, Format3RbNum=RB16;
//Setting the number of RBs used in format 4 to six when the long PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, Format4RbNum=RB6;
//Setting the number of CSI-dedicated RBs used in format 3 to two and that used in format 4 to zero when
the long PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, CsiDedicatedRbNum=RB2, Format4CsiDedicatedRbNum=RB0;
//Enabling cell-level long PUCCH RB adaptation when the long PUCCH is used for the cell
MOD NRDUCCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RBRES_ADAPTIVE_SWITCH-1;
//Enabling PUCCH RB adaptation optimization
MOD NRDUCCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RBRES_ADAPT_OPT_SW-1;
```

```
//Enabling quadruple code division for format-4 resource allocation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT4_OCC_LENGTH_N4_SWITCH-1;
//Enabling format-1 SR code channel allocation optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_SR_CHN_ALLOC_OPT_SW-1;
//Enabling format-1 SR code channel allocation in RB saving mode
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_CHN_RB_SAVING_MODE_ALLOC_SW-1;
//Enabling format-1 HARQ-ACK resource scheduling in RB saving mode
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=HARQ_ACK_RB_SAVING_SCH_SW-1;

//Configuring a static SR period or enabling SR period adaptation as required
//Setting the SR period to 40 slots
MOD NRDUCELLPUCCH: NrDuCellId=0, SrPeriod=SLOT40;
//Enabling SR period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-1;
//Enabling enhanced SR period adaptation when SR period adaptation is enabled
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=ENHANCED_SR_PERIOD_ADAPT_SW-1;
//Enabling SR period adaptation optimization when SR period adaptation is enabled
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_OPT_SWITCH-1;

//Setting the SR prohibit timer to 16 ms
MOD NRDUCELLPUCCH: NrDuCellId=0, SrProhibitTimer=MS16;

//Configuring a static CSI reporting period or enabling CSI reporting period adaptation as required
//Setting the CSI reporting period to 80 slots
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiReportPeriod=SLOT80;
//Enabling CSI reporting period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-1;

//Enabling scheduling optimization for HARQ-ACK resource set 0
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchPerformanceSw=HARQ_ACK_RES_SET0_SCH_OPT_SW-1;
//Enabling PUCCH MRC/IRC adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchReceiveEnhSwitch=PUCCH_MRC_IRC_SW-1;
//Enabling format-4 PUCCH capability adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, UePucchCapabilityAlgoSw=PUCCH_F4_WITHFH_SW-1;
//Enabling frequency-domain interference optimization for PUCCH format 1
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_ALLOC_INTRF_OPT_SW-1;
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_FREQ_INTRF_OPT_SW-1;
//Enabling PUCCH format-1 RB resource control
MOD GNODEBALGO: PucchSwitch=FORMAT1_RB_RES_CTRL_SW-1;
//Enabling inter-TRP RB resource sharing
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-1;
//Enabling format-3 and format-4 ACK RB resource adaptation and configuring related thresholds
MOD NRDUCELLPUCCH: NrDuCellId=0, F3AndF4HarqResUseRatio=90, F3AndF4HarqResFailRatio=5,
PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-1;
//Enabling inward frequency shifting of the common PUCCH
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=COMMON_PUCCH_POS_INWARD_SW-1;
//Disabling frequency hopping for format-4 resources and setting the format-4 and format-3 HARQ SINR
threshold
MOD NRDUCELLPUCCH: NrDuCellId=0, F3AndF4HarqSinrThld=5,
PucchAlgoExtSwitch=F4_NO_FREQ_HOPPING_SW-1;
//Enabling refined control of format-1 resources
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F1_RES_REFINED_CTRL_SW-1;
//Enabling HARQ resource increase
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=HARQ_SPEC_INCREASE_SW-1;
//Enabling maximized search within HARQ resource set 1
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=HARQ_RES_SET1_MAX_SRCH_SW-1;
//Enabling PUCCH symbol length number increase
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_SYM_TYPE_NUM_INCR_SW-1;
```

High-frequency TDD command examples

```
//Setting the cell-level PUCCH format to short format
MOD NRDUCELLPUCCH: NrDuCellId=0, StructureType=SHORT_STRUCTURE;
//Configuring a static number of symbols for the short PUCCH or enabling cell-level PUCCH symbol number
adaptation as required when the short PUCCH is used for the cell
//Setting the number of symbols for the short PUCCH to one for the cell
MOD NRDUCELLPUCCH: NrDuCellId=0, ShortPucchSymbolNum=SYMBOL1;
//Enabling cell-level PUCCH symbol number adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RES_ADAPTIVE_SWITCH-1;
```

```
//Configuring a static SR period or enabling SR period adaptation as required
//Setting the SR period to 40 slots
MOD NRDUCELLPUCCH: NrDuCellId=0, SrPeriod=SLOT40;
//Enabling SR period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-1;
//Setting the SR prohibit timer to 16 ms
MOD NRDUCELLPUCCH: NrDuCellId=0, SrProhibitTimer=MS16;
//Enabling scheduling optimization for HARQ-ACK resource set 0
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchPerformanceSw=HARQ_ACK_RES_SET0_SCH_OPT_SW-1;
//Enabling format-4 PUCCH capability adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, UePucchCapabilityAlgoSw=PUCCH_F4_WITHFH_SW-1;
//Enabling PUCCH resource reduction
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchResrcRednMthd=MASS_FLOW;
//Enabling PUCCH symbol length number increase
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_SYM_TYPE_NUM_INCR_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples for the sub-functions. You can determine whether to restore the settings of other parameters based on actual network conditions.

Low-frequency TDD command examples

```
//Disabling PUCCH symbol length number increase
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_SYM_TYPE_NUM_INCR_SW-0;
//Disabling refined control of format-1 resources
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F1_RES_REFINED_CTRL_SW-0;
//Disabling HARQ resource increase
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=HARQ_SPEC_INCREASE_SW-0;
//Disabling maximized search within HARQ resource set 1
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=HARQ_RES_SET1_MAX_SRCH_SW-0;
//Enabling frequency hopping for format-4 resources
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F4_NO_FREQ_HOPPING_SW-0;
//Disabling format-3 and format-4 ACK RB resource adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-0;
//Disabling inward frequency shifting of the common PUCCH
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=COMMON_PUCCH_POS_INWARD_SW-0;
//Disabling PUCCH RB adaptation optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RBRES_ADAPT_OPT_SW-0;
//Disabling cell-level long PUCCH RB adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RBRES_ADAPTIVE_SWITCH-0;
//Disabling quadruple code division for format-4 resource allocation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT4_OCC_LENGTH_N4_SWITCH-0;
//Disabling format-1 SR code channel allocation optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_SR_CHN_ALLOC_OPT_SW-0;
//Disabling format-1 SR code channel allocation in RB saving mode
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_CHN_RB_SAVING_MODE_ALLOC_SW-0;
//Disabling format-1 HARQ-ACK resource scheduling in RB saving mode
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=HARQ_ACK_RB_SAVING_SCH_SW-0;
//Disabling enhanced SR period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=ENHANCED_SR_PERIOD_ADAPT_SW-0;
//Disabling SR period adaptation optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_OPT_SWITCH-0;
//Disabling SR period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-0;
//Disabling CSI reporting period adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-0;
//Disabling scheduling optimization for HARQ-ACK resource set 0
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchPerformanceSw=HARQ_ACK_RES_SET0_SCH_OPT_SW-0;
//Disabling PUCCH MRC/IRC adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchReceiveEnhSwitch=PUCCH_MRC_IRC_SW-0;
//Disabling format-4 PUCCH capability adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, UePucchCapabilityAlgoSw=PUCCH_F4_WITHFH_SW-0;
//Disabling frequency-domain interference optimization for PUCCH format 1
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_ALLOC_INTRF_OPT_SW-0;
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_FREQ_INTRF_OPT_SW-0;
//Disabling PUCCH format-1 RB resource control
```

```
MOD GNODEBALGO: PucchSwitch=FORMAT1_RB_RES_CTRL_SW-0;  
//Disabling inter-TRP RB resource sharing  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-0;
```

High-frequency TDD command examples

```
//Disabling PUCCH symbol length number increase  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_SYM_TYPE_NUM_INCR_SW-0;  
//Disabling cell-level PUCCH symbol number adaptation  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_RES_ADAPTIVE_SWITCH-0;  
//Disabling SR period adaptation  
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=SR_PERIOD_ADAPT_SWITCH-0;  
//Disabling scheduling optimization for HARQ-ACK resource set 0  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchPerformanceSw=HARQ_ACK_RES_SET0_SCH_OPT_SW-0;  
//Disabling format-4 PUCCH capability adaptation  
MOD NRDUCELLPUCCH: NrDuCellId=0, UePucchCapabilityAlgoSw=PUCCH_F4_WITHFH_SW-0;  
//Disabling PUCCH resource reduction  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchResrcRednMthd=OFF;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

- Step 1** Start Uu signaling tracing on the MAE-Access as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Enable a UE to access the network. In SA networking, check the traced RRCReconfiguration message, as shown in [Figure 6-6](#), and check the fields in the *pucch-Config* IE. If format information is present in the PUCCH-Resource field, the indicated format has taken effect.

NOTE

In NSA networking, check the SgNB Addition Request Acknowledge message for the IEs described in SA networking.

Figure 6-6 RRCReconfiguration message tracing report (SA networking)



----End

Observe the following performance indicators to check whether each sub-function of PUCCH channel management has taken effect.

Table 6-6 Indicators related to the sub-functions involved in PUCCH channel management

Indicator	Formula
Average uplink cell throughput	Cell Uplink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink UE throughput	User Downlink Average Throughput (DU)
Random access success rate	$\text{N.RA.Contention.Resolution.Succ}/\text{N.RA.Contention.Att} \times 100\%$
Average uplink CCE aggregation level	$(1 \times \text{N.CCE.UL.AggLvl1Num} + 2 \times \text{N.CCE.UL.AggLvl2Num} + 4 \times \text{N.CCE.UL.AggLvl4Num} + 8 \times \text{N.CCE.UL.AggLvl8Num} + 16 \times \text{N.CCE.UL.AggLvl16Num}) / (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl16Num})$

Indicator	Formula
Average downlink CCE aggregation level	$(1 \times N.CCE.DL.AggLvl1Num + 2 \times N.CCE.DL.AggLvl2Num + 4 \times N.CCE.DL.AggLvl4Num + 8 \times N.CCE.DL.AggLvl8Num + 16 \times N.CCE.DL.AggLvl16Num) / (N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)$
Duration in which BWP2 is in effect in a cell	N.Bwp2.Dur.Total
Proportion of DRX sleep time in a cell	$N.Cdrx.Sleep.Dur.Total / (N.Cdrx.Sleep.Dur.Total + N.Cdrx.Active.Dur.Total)$
Average number of available PRBs for the PUSCH	N.PR.B.UL.PUSCH.Avail.Avg
Total number of PRBs that are paired for uplink MIMO	N.ChMeas.MIMO.UL.Pair.PR.B
Number of times the measured PUCCH SINR values fall into different ranges	N.UL.SINR.PUCCH.Index0 to N.UL.SINR.PUCCH.Index7
Average interference and noise received on each uplink PRB	N.UL.NI.Avg
Number of successful PUCCH HARQ-ACK resource requests	N.PUCCH.ACK.AllocReq.Succ

6.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

7 CSI-RS Resource Management

7.1 Principles

7.1.1 Overview

CSI-RS mainly includes non-zero-power CSI-RS (NZP CSI-RS) and zero-power CSI-RS (ZP CSI-RS). UEs measure NZP CSI-RS to obtain related channel information, whereas they consider that the time-frequency resources occupied by ZP CSI-RS are not used for PDSCH transmission.

- NZP CSI-RS is used for channel state measurement, beam measurement, and other functions. It is classified into the following types:
 - CSI-RS for CM: used for downlink channel state measurement. The gNodeB transmits CSI-RS for CM within the active BWP. UEs receive and process CSI-RS for CM to obtain corresponding information such as the CQI, RI, and PMI, and report the information to the gNodeB.
 - CSI-RS for BM: used for downlink beam measurement. The gNodeB transmits CSI-RS for BM within the active BWP. UEs receive and measure CSI-RS for BM to obtain the corresponding reference signal received power (RSRP), and report the RSRP and CSI-RS resource indicator (CRI) to the gNodeB. The gNodeB then obtains the RSRP information about the corresponding beam.
 - Tracking reference signal (TRS): used for time and frequency offset tracking. The gNodeB transmits TRS within the active BWP. UEs receive TRS for tracking time and frequency offsets.
- ZP CSI-RS is designed for NR to measure inter-cell interference, fully utilize PDSCH resources, and avoid interference caused by NZP CSI-RS of neighboring cells. The time-frequency resources occupied by ZP CSI-RS are not used for PDSCH transmission.
 - CSI-RS for interference measurement (CSI-RS for IM) is the ZP CSI-RS used for inter-cell interference measurement. UEs measure the interference from neighboring cells at the time-frequency position of CSI-RS for IM and report the measurement results to the gNodeB.
 - ZP CSI-RS configured for CSI-RS for CM and CSI-RS for IM is determined based on the status of CSI-RS rate matching, which is specified by the

CSIRS_RATEMATCH_SW option of the **NRDUCellPdsch.RateMatchSwitch** parameter. If this option is deselected, ZP CSI-RS is not configured for CSI-RS for CM and CSI-RS for IM. If this option is selected, ZP CSI-RS is configured for CSI-RS for CM and CSI-RS for IM, and occupies the time-frequency resources of CSI-RS for IM and CSI-RS for CM. When ZP CSI-RS is configured, PDSCH resources can be fully utilized. For details about CSI-RS rate matching, see *Downlink Scheduling*.

- ZP CSI-RS configured for TRS is determined based on the status of TRS rate matching, which is specified by the **TRS_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter. If this option is deselected, ZP CSI-RS is not configured for TRS. If this option is selected, ZP CSI-RS is configured for TRS, and occupies the time-frequency resources of TRS. When ZP CSI-RS is configured, interference caused by TRS of neighboring cells can be avoided. For details about TRS rate matching, see *Downlink Scheduling*.

NOTE

Section 7.4.1.5 "CSI reference signals" in 3GPP TS 38.211 V16.3.0 provides details about NZP CSI-RS and ZP CSI-RS.

For more information on each type of NZP CSI-RS, see the following sections in 3GPP TS 38.214 V15.4.0:

- CSI-RS for CM: see section 5.2 "UE procedure for reporting channel state information (CSI)"
- CSI-RS for BM: see section 5.1.6.1.2 "CSI-RS for L1-RSRP computation"
- TRS: see section 5.1.6.1.1 "CSI-RS for tracking"

For more information on ZP CSI-RS, see section 5.1.4 "PDSCH resource mapping" in 3GPP TS 38.214 V16.3.0.

This section does not differentiate "CSI-RS" between NZP CSI-RS (including CSI-RS for CM, CSI-RS for BM, and TRS) and ZP CSI-RS. Instead, it uses the following in cases where differentiation is required: NZP CSI-RS, CSI-RS for CM, CSI-RS for BM, TRS, ZP CSI-RS, CSI-RS for IM, or ZP CSI-RS configured for CSI-RS for IM or for NZP CSI-RS.

CSI-RS can be periodic, aperiodic, or semi-persistent.

Table 7-1 Periodic CSI-RS, aperiodic CSI-RS, and semi-persistent CSI-RS

Type	Description	Support in This Version
Periodic CSI-RS	For NZP CSI-RS: The gNodeB periodically transmits CSI-RS to UEs in accordance with the configured period, and UEs receive CSI-RS in the same period. For ZP CSI-RS: The gNodeB does not schedule the PDSCH at the time-frequency position of ZP CSI-RS in accordance with the configured period, and UEs do not demodulate the PDSCH at the corresponding time-frequency position in the same period.	For NZP CSI-RS: - In low frequency bands, CSI-RS for CM is periodically transmitted by default. The gNodeB periodically transmits CSI-RS to UEs at an interval specified by the NRDUCellCsirs.CsiPeriod parameter. Changing the value of this parameter will cause batch reconfiguration for all UEs in the cell so that the latest value takes effect. As such, you are advised to change the value of this parameter only when there are a small number of UEs in the cell. - The number of resources for periodic CSI-RS transmission is specified by the NRDUCellCsirs.CsirsCellResourceNum parameter. UEs send measurement reports to the gNodeB. - In high frequency bands, CSI-RS for CM cannot be periodically transmitted. - TRS can be periodically transmitted. The gNodeB periodically transmits TRS to UEs at an interval specified by the NRDUCellCsirs.TrsPeriod parameter. - CSI-RS for BM can be periodically transmitted. For ZP CSI-RS: - ZP CSI-RS configured for TRS can be periodically transmitted in accordance with the TRS period. - Other types of ZP CSI-RS cannot be periodically transmitted.

Type	Description	Support in This Version
Semi-persistent CSI-RS	For NZP CSI-RS: Once semi-persistent CSI-RS is activated through the MAC CE, the gNodeB periodically transmits CSI-RS to UEs in accordance with the configured period, and UEs receive CSI-RS in the same period. For ZP CSI-RS: Once semi-persistent CSI-RS is activated through the MAC CE, the gNodeB does not schedule the PDSCH at the time-frequency position of ZP CSI-RS in accordance with the configured period, and UEs do not demodulate the PDSCH at the corresponding time-frequency position in the same period.	The current version does not support semi-persistent CSI-RS.

Type	Description	Support in This Version
Aperiodic CSI-RS	For NZP CSI-RS: The gNodeB sends DCI to UEs and transmits NZP CSI-RS to UEs at the time-frequency position indicated in the DCI. UEs receive NZP CSI-RS from the gNodeB at the time-frequency position. For ZP CSI-RS: The gNodeB sends DCI to UEs and does not schedule the PDSCH at the time-frequency position indicated in the DCI. UEs do not demodulate the PDSCH at the time-frequency position.	For NZP CSI-RS: • In low frequency bands, CSI-RS for CM can be aperiodically transmitted in specific scenarios (such as DL CoMP). • In high frequency bands, CSI-RS for CM can be aperiodically transmitted. • TRS cannot be aperiodically transmitted. • CSI-RS for BM cannot be aperiodically transmitted. All types of ZP CSI-RS can be aperiodically transmitted.

NZP CSI-RS and CSI-RS for IM measurement results are reported periodically or aperiodically. For details about periodic reporting, see [6 PUCCH Channel Management](#). For details about aperiodic reporting, see *Uplink Scheduling*.

7.1.2 NZP CSI-RS Resource Management

7.1.2.1 CSI-RS for CM Resource Management

CSI-RS for CM resource management includes the following functions.

CSI-RS for CM Port Number Management (Low-Frequency TDD)

When transmitting CSI-RS for CM, the gNodeB can notify UEs of the number of ports used for such transmission through RRC reconfiguration messages. The number of ports can be configured based on the UE capability and actual scenario. UEs receive CSI-RS for CM based on the number of ports.

In low-frequency TDD, the maximum number of ports used to transmit CSI-RS for CM can be configured using the **NRDUCellCsirs.FR1MaxCellCsirsPortNum** parameter.

In low-frequency TDD, if the gNodeB uses four or eight ports to transmit CSI-RS for CM, the number of ports can be adaptively adjusted based on the spectral efficiency of UEs.

- When the spectral efficiency of UEs is less than the value of **NRDUCellCsirs.Csirs4P8PAdaptiveSwThld**, the gNodeB changes the number of ports used to transmit CSI-RS for CM from eight to four.
- When the spectral efficiency of UEs is greater than or equal to the value of **NRDUCellCsirs.Csirs4P8PAdaptiveSwThld**, the gNodeB changes the number of ports used to transmit CSI-RS for CM from four to eight.

In this function:

- If the **NRDUCellCsirs.Csirs4P8PAdaptiveSwThld** parameter is set to **0**, port number adaptation for CSI-RS for CM does not take effect.
- After port number adaptation for CSI-RS for CM is enabled, the gNodeB notifies UEs of the adjusted number of ports they will use to receive CSI-RS for CM by sending RRC reconfiguration messages. This may increase the number of RRC reconfigurations.
- Port number adaptation for CSI-RS for CM takes effect only for non-antenna-selection UEs served by 8T8R, 32T32R, and 64T64R TDD RF modules operating in low frequency bands.

Inter-Cell CSI-RS for CM Interference Avoidance (Low-Frequency TDD)

Because the CSI-RS for CM time-frequency resources configured for the serving cell and its neighboring cells may be different, the PDSCH of the serving cell and the CSI-RS for CM of the neighboring cells may interfere with each other. To address this issue, inter-cell CSI-RS for CM interference avoidance is introduced, and is controlled by the **NRDUCellCsirs.NCellCsirsIntrfAvoid** parameter.

- If this parameter is set to **NO_AVOID**, inter-cell CSI-RS for CM interference avoidance is disabled. The serving cell does not avoid interference at the time-frequency position of CSI-RS for CM transmitted in neighboring cells.
- If this parameter is set to a value other than **NO_AVOID**, inter-cell CSI-RS for CM interference avoidance is enabled. The serving cell avoids interference at the time-frequency positions of CSI-RS for CM transmitted in neighboring cells based on the value of this parameter and does not schedule the PDSCH. Specifically, the serving cell avoids interference only at the time-frequency positions of specific types of CSI-RS for CM transmitted in neighboring cells. CSI-RS for CM has the following types: periodic CSI-RS for CM (including common periodic CSI-RS for CM and full-buffer UE-specific periodic CSI-RS for CM) and aperiodic CSI-RS for CM.
 - If this parameter is set to **P_CELL**, the serving cell avoids interference at the time-frequency position of common periodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.
 - If this parameter is set to **P_CELL_AND_AP**, the serving cell avoids interference at the time-frequency positions of common periodic CSI-RS for CM and aperiodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.
 - If this parameter is set to **P_USER**, the serving cell avoids interference at the time-frequency position of full-buffer UE-specific periodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.

- If this parameter is set to **P_USER_AND_AP**, the serving cell avoids interference at the time-frequency positions of full-buffer UE-specific periodic CSI-RS for CM and aperiodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.
- If this parameter is set to **P_CELL_AND_P_USER**, the serving cell avoids interference at the time-frequency positions of common periodic CSI-RS for CM and full-buffer UE-specific periodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.
- If this parameter is set to **P_CELL_AND_P_USER_AND_AP**, the serving cell avoids interference at the time-frequency positions of common periodic CSI-RS for CM, full-buffer UE-specific periodic CSI-RS for CM, and aperiodic CSI-RS for CM transmitted in neighboring cells and does not schedule the PDSCH.

Enabling inter-cell CSI-RS for CM interference avoidance has the following impacts on network performance:

- When a neighboring cell is transmitting CSI-RS for CM, the downlink UE throughput in the serving cell increases. This has no impact on the neighboring cell.
- When a neighboring cell is not transmitting CSI-RS for CM, the downlink UE throughput in the serving cell decreases. This also has no impact on the neighboring cell.

Configuration of the Minimum Scheduling Interval for Aperiodic CSI-RS for CM

The gNodeB sends DCI to UEs and transmits CSI-RS for CM to UEs at the time-frequency position indicated in the DCI. UEs receive CSI-RS for CM at the same time-frequency position. For Huawei gNodeBs, the minimum scheduling interval for aperiodic CSI-RS for CM can be configured by using the **NRDUCellCsirs.AperiodicCsiMinSchIntvl** parameter.

Full-Buffer UE-Specific CSI-RS for CM (Low-Frequency TDD)

In low-frequency TDD, narrower CSI-RS for CM beams (compared with common CSI-RS for CM beams) are introduced so that full-buffer UEs can obtain more accurate channel measurement results. The corresponding CSI-RS for CM resources are full-buffer UE-specific CSI-RS for CM.

Full-buffer UE-specific CSI-RS for CM can be transmitted only periodically as specified by the **NRDUCellCsirs.CsiUserPeriod** parameter.

- If this parameter is set to **NOT_CONFIG**, dedicated CSI-RS for CM resources are not allocated to full-buffer UEs.
- If this parameter is set to a value other than **NOT_CONFIG**, dedicated CSI-RS for CM resources are allocated to full-buffer UEs, and the value indicates the period of transmitting UE-specific CSI-RS for CM.

If a UE is incompatible with CSI-RS for CM resource reconfiguration, service drops may occur on the UE, resulting in a higher service drop rate in the cell. To address this issue, optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration is introduced. This function enables Huawei base stations to perform special processing for such UEs. It is controlled by the

CSIRS_INCOMPATIBLE_UE_SW option of the **NRDUCellCsirs.CsiSwitch** parameter, and takes effect only in low-frequency TDD.

When this function is enabled and the gNodeB identifies a UE incompatible with CSI-RS for CM resource reconfiguration, the gNodeB considers the UE as a full-buffer UE and allocates dedicated CSI-RS for CM resources to it. The number of incompatible UEs that can use dedicated CSI-RS for CM resources in a cell is specified by the **NRDUCellCsirs.CsirsResFullBufferUeNum** parameter, and the corresponding UE-specific CSI-RS for CM period is specified by the **NRDUCellCsirs.IncompatibleUeCsirsPeriod** parameter.

In low-frequency TDD, incompatible UEs occupy the CSI-RS for CM resources for full-buffer UEs. Consequently, more incompatible UEs can use dedicated CSI-RS for CM resources in a cell if a longer period is configured for full-buffer UE-specific CSI-RS for CM. Their configuration constraints are as follows:

- When **NRDUCellCsirs.CsiUserPeriod** is set to **SLOT20**, **NRDUCellCsirs.CsirsResFullBufferUeNum** can only be set to a value ranging from **0** to **2**.
- When **NRDUCellCsirs.CsiUserPeriod** is set to **SLOT40**, **NRDUCellCsirs.CsirsResFullBufferUeNum** can only be set to a value ranging from **0** to **4**.
- When **NRDUCellCsirs.CsiUserPeriod** is set to **SLOT80**, **NRDUCellCsirs.CsirsResFullBufferUeNum** can only be set to a value ranging from **0** to **8**.
- When **NRDUCellCsirs.CsiUserPeriod** is set to **SLOT160**, **NRDUCellCsirs.CsirsResFullBufferUeNum** can only be set to a value ranging from **0** to **16**.
- When **NRDUCellCsirs.CsiUserPeriod** is set to **SLOT320**, **NRDUCellCsirs.CsirsResFullBufferUeNum** can only be set to a value ranging from **0** to **32**.

Optimized UE Capability Identification Based on Configurable CSI-RS for CM Resources

According to 3GPP Release 15, the *codebookParameters* IE reported by a UE indicates the maximum configurable CSI-RS for CM resources for the UE. Specifically, it indicates only the maximum configurable CSI-RS for CM resources for UEs operating in a single frequency band, but cannot define the maximum for UEs operating in multiple frequency bands. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V15.9.0.

3GPP Release 16 introduces two new IEs *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* to indicate the maximum configurable CSI-RS for CM resources for UEs. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V16.2.0.

- The *codebookParameters* IE indicates the maximum configurable CSI-RS for CM resources for UEs operating in a single frequency band. It is always reported by UEs and does not need to be queried by the base station.
- The *codebookParametersPerBand-r16* IE also indicates the maximum configurable CSI-RS for CM resources for UEs operating in a single frequency band. It is reported by UEs only after the base station delivers the *codebookTypeRequest-r16* IE.

- The *codebookParametersPerBC-r16* IE indicates the maximum configurable CSI-RS for CM resources for UEs operating in multiple frequency bands. It is reported by UEs only after the base station delivers the *codebookTypeRequest-r16* IE.

The base station determines whether to use the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs to identify the maximum configurable CSI-RS for CM resources for UEs based on the following switches.

In SA networking, this function is controlled by the **CSIRS_UE_CAP_R16_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter.

- When this option is selected, the gNodeB delivers the *codebookTypeRequest-r16* IE. After receiving the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs reported by a UE, the gNodeB identifies the maximum configurable CSI-RS for CM resources for the UE based on the IEs.
- When this option is deselected, the gNodeB does not deliver the *codebookTypeRequest-r16* IE and identifies the maximum configurable CSI-RS for CM resources for a UE only based on the *codebookParameters* IE, regardless of whether the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs are reported by the UE.

In NSA networking, this function is controlled by the **CSIRS_UE_CAP_R16_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter on the NR side and the **R16_CSI_RS_UE_CAPB_QUERY_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter on the LTE side.

- When both options on NR and LTE sides are selected, the eNodeB delivers the *codebookTypeRequest-r16* IE. After receiving the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs reported by a UE, the gNodeB identifies the maximum configurable CSI-RS for CM resources for the UE based on the IEs.
- When the option on the LTE side is selected but the option on the NR side is deselected, the eNodeB delivers the *codebookTypeRequest-r16* IE. The gNodeB identifies the maximum configurable CSI-RS for CM resources for a UE based only on the *codebookParameters* IE regardless of whether the UE reports the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs.
- When the option on the NR side is selected but the option on the LTE side is deselected, the eNodeB does not deliver the *codebookTypeRequest-r16* IE, and a UE does not report the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs. The gNodeB identifies the maximum configurable CSI-RS for CM resources for the UE based only on the *codebookParameters* IE.
- When both options on NR and LTE sides are deselected, the eNodeB does not deliver the *codebookTypeRequest-r16* IE, and a UE does not report the *codebookParametersPerBand-r16* and *codebookParametersPerBC-r16* IEs. The gNodeB identifies the maximum configurable CSI-RS for CM resources for the UE based only on the *codebookParameters* IE.

UE Capability Adaptation Based on Configurable CSI-RS for CM Resources (Low-Frequency TDD)

The maximum configurable CSI-RS for CM resources reported by a UE may be inconsistent with that supported by the UE. If the reported value is less than the UE's capability, functions such as DL CoMP and spectrum sharing cannot work together. As such, UE capability adaptation based on configurable CSI-RS for CM resources is introduced. This function is controlled by the **CSIRS_CONFIG_CAP_ADAPTIVE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. This function is supported only in low frequency bands.

- If this option is selected, the gNodeB adaptively allocates CSI-RS for CM resources based on the maximum configurable CSI-RS for CM resources reported by the UE and the capability required by the system.
- If this option is deselected, the gNodeB allocates CSI-RS for CM resources only based on the maximum configurable CSI-RS for CM resources reported by the UE.

Adaptation of CSI-RS for CM Capabilities in Codebooks (Low-Frequency TDD)

The maximum number of ports used to receive CSI-RS for CM reported by a UE may be inconsistent with that supported by the UE. If the reported value is less than the UE's capability, functions such as DL CoMP and spectrum sharing cannot work together. As such, adaptation of CSI-RS for CM capabilities in codebooks is introduced. This function is controlled by the **CSIRS_CB_CAP_ADAPTIVE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. This function is supported only in low frequency bands.

- If this option is selected, the gNodeB adaptively allocates CSI-RS for CM resources based on the maximum number of ports used to receive CSI-RS for CM in the codebook reported by the UE and the capability required by the system.
- If this option is deselected, the gNodeB allocates CSI-RS for CM resources only based on the maximum number of ports used to receive CSI-RS for CM in the codebook reported by the UE.

CSI-RS for CM Reuse 3 (Low-Frequency TDD)

Because the CSI-RS for CM time-frequency resources configured for the serving cell and its neighboring cells may be the same in wide-beam scenarios, the CSI-RS for CM of the serving cell may be interfered by the CSI-RS for CM of neighboring cells. To address this issue, CSI-RS for CM reuse 3 is introduced. It is controlled by the **CSIRS_REUSE3_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. This function is supported only in low frequency bands.

- If this option is selected, CSI-RS for CM reuse 3 is enabled. In this case, the CSI-RS for CM time-frequency resources configured for the serving cell and its neighboring cells are used based on reuse 3. That is, CSI-RSs for CM are transmitted in staggered positions in different cells.
- If this option is deselected, CSI-RS for CM reuse 3 is disabled.

CSI-RS for CM reuse 3 may cause interference from the CSI-RS for CM to the PDSCH. To avoid such interference, at least one of the following functions must be enabled before this function is enabled:

- Inter-cell CSI-RS for CM interference avoidance (with the **NRDUCellCsirs.NCellCsirsIntrfAvoid** parameter set to **P_CELL**, **P_CELL_AND_AP**, **P_CELL_AND_P_USER**, or **P_CELL_AND_P_USER_AND_AP**)
- Static avoidance of CSI-RS for CM interference (controlled by the **CSIRS_INTRF_STATIC_AVOID_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter)
- Intra-frequency CSI-RS for CM interference avoidance (controlled by the **INTRA_FREQ_CSIRS_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter)

CSI-RS Resource Allocation Optimization (Low-Frequency TDD)

In low frequency bands, some CSI-RS resources are effective only when CSI-RS measurement results are reported. If CSI reporting resources are insufficient, the allocated CSI-RS resources will be ineffective, resulting in a waste of CSI-RS resources. To address this issue, CSI-RS resource allocation optimization is introduced, and is controlled by the **CSIRS_ALLOC_OPT_SW** option of the **NRDUCellCsirs.CsirsExtSwitch** parameter. With this function enabled, the allocation of CSI-RS resources is optimized based on a UE's CSI reporting capability. In other words, CSI-RS resources are allocated based on the estimated UE capability, which increases the CSI-RS resource usage.

UE-specific CSI-RS Reconfiguration Reduction (Low-Frequency TDD)

In low frequency bands, changes in the CSI-RS measurement results carried on the PUCCH during UE handovers can trigger repeated reconfigurations of UE-specific CSI-RS resources, increasing the uplink BLER and affecting user experience. To address this issue, UE-specific CSI-RS reconfiguration reduction is introduced, and is controlled by the **USER_CSI_RECFG_TIMES_RDC_SW** option of the **NRDUCellCsirs.CsirsExtSwitch** parameter. When enabled, this function reduces the number of UE-specific CSI-RS reconfigurations caused by changes in the $n1-n2$ value for the 8-port Type I codebook. For details about the $n1-n2$ value for the 8-port Type I codebook, see section 5.2.2.2.1 "Type I Single-Panel Codebook" in 3GPP TS 38.214 V15.16.0.

CSI Measurement Optimization (Low-Frequency TDD)

In low frequency bands, when certain functions (such as distributed massive MIMO and VMIMO in DAS-based indoor coverage) take effect and aperiodic CSI-RS resources are insufficient for UEs in non-overlapping areas, CSI measurement optimization can be enabled to adaptively adjust periodic CSI-RS resources, preventing CSI measurement failures for UEs in these areas. This function is controlled by the **CSI_MEAS_OPT_SW** option of the **NRDUCellCsirs.CsirsExtSwitch** parameter.

It is recommended that CSI measurement optimization be enabled under any of the following conditions:

- The **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter is selected.

- The **NRDUCellAlgoSwitch.VmimoSwitch** parameter is set to **ON**.
- The **RF_COMBINING_NOISE_SUPPR_SW** option of the **NRDUCellUISch.RfCombSwitch** parameter is selected.
- The **DISTRIBUTED_ANT_PMI_WT_OPT_SW** option of the **NRDUCellAlgoSwitch.BeamOptSwitch** parameter is selected.

7.1.2.2 CSI-RS for BM Resource Management (High-Frequency TDD)

In high-frequency TDD, CSI-RS for BM resource management includes the following functions.

Configuration of the CSI-RS for BM Measurement Period

The gNodeB transmits CSI-RS for BM within the active BWP. UEs receive and measure CSI-RS for BM to obtain the corresponding RSRP, and report the RSRP and CRI to the gNodeB. The gNodeB then obtains the RSRP information about the corresponding beam. The CSI-RS for BM measurement period is specified by the **NRDUCellCsirs.CsirsBeamMeasPeriod** parameter.

Full-Buffer UE-specific CSI-RS for BM

CSI-RS for BM beams dedicated to full-buffer UEs are introduced so that CSI-RS for BM resources do not need to be reconfigured for full-buffer UEs during SSB beam switching, enabling fast beam measurement. The corresponding CSI-RS for BM resources are full-buffer UE-specific CSI-RS for BM. This function is controlled by the **USER_BM_CSIRS_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. When this option is selected, this function takes effect only for the first identified UE running downlink full-buffer services in the cell.

NOTE

The changed setting of this option takes effect only for UEs that access the network afterwards. As such, it is recommended that target UEs re-access the network after the setting of this option is changed.

Ignoring CSI-RS Capabilities

The function of ignoring CSI-RS capabilities is controlled by the **CSIRS_CAPB_IGNORE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter.

- If this option is selected, the function takes effect, and the gNodeB configures CSI-RS resources for UEs based on feature requirements on the gNodeB side without considering the CSI-RS capabilities reported by them.
 - For UEs whose CSI-RS capabilities are greater than the reported CSI-RS capabilities, their downlink performance is improved.
 - For other UEs, they are possibly configured with CSI-RS resources beyond their CSI-RS capabilities. As a result, CSI-RS measurement results may be incorrect or service drops may occur.
- If this option is deselected, the function does not take effect, and the gNodeB configures CSI-RS resources for UEs based on their reported CSI-RS capabilities. In this case, UEs will not be configured with CSI-RS resources beyond their CSI-RS capabilities.

If insufficient CSI-RS resources can be configured due to limited CSI-RS capabilities of UEs, the gNodeB adaptively configures CSI-RS resources for UEs based on the required resources and UE capabilities. Adaptive configuration may increase the number of RRC reconfiguration messages and change the scheduling priority of periodic CSI. As a result, the average downlink UE throughput (**User Downlink Average Throughput (DU)**), average uplink UE throughput (**User Uplink Average Throughput (DU)**), average downlink cell throughput (**Cell Downlink Average Throughput (DU)**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) may fluctuate.

7.1.2.3 TRS Resource Management (High-Frequency TDD)

UE-specific TRS Resource Management

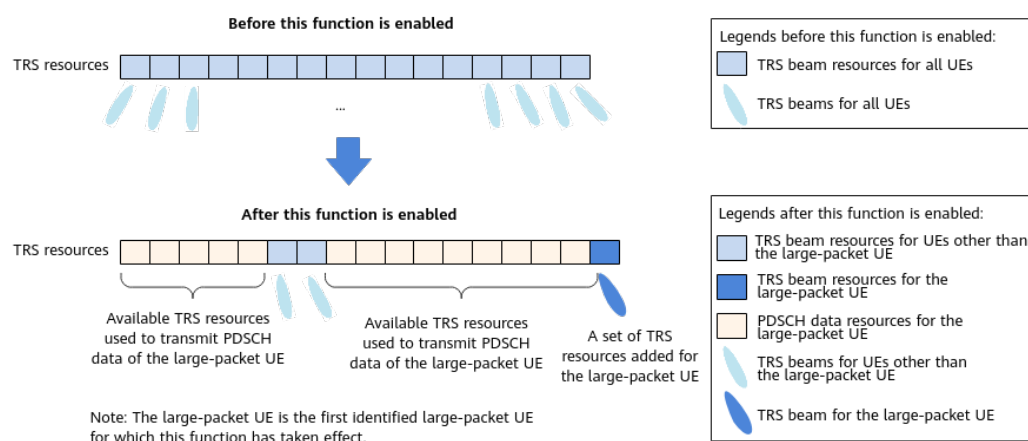
All TRS resources in a high-frequency TDD cell are used to transmit TRS beams regardless of the number of UEs in the cell. When the cell serves a small number of UEs, there may be no UEs in some TRS beams, causing a waste of TRS resources. As such, UE-specific TRS resource management is introduced, and is controlled by the **USER_TRS_BEAM_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter.

- If this option is deselected, UE-specific TRS resource management is disabled.
- If this option is selected, UE-specific TRS resource management is enabled. In a cell serving no UEs performing downlink large-packet services, the TRS resource allocation mode remains unchanged. Once the base station identifies the first UE performing downlink large-packet services in the cell, the TRS resource allocation mode changes, as shown in [Figure 7-1](#).
 - A set of TRS resources is added to transmit the TRS beam serving this large-packet UE.
 - Among other TRS resources, if a TRS beam corresponding to a set of TRS resources serves no UE, the set of TRS resources will be used to transmit PDSCH data of this large-packet UE rather than transmit a TRS beam. If a TRS beam corresponding to a set of TRS resources serves a UE, the set of TRS resources is still used to transmit a TRS beam.

A UE is considered to be performing downlink large-packet services when it meets all the following conditions:

- $\text{Proportion of TTIs in which the UE is scheduled in the downlink within 1s} = \frac{\text{Number of TTIs in which the UE is scheduled in the downlink within 1s}}{\text{Number of TTIs in the downlink in the cell within 1s}} \geq 40\%$
- Downlink PRB usage of the UE $\geq 30\%$
- Average downlink UE throughput ≥ 10 Mbit/s

Figure 7-1 UE-specific TRS resource management



7.1.2.4 QCL Relationship Management

To increase the accuracy of channel measurement results, the concept of quasi co-location (QCL) is introduced into NR. If the channel characteristics (for example, delay spread and Doppler spread) of a signal can be obtained according to those of another signal, a QCL relationship exists between the two signals. Optional QCL relationships between TRS and SSB, TRS and PDSCH, and TRS and CSI-RS for CM are described in section 5.1.5 "Antenna ports quasi co-location" in 3GPP TS 38.214.

QCL Relationship Management Between TRS and SSB (Low-Frequency TDD)

In low frequency bands, TRS is quasi co-located only with the optimal SSB beam reported by a UE. When the optimal SSB beam of the UE changes, the QCL relationship between TRS and SSB needs to be updated. This is known as SSB beam switching.

For Huawei gNodeBs, the **NRDUCellCsirs.SsbBeamSwitchingMode** parameter is used to specify whether to enable SSB beam switching and indicate the SSB beam switching mode.

- If this parameter is set to **NO_SWITCHING**, SSB beam switching is disabled. In this case, the QCL relationship between TRS and SSB does not take effect.
- If this parameter is set to **MAC_CE_MODE**, SSB beam switching is indicated through the MAC CE.

QCL Relationship Management Between TRS and CSI-RS for CM (Low-Frequency TDD)

The QCL relationship between TRS and CSI-RS for CM enables UEs to obtain channel characteristics of CSI-RS for CM according to those of TRS, increasing the accuracy of CSI-RS for CM measurement. The measurement results of semi-persistent CSI-RS for CM (not supported in the current version) and aperiodic CSI-RS for CM are less accurate than those of periodic CSI-RS for CM. As such, TRS is always quasi co-located with semi-persistent CSI-RS for CM (not supported in the current version) and aperiodic CSI-RS for CM according to 3GPP specifications. The **CSIRS_QCL_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter specifies whether the QCL relationship between TRS and periodic CSI-RS for CM takes effect.

- If this option is selected, the QCL relationship between TRS and periodic CSI-RS for CM takes effect, leading to a high accuracy of periodic CSI-RS for CM measurement.
- If this option is deselected, the QCL relationship between TRS and periodic CSI-RS for CM does not take effect, leading to a low accuracy of periodic CSI-RS for CM measurement.

When the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL** (indicating a hyper cell) or the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** (indicating a combined cell) for an NR DU cell:

- If one set of TRS resources is configured for UEs that initially access the cell (with **NRDUCellCsirs.TrsAllocationStrategy** set to **SINGLE_TRS**), the TRS resources change with handovers between TRPs. After this function is enabled, the RRC reconfiguration message for an inter-TRP handover contains an indication for the UE to update the QCL relationship between TRS and periodic CSI-RS for CM. In this way, the number of RRC reconfigurations is not increased.
- If multiple sets of TRS resources are configured for UEs that initially access the cell (with **NRDUCellCsirs.TrsAllocationStrategy** set to **MULTI_TRS**), the QCL relationship between TRS and periodic CSI-RS for CM cannot take effect.

7.1.3 ZP CSI-RS Resource Management

CSI-RS for IM resource management includes the following function.

Interference Measurement Resource Combination

Interference measurement resource combination is introduced to reduce CSI-RS for IM resource consumption. This function is controlled by the **IM_RESOURCE_MERGE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter.

- If this option is deselected, multiple sets of CSI-RS for IM resources are configured for each cell to measure interference from neighboring cells.
- If this option is selected, only one set of CSI-RS for IM resource is configured for each cell to measure interference from neighboring cells, reducing resource consumption and increasing the downlink cell throughput.

7.2 Network Analysis

7.2.1 Benefits

CSI-RS resource management improves network performance by properly utilizing CSI-RS resources.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.2.2 Impacts](#).

If a large number of UEs are served, UEs that require high data rates are preferentially allocated CSI-RS resources. If CSI-RS resources are insufficient, these

UEs preempt such resources of other UEs that do not require high data rates. This mechanism increases the possibility of service drops for UEs that do not require high data rates.

Table 7-2 describes the network impacts of each sub-function involved in CSI-RS resource management. For the definitions of related indicators, see [7.4.2 Activation Verification](#).

Table 7-2 Network impacts of the sub-functions involved in CSI-RS resource management

Sub-Function Name	Network Impacts
Periodic CSI-RS	A shorter CSI-RS transmission period increases the accuracy of channel measurement and helps guarantee the performance of features that depend on CSI-RS measurement. However, it also increases CSI-RS resource consumption. A longer CSI-RS transmission period results in less CSI-RS resource consumption. However, it also decreases the accuracy of channel measurement, negatively affects the performance of features that depend on CSI-RS measurement, and possibly increases the uplink RBLER.
Port number adaptation for CSI-RS for CM	In low-frequency TDD, after this function is enabled, the number of RRC reconfigurations may increase.
Inter-cell CSI-RS for CM interference avoidance	In low-frequency TDD, after this function is enabled: • When a neighboring cell is transmitting CSI-RS for CM, the downlink UE throughput in the serving cell increases. This has no impact on the neighboring cell. • When a neighboring cell is not transmitting CSI-RS for CM, the downlink UE throughput in the serving cell decreases. This also has no impact on the neighboring cell.

Sub-Function Name	Network Impacts
Configuration of the minimum scheduling interval for aperiodic CSI-RS for CM	When the minimum scheduling interval for aperiodic CSI-RS for CM is configured in a high-frequency TDD cell serving a large number of UEs but with light traffic load, the slots in which the gNodeB configures aperiodic CSI-RS for CM for UEs may change. As a result, the cell throughput may fluctuate. The degree of fluctuation depends on the capability of the UE to measure aperiodic CSI-RS for CM. When the minimum scheduling interval for aperiodic CSI-RS for CM is configured, the actual scheduling interval is the configured value in single-UE scenarios, but may be greater than the configured value in multiple-UE scenarios. Its network impacts are as follows: When the channel environment changes rapidly (for example, in high-speed mobility scenarios): • A smaller minimum scheduling interval for aperiodic CSI-RS for CM results in a higher channel matching degree, higher downlink measurement resource consumption, and increased downlink UE throughput. • A larger minimum scheduling interval for aperiodic CSI-RS for CM results in a lower channel matching degree, lower downlink measurement resource consumption, and decreased downlink UE throughput. When the channel environment changes slowly (for example, in stationary scenarios): • A smaller minimum scheduling interval for aperiodic CSI-RS for CM results in a higher channel matching degree, higher downlink measurement resource consumption, and decreased downlink UE throughput. • A larger minimum scheduling interval for aperiodic CSI-RS for CM results in a lower channel matching degree, lower downlink measurement resource consumption, and increased downlink UE throughput.

Sub-Function Name	Network Impacts
Full-buffer UE-specific CSI-RS for CM	In low frequency bands, after this function is enabled, the downlink throughput of full-buffer UEs increases. Adjusting the period of transmitting full-buffer UE-specific CSI-RS for CM after this function is enabled has the following impacts: • A longer period results in a longer measurement time for full-buffer UEs, decreased occupation of CSI-RS for CM resources, and larger number of allowed full-buffer UEs. However, it also results in poorer timeliness of channel tracing and decreased downlink throughput of full-buffer UEs. • A shorter period results in a shorter measurement time for full-buffer UEs, increased occupation of CSI-RS for CM resources, and smaller number of allowed full-buffer UEs. However, it also results in better timeliness of channel tracing and increased downlink throughput of full-buffer UEs.
Optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration	In low-frequency TDD, after this function is enabled, the service drop rate in the cell decreases but the CSI-RS for CM resource consumption increases if the cell serves incompatible UEs. Adjusting the number of incompatible UEs that can use dedicated CSI-RS for CM resources after this function is enabled in a cell has the following impacts: • A smaller number results in a lower probability that a short period is used for transmitting full-buffer UE-specific CSI-RS for CM and lower downlink throughput of full-buffer UEs. However, more common UEs in the cell can be allocated periodic CSI-RS for CM, increasing the downlink throughput of common UEs. • A larger number results in a higher probability that a short period is used for transmitting full-buffer UE-specific CSI-RS for CM and higher downlink throughput of full-buffer UEs. However, fewer common UEs in the cell can be allocated periodic CSI-RS for CM, decreasing the downlink throughput of common UEs.
Optimized UE capability identification based on configurable CSI-RS for CM resources	After this function is enabled, the gNodeB operates based on the highest capability reported by UEs. In this way, the network performance matches the UE capability more closely. If a UE supports the <i>codebookParametersPerBand-r16</i> and <i>codebookParametersPerBC-r16</i> IEs but does not support the CSI-RS for CM capability indicated by the IEs, the downlink throughput of the UE decreases, and service drops may occur.

Sub-Function Name	Network Impacts
UE capability adaptation based on configurable CSI-RS for CM resources	After this function is enabled, if the maximum configurable CSI-RS for CM resources reported by a UE are less than its capability, the gNodeB adaptively allocates resources based on the estimated capability of the UE to meet service requirements. In this way, the downlink UE throughput increases.
Adaptation of CSI-RS for CM capabilities in codebooks	After this function is enabled, if the maximum number of ports used to receive CSI-RS for CM reported by the UE is less than its capability, the gNodeB adaptively allocates resources based on the estimated capability of the UE to meet service requirements. In this way, the downlink UE throughput increases.
CSI-RS for CM reuse 3	After this function is enabled, the CSI-RS for CM of the serving cell suffers from less interference from the CSI-RS for CM of its neighboring cells, the accuracy of channel measurement increases, and the downlink cell and UE throughputs increase.
CSI-RS resource allocation optimization	After this function takes effect, the allocated CSI-RS resources will not become ineffective, leading to an increase in the CSI-RS resource usage and fluctuations in the average downlink cell throughput.
UE-specific CSI-RS reconfiguration reduction	After this function takes effect, uplink retransmissions due to reconfigurations decrease, along with the uplink IBLER and RBLER.
CSI measurement optimization	After this function takes effect, the base station adaptively allocates aperiodic or periodic CSI-RS resources to UEs in non-overlapping areas, leading to fluctuations in the CSI measurement result and average downlink UE throughput.
Configuration of the CSI-RS for BM measurement period	In high-frequency TDD, setting the CSI-RS for BM measurement period to different values has the following impacts: • A shorter beam measurement period results in a shorter beam update period, higher downlink beam measurement resource consumption, and lower average downlink cell throughput. • A longer period results in a longer beam update period, lower downlink beam measurement resource consumption, and higher average downlink cell throughput.

Sub-Function Name	Network Impacts
Full-buffer UE-specific CSI-RS for BM	In high-frequency TDD, after this function is enabled: - For full-buffer UEs for which this function has taken effect, the average downlink throughput and average downlink MCS index increase, the downlink IBLER may fluctuate, and the average uplink throughput and average uplink MCS index fluctuate. - In light-load scenarios, the average uplink and downlink throughputs, average uplink and downlink MCS indexes, and downlink IBLER of the UEs for which this function does not take effect (background UEs) may fluctuate. In heavy-load scenarios, the average uplink and downlink throughputs of background UEs decrease, and the average uplink and downlink MCS indexes and downlink IBLER of background UEs may fluctuate.
Ignoring CSI-RS capabilities	In high-frequency TDD, after this function is enabled: - For UEs whose CSI-RS capabilities are greater than the reported CSI-RS capabilities, their downlink performance is improved. - For other UEs, they are possibly configured with CSI-RS resources beyond their CSI-RS capabilities. As a result, CSI-RS measurement results may be incorrect or service drops may occur.
UE-specific TRS resource management	After this function is enabled for a high-frequency TDD cell and UE-specific TRS resources are allocated to UEs performing downlink large-packet services in the cell, the average downlink MCS index of the cell increases, the downlink throughput of UEs performing downlink large-packet services increases, and the average downlink cell throughput increases. If there are no UEs performing downlink large-packet services in the cell after this function is enabled, there is no impact.
SSB beam switching	After this function is enabled, UEs must report optimal SSB beams, increasing uplink overheads and decreasing the average uplink UE throughput and average uplink cell throughput. In addition, UEs that have compatibility issues with this function may experience an increased service drop rate. The compatibility issues result from UE vendors' different interpretations on this function (though in compliance with 3GPP specifications).
QCL relationship management between TRS and CSI-RS for CM	After the QCL relationship between TRS and CSI-RS for CM takes effect, the accuracy of the measurement results of CSI-RS for CM increases. When the channel environment changes rapidly (for example, in high-speed mobility scenarios), the downlink UE throughput increases.

Sub-Function Name	Network Impacts
Interference measurement resource combination	After this function is enabled, the resource consumption decreases, and the downlink cell throughput increases.

7.3 Requirements

7.3.1 Licenses

This basic function is not under license control.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Periodic TRS transmission	NRDUCellCsirs.TrsPeriod	<i>Channel Management</i>	SSB beam switching can be enabled (with NRDUCellCsirs.SsbBeamSwitchingMode set to MAC_CE_MODE) only when NRDUCellCsirs.TrsPeriod is set to a value other than MS0 .
Low-frequency TDD	Long PUCCH format	NRDUCellPucch.StructureType	<i>Channel Management</i>	SSB beam switching can be enabled (with NRDUCellCsirs.SsbBeamSwitchingMode set to MAC_CE_MODE) only when a long PUCCH time-frequency format (with NRDUCellPucch.StructureType set to LONG_STRUCTURE) is used in the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Periodic CSI-RS	NRDUCellCsirs.CsiPeriod	<i>Channel Management</i>	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) can be enabled only when NRDUCellCsirs.CsiPeriod is not set to SLOT10 or SLOT20 .
Low-frequency TDD	Configuration of the maximum port number used for CSI-RS for CM	NRDUCellCsirs.FR1MaxCellCsirsPortNum	<i>Channel Management</i>	When CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) is enabled and NRDUCellCsirs.CsiPeriod is set to SLOT10 , SLOT20 , or SLOT40 , the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter (specifying the maximum number of ports used to transmit CSI-RS for CM) cannot be set to 32PORT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-cell CSI-RS for CM interference avoidance	NRDUCellCsirs.NCellCsirsIntrfAvoid	<i>Channel Management</i>	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) can be enabled only when at least one of the following functions is enabled: • Inter-cell CSI-RS for CM interference avoidance (with the NRDUCellCsirs.NCellCsirsIntrfAvoid parameter set to P_CELL, P_CELL_AND_AP, P_CELL_AND_P_USER, or P_CELL_AND_P_USER_AND_AP) • Static avoidance of CSI-RS for CM interference (controlled by the CSIRS_INTRF_STATIC_AVOID_SW option of the NRDUCellCsirs.CsiSwitch parameter) • Intra-frequency CSI-RS for CM interference avoidance (controlled by the INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoid-Switch parameter)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency CSI-RS for CM interference avoidance	INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) can be enabled only when at least one of the following functions is enabled: • Inter-cell CSI-RS for CM interference avoidance (with the NRDUCellCsirs.NCellCsirsIntrfAvoid parameter set to P_CELL, P_CELL_AND_AP, P_CELL_AND_P_USER, or P_CELL_AND_P_USER_AND_AP) • Static avoidance of CSI-RS for CM interference (controlled by the CSIRS_INTRF_STATIC_AVOID_SW option of the NRDUCellCsirs.CsiSwitch parameter) • Intra-frequency CSI-RS for CM interference avoidance (controlled by the INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Static avoidance of CSI-RS for CM interference	CSIRS_INTRF_STATIC_AVOID_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) can be enabled only when at least one of the following functions is enabled: - Inter-cell CSI-RS for CM interference avoidance (with the NRDUCellCsirs.NCellCsirsIntrfAvoid parameter set to P_CELL, P_CELL_AND_AP, P_CELL_AND_P_USER, or P_CELL_AND_P_USER_AND_AP) - Static avoidance of CSI-RS for CM interference (controlled by the CSIRS_INTRF_STATIC_AVOID_SW option of the NRDUCellCsirs.CsiSwitch parameter) - Intra-frequency CSI-RS for CM interference avoidance (controlled by the INTRAFREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoid-Switch parameter)
High-frequency TDD	None	None	None	None

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter (specifying the maximum number of ports used to transmit CSI-RS for CM) cannot be set to 16PORT .
Low-frequency TDD	Amount of CSI-RS resources in a cell	NRDUCellCsirs.CsirsCellResourceNum	<i>Channel Management</i>	When NRDUCellTrp.TxRxMode is set to 32T32R or 64T64R and NRDUCellCsirs.CsirsCellResourceNum is not set to 1_RESOURCE , CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	SSB beam sweeping period	NRDUCell.SsbPeriod	<i>Channel Management</i>	- When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected) and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT160, the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) can only be set to MS20. - When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected) and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT320, the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) can only be set to MS20 or MS40. - When CSI-RS for BM is transmitted in SSB slots (with the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter selected)

RAT	Function Name	Function Switch	Reference	Description
				and the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter (specifying the CSI-RS for BM measurement period) is set to SLOT640 , the NRDUCell.SsbPeriod parameter (specifying the SSB beam sweeping period) cannot be set to MS160 .
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Interference measurement resource combination (controlled by the IM_RESOURCE_MERGE_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled in hyper cells (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL).
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled in hyper cells (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL).
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled in hyper cells (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Interference measurement resource combination (controlled by the IM_RESOURCE_MERGE_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled in combined cells (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE).
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	CSI-RS for CM reuse 3 (controlled by the CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled in combined cells (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE) for which NRDUCellMultiTrp.DataTrasMode is not set to JOINT_MODE .
Low-frequency TDD	TRS rate matching optimization	TRS_RATEMATCHCH_OPT_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When NRDUCellCsirs.TrsPeriod is set to MS10 , TRS rate matching optimization cannot be enabled.
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	UE-specific TRS resource management cannot be enabled when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103 or SCENARIO_104 .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	When the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103 or SCENARIO_104 , full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
High-frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	When the Flexible Dense Beam function is enabled, the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter cannot be set to SLOT160 .
High-frequency TDD	CSI Beam Allocation Policy	NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy	<i>mmWave Beam Management (High-Frequency TDD)</i>	When the NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy parameter is set to CSI_OPT_SEL_BEAM , full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Connected mode coverage pattern for emergency and normal services	NRDUCellTrpBeam.ConnModeCovScenarioExt	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	• If the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT or SCENARIO_1 for the cell. • If the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT, SCENARIO_1, SCENARIO_2, or SCENARIO_3 for the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Connected mode coverage pattern for emergency and normal services	NRDUCellTrpBeam.ConnModeCovScenarioExt	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	- If the NRDUCellCsirs.CsirsCellResourceNum parameter (specifying the number of resources for periodic CSI-RS transmission) is set to 1_RESOURCE, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT, and the NRDUCellCsirs.CsirsBeamType parameter is set to TYPE0 for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell. - If the NRDUCellCsirs.CsirsCellResourceNum parameter (specifying the number of resources for periodic CSI-RS transmission) is set to FD_RESOURCE for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter cannot be set to SCENARIO_3 or SCENARIO_4 for the cell.
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceUISchSwitch parameter	<i>VoNR</i>	When 3GPP R15-compliant uplink slot aggregation is enabled, optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration (controlled by the CSIRS_INCOMPATIBLE_UE_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI-RS beam sweeping for carriers	CARR_CSI_BEAM_SWEEPING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	When the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE , CSI-RS beam sweeping for carriers (controlled by the CARR_CSI_BEAM_SWEEPING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter) cannot be enabled.
Low-frequency TDD	32-port CSI-RS measurement on SCCs	CA_SCC_CSIRS_32PORT_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Carrier Aggregation</i>	32-port CSI-RS measurement on SCCs cannot be enabled when NRDUCellCsirs.CsiPeriod is set to SLOT10 .
Low-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	PUCCH grouping adaptation for inter-FR CA and SSB beam switching cannot be both enabled as they cannot work together. The NRDUCellCsirs.SsbBeamSwitchingMode parameter must be set to NO_SWITCHING before PUCCH grouping adaptation for inter-FR CA can be enabled.
High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	When PUCCH grouping adaptation for inter-FR CA is enabled, full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INT RF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When proactive avoidance of remote interference (controlled by the SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, NRDUCellCsirs.FR1MaxCellCsirsPortNum cannot be set to 16PORT .
High-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	When basic functions of low latency and high reliability are enabled, full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When distributed massive MIMO is enabled, full-buffer UE-specific CSI-RS for CM cannot be enabled (that is, NRDUCellCsirs.CsiUserPeriod must be set to NOT_CONFIG).
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	In TDD: When the NRDUCellCsirs.TrsPeriod parameter is set to MS0 or MS10 , RedCap cannot be enabled.
High-frequency TDD	Slot configuration	NRDUCell.SlotAssignment set to 2_3_DSUUU	None	When the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , UE-specific TRS resource management (controlled by the USER_TRS_BEAM_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Slot configuration	NRDUCell.SlotAssignment set to 2_3_DSUUU	None	When the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
High-frequency TDD	Transmit and Receive Mode	NRDUCellTrp.TxRxMode	None	When the NRDUCellTrp.TxRxMode parameter is set to 2T , full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
High-frequency TDD	TRS Beam Type	NRDUCellCsirs.TrsBeamType	None	When the NRDUCellCsirs.TrsBeamType parameter is set to WIDE_BEAM , full-buffer UE-specific CSI-RS for BM (controlled by the USER_BM_CSIRS_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
High-frequency TDD	TRS beam type configuration	NRDUCellCsirs.TrsBeamType	None	When the NRDUCellCsirs.TrsBeamType parameter (specifying the TRS beam type) is set to WIDE_BEAM , UE-specific TRS resource management (controlled by the USER_TRS_BEAM_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	None	If multiple sets of TRS resources are configured for UEs that initially access the cell (with NRDUCellCsirs.TrsAllocationStrategy set to MULTI_TRS), the QCL relationship between TRS and CSI-RS for CM (controlled by the CSIRS_QCL_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled.
Low-frequency TDD	PMI-based weight optimization for distributed antennas	DISTRIBUTED_ANT_PMI_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	Optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration (controlled by the CSIRS_INCOMPATIBLE_UE_SW option of the NRDUCellCsirs.CsiSwitch parameter) cannot be enabled together with PMI-based weight optimization for distributed antennas.

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED) do not support TRS periods longer than 20 ms. This prevents delayed tracking of time and frequency offsets when UEs are moving at high speeds.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	For cells with intra-base-station DL CoMP enabled, if periodic measurement of CSI-RS for CM is used and NRDUCellCsirs.CsiPeriod is set to SLOT10 , UEs receive CSI-RS for CM and report CSI-RS for CM measurement results according to the same period. This will decrease the probability that intra-base-station DL CoMP takes effect. You are advised to change the CSI-RS for CM measurement period to another value. When aperiodic measurement of CSI-RS for CM is used in a single cell, intra-base-station DL CoMP cannot be performed for UEs that support only one set of aperiodic CSI-RS for CM resources.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	TRS is configured in a staggered manner for the TRPs in a hyper cell to prevent mutual interference based on the different results of TRP ID modulo 6. CSI-RS for CM is configured in the same manner as that before TRP combination.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	TRS is configured in a staggered manner for the TRPs in a combined cell to prevent mutual interference based on the different results of TRP ID modulo 6. CSI-RS for CM is configured in the same manner as that before the combination.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	For low-frequency cells: - Periodic CSI-RS for CM: The gNodeB does not check CSI-RS for CM measurement reports from UEs in DRX sleep time. As a result, the number of valid CSI-RS for CM measurement reports received by the gNodeB decreases. - Aperiodic CSI-RS for CM: UEs send CSI-RS for CM measurement reports on the PUSCH, which may cause continuous scheduling of uplink data for these UEs. As a result, these UEs cannot enter DRX sleep time. For high-frequency cells: - Periodic CSI-RS for BM: The gNodeB does not check CSI-RS measurement reports from UEs in DRX sleep time. As a result, the number of valid CSI-RS measurement reports received by the gNodeB decreases. - Aperiodic CSI-RS for BM: The gNodeB schedules the aperiodic CSI-RS measurement results reported by UEs only within the length of the Inactivity Timer triggered by common data transmission or the On Duration Timer. As a result, the number of valid CSI reports received by the gNodeB decreases and the reporting interval increases.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	The Flexible Dense Beam function involves beam densification and flexible sweeping. The two sub-functions take effect simultaneously by selecting the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter. After beam densification takes effect, the resource consumption of beam sweeping increases with the number of CSI-RS for BM beams. Meanwhile, flexible sweeping can reduce the resource consumption in a cell serving a small number of UEs, but significantly increase the resource consumption in a cell serving a large number of UEs. As prolonging the beam sweeping period can reduce the resource consumption, you are advised to modify the CSI-RS for BM measurement period configuration when the Flexible Dense Beam function is enabled. • In eMBB scenarios, you are advised to set the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter to SLOT320. • In FWA scenarios, you are advised to set the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter to SLOT640.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When the NRDUCellComp.ItCsirsTxMode parameter is set to PERIODIC_TD_MODE_ABN or PERIODIC_TD_MODE_ALL , Co-MM takes effect only if the NRDUCellCsirs.CsiUserPeriod parameter is set to SLOT20 , SLOT40 , or SLOT80 .
High-frequency TDD Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	UE-specific TRS resource management (controlled by the USER_TRS_BEAM_SW option of the NRDUCellCsirs.CsiSwitch parameter) does not take effect on high-frequency SCCs.
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode	<i>MIMO (TDD)</i>	When MetaAAU beam sensing is enabled, you are advised to disable UE capability adaptation based on configurable CSI-RS for CM resources (controlled by the CSIRS_CONFIG_CAP_ADAPTIVE_SW option of the NRDUCellCsirs.CsiSwitch parameter).
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	A distributed massive MIMO cell supports only one CSI-RS resource (with the value 1_RESOURCE always taking effect for the NRDUCellCsirs.CsirsCellResourceNum parameter).
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDirank.DlRankSelAlgoSw parameter	<i>FWA</i>	When downlink 8-layer transmission for a single FWA user takes effect, the NRDUCellCsirs.Csirs4P8PAdaptiveSwThld parameter must be set to 0 .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWNSW option of the NRDUCellAlgorithmPowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If full-buffer UE-specific CSI-RS for BM has taken effect for UEs, RF channel intelligent shutdown cannot take effect. If RF channel intelligent shutdown has taken effect for UEs, full-buffer UE-specific CSI-RS for BM cannot take effect unless the UEs exit the RF channel intelligent shutdown state.
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWNSW option of the NRDUCellAlgorithmPowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If UE-specific TRS resource management has taken effect for UEs, RF channel intelligent shutdown cannot take effect. If RF channel intelligent shutdown has taken effect for UEs, UE-specific TRS resource management cannot take effect unless the UEs exit the RF channel intelligent shutdown state.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

 NOTE

3900 series base stations do not support high-frequency TDD functions.

Boards

Except for the RAT, the sub-functions of this function have no other requirements on main control boards and baseband processing units. All main control boards and baseband processing units of the corresponding RAT support these sub-

functions. For details about the RATs supported by different main control boards and baseband processing units, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Configuration of the maximum port number used for CSI-RS for CM
All NR TDD-capable low-frequency RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Port number adaptation for CSI-RS for CM
All NR TDD-capable low-frequency RF modules with at least 8T8R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Interference measurement resource combination
All NR-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Other sub-functions of CSI-RS resource management
Except for the RAT, the sub-functions of this function have no other requirements on RF modules. All RF modules of the corresponding RAT support these sub-functions. For details about the RATs supported by different RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Cells

CSI-RS for CM reuse 3 (controlled by the **CSIRS_REUSE3_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter) cannot be enabled in SUL cells (with **NRDUCell.DuplexMode** set to **CELL_SUL**).

CSI-RS for CM reuse 3 (controlled by the **CSIRS_REUSE3_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter) cannot be enabled in high-frequency cells (with **NRDUCell.FrequencyBand** set to **N257**, **N258**, **N260**, or **N261**).

UE-specific TRS resource management (controlled by the **USER_TRS_BEAM_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter) cannot be enabled in NR cells with **NRDUCell.SlotAssignment** set to **2_3_DSUUU**.

SSB beam switching can be enabled (with **NRDUCellCsirs.SsbBeamSwitchingMode** set to a value other than **NO_SWITCHING**) only in NR cells with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**.

Interference measurement resource combination (controlled by the **IM_RESOURCE_MERGE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter) can be enabled only in NR cells with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL** or **FUSION_CELL**, **NRDUCellCsirs.CsirsCellResourceNum** set to **4_RESOURCE**, and **NRDUCellCsirs.FR1MaxCellCsirsPortNum** set to **8PORT**.

UE-specific CSI-RS reconfiguration reduction (controlled by the **USER_CSI_RECFG_TIMES_RDC_SW** option of the **NRDUCellCsirs.CsiRsExtSwitch** parameter) cannot be enabled in NR cells with

NRDUCell.NrDuCellNetworkingMode set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**.

7.3.5 Others

Port number adaptation for CSI-RS for CM is supported only by non-antenna-selection UEs.

UE-specific CSI-RS reconfiguration reduction requires UEs to support 8-port CSI-RS measurement on SCCs. That is, UEs must report the *maxNumberTxPortsPerResource* IE with values that include **p8**.

UE-specific TRS resource management must be either enabled or disabled for all cells belonging to the same sector.

Other functions have no further requirements.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

This function takes effect by default and no parameters are used for activation. [Table 7-3](#) describes the parameters used for function optimization.

Table 7-3 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
CSI Period	NRDUCellCsirs.CsiPeriod	None	You are advised to retain the default value. Changing the value of this parameter will cause batch reconfiguration for all UEs in the cell so that the latest value takes effect. As such, you are advised to change the value of this parameter only when there are a small number of UEs in the cell. This parameter takes effect only on CSI-RS for CM in low frequency bands. It is inapplicable to high frequency bands.

Parameter Name	Parameter ID	Option	Setting Notes
TRS Period	NRDUCellCsirs.TrsPeriod	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. You are advised to retain the default value.
CSIRS Cell Resource Number	NRDUCellCsirs.CsirsCellResourceNum	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. You are advised to retain the default value. This parameter takes effect only on CSI-RS for CM in low frequency bands. It is inapplicable to high frequency bands.
FR1 Max Cell CSI-RS Port Num	NRDUCellCsirs.FR1MaxCellCsirsPortNum	None	You are advised to retain the default value.
CSI-RS 4P/8P Adaptive Switching Thld	NRDUCellCsirs.Csirs4P8PAdaptiveSwThld	None	You are advised to retain the default value.
NCell CSI-RS Interference Avoidance	NRDUCellCsirs.NCellCsirsInterfAvoid	None	You are advised to retain the default value.
Aperiodic CSI Min Scheduling Interval	NRDUCellCsirs.AperiodicCsiMinSchIntvl	None	You are advised to retain the default value.
User CSI Period	NRDUCellCsirs.CsiUserPeriod	None	You are advised to retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_INCOMPATIBLE_UE_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Select the CSIRS_INCOMPATIBLE_UE_SW option if optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_CB_CAP_ADAPTIVE_SW	Select the CSIRS_CB_CAP_ADAPTIVE_SW option if adaptation of CSI-RS for CM capabilities in codebooks is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_CONFIG_CAP_ADAPTIVE_SW	Select the CSIRS_CONFIG_CAP_ADAPTIVE_SW option if UE capability adaptation based on configurable CSI-RS for CM resources is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	USER_TRS_BEAM_SW	Select the USER_TRS_BEAM_SW option if UE-specific TRS resource management is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_QCL_SW	Select the CSIRS_QCL_SW option if the QCL relationship between TRS and periodic CSI-RS for CM is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	IM_RESOURCE_MERGE_SW	Select the IM_RESOURCE_MERGE_SW option if interference measurement resource combination is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_REUSE3_SW	Select the CSIRS_REUSE3_SW option if CSI-RS for CM reuse 3 is required. You are advised to use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
CSI Switch	NRDUCellCsirs.CsiSwitch	USER_BM_CSIRS_SW	Select the USER_BM_CSIRS_SW option if full-buffer UE-specific CSI-RS for BM is required. You are advised to use the recommended value.
CSI Switch	NRDUCellCsirs.CsiSwitch	CSIRS_CAPB_IGNORE_SW	Select the CSIRS_CAPB_IGNORE_SW option if the function of ignoring CSI-RS capabilities is required. You are advised to use the recommended value.
Compatibility Algorithm Switch	gNodeBParam.CompatibilityAlgoSwitch	CSIRS_UE_CAP_R16_SW	Select the CSIRS_UE_CAP_R16_SW option if optimized UE capability identification based on configurable CSI-RS for CM resources in accordance with 3GPP Release 16 is required. In NSA networking, the R16_CSI_RS_UE_CAPB_QUERY_SW option of the NsaDcAlgoParam.NsaDcAlgoSwitch parameter on the LTE side also needs to be selected.
CSI-RS Res Full-Buffer UE Num	NRDUCellCsirs.CsirsResFullBufferUeNum	None	You are advised to retain the default value.
Incompatible UE CSI-RS Period	NRDUCellCsirs.IncompatibleUeCsirsPeriod	None	You are advised to retain the default value.
CSI-RS Beam Measurement Period	NRDUCellCsirs.CsirsBeamMeasurementPeriod	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. It is recommended that this parameter be set to SLOT320 in eMBB mobility scenarios and SLOT640 in fixed wireless access (FWA) scenarios.
SSB Beam Switching Mode	NRDUCellCsirs.SsbBeamSwitchingMode	None	You are advised to retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
CSI-RS Extend Switch	NRDUCellCsirs.CsirsExtSwitch	CSIRS_ALLOC_OPT_SW	Select the CSIRS_ALLOC_OPT_SW option if CSI-RS resource allocation optimization is required. You are advised to use the recommended value. This option is selected by default, which is also the recommended setting.
CSI-RS Extend Switch	NRDUCellCsirs.CsirsExtSwitch	USER_CSI_RECFG_TIMES_RDC_SW	Select the USER_CSI_RECFG_TIMES_RDC_SW option if UE-specific CSI-RS reconfiguration reduction is required. You are advised to use the recommended value.
CSI-RS Extend Switch	NRDUCellCsirs.CsirsExtSwitch	CSI_MEAS_OPT_SW	Select the CSI_MEAS_OPT_SW option if CSI measurement optimization is required. You are advised to use the recommended value.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following provides activation command examples for the sub-functions of CSI-RS resource management.

Low-frequency TDD command examples

```
//Setting the period of periodic CSI-RS for CM transmission to 40 slots
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiPeriod=SLOT40;
//Setting the number of resources for periodic CSI-RS for CM to four
MOD NRDUCELLCSIRS: NrDuCellId=0, CsirsCellResourceNum=4_RESOURCE;
//Setting the period of periodic TRS transmission to 20 ms
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsPeriod=MS20;

//CSI-RS for CM port number management
//Setting the FR1MaxCellCsirsPortNum parameter to 8PORT (only after the CsirsCellResourceNum
parameter is configured)
MOD NRDUCELLCSIRS: NrDuCellId=0, CsirsCellResourceNum=4_RESOURCE,
FR1MaxCellCsirsPortNum=8PORT;
```

```
//Enabling port number adaptation for CSI-RS for CM (by setting the Csirs4P8PAdaptiveSwThld parameter
to a non-zero value, such as 5)
MOD NRDUCELLCSIRS: NrDuCellId=0, Csirs4P8PAdaptiveSwThld=5;
//Enabling inter-cell CSI-RS for CM interference avoidance (by setting the NCellCsirsIntrfAvoid parameter to
a value other than NO_AVOID, such as P_CELL, which indicates that the serving cell avoids interference at
the time-frequency position of common periodic CSI-RS for CM transmitted in neighboring cells)
MOD NRDUCELLCSIRS: NrDuCellId=0, NCellCsirsIntrfAvoid=P_CELL;
//Setting the minimum scheduling interval for aperiodic CSI reports to 40 ms
MOD NRDUCELLCSIRS: NrDuCellId=0, AperiodicCsiMinSchIntvl=MS40;

//Full-buffer UE-specific CSI-RS for CM
//Setting the CsiUserPeriod parameter to SLOT40
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiUserPeriod=SLOT40;
//Selecting the CSIRS_INCOMPATIBLE_UE_SW option
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_INCOMPATIBLE_UE_SW-1;
//Setting the CsirsResFullBufferUeNum parameter to 1
MOD NRDUCELLCSIRS: NrDuCellId=0, CsirsResFullBufferUeNum=1;
//Setting the IncompatibleUeCsirsPeriod parameter to SLOT320
MOD NRDUCELLCSIRS: NrDuCellId=0, IncompatibleUeCsirsPeriod=SLOT320;

//Optimized UE capability identification based on configurable CSI-RS for CM resources
//((SA networking) Selecting the CSIRS_UE_CAP_R16_SW option if optimized UE capability identification
based on configurable CSI-RS for CM resources is required
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-1;
//((NSA networking) Selecting the CSIRS_UE_CAP_R16_SW option on the NR side and the
R16_CSI_RS_UE_CAPB_QUERY_SW option on the LTE side if optimized UE capability identification based on
configurable CSI-RS for CM resources is required
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-1;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=R16_CSI_RS_UE_CAPB_QUERY_SW-1;

//Enabling UE capability adaptation based on configurable CSI-RS for CM resources
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CONFIG_CAP_ADAPTIVE_SW-1;
//Enabling adaptation of CSI-RS for CM capabilities in codebooks
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CB_CAP_ADAPTIVE_SW-1;
//Enabling SSB beam switching
MOD NRDUCELLCSIRS: NrDuCellId=0, SsbBeamSwitchingMode=MAC_CE_MODE;
//Selecting the CSIRS_QCL_SW option
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_QCL_SW-1;
//Enabling interference measurement resource combination
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=IM_RESOURCE_MERGE_SW-1;
//Enabling CSI-RS for CM reuse 3
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_REUSE3_SW-1;
//Enabling CSI-RS resource allocation optimization
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=CSIRS_ALLOC_OPT_SW-1;
//Enabling UE-specific CSI-RS reconfiguration reduction
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=USER_CSI_RECFG_TIMES_RDC_SW-1;
//Enabling CSI measurement optimization
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=CSI_MEAS_OPT_SW-1;
```

High-frequency TDD command examples

```
//Setting the period of periodic TRS transmission to 20 ms
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsPeriod=MS20;
//Setting the minimum scheduling interval for aperiodic CSI reports to 40 ms
MOD NRDUCELLCSIRS: NrDuCellId=0, AperiodicCsiMinSchIntvl=MS40;

//Optimized UE capability identification based on configurable CSI-RS for CM resources
//((SA networking) Selecting the CSIRS_UE_CAP_R16_SW option if optimized UE capability identification
based on configurable CSI-RS for CM resources is required
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-1;
//((NSA networking) Selecting the CSIRS_UE_CAP_R16_SW option on the NR side and the
R16_CSI_RS_UE_CAPB_QUERY_SW option on the LTE side if optimized UE capability identification based on
configurable CSI-RS for CM resources is required
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-1;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=R16_CSI_RS_UE_CAPB_QUERY_SW-1;

//Setting the CsiRsBeamMeasPeriod parameter to SLOT320 (in eMBB mobility scenarios)
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsBeamMeasPeriod=SLOT320;
//Enabling UE-specific TRS resource management
```

```
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=USER_TRS_BEAM_SW-1;  
//Enabling interference measurement resource combination  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=IM_RESOURCE_MERGE_SW-1;  
//Enabling full-buffer UE-specific CSI-RS for BM  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=USER_BM_CSIRS_SW-1;  
//Enabling the function of ignoring CSI-RS capabilities  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CAPB_IGNORE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples for the sub-functions. You can determine whether to restore the settings of other parameters based on actual network conditions.

Low-frequency TDD command examples

```
//Disabling port number adaptation for CSI-RS for CM (by setting the Csirs4P8PAdaptiveSwThld parameter to 0)  
MOD NRDUCELLCSIRS: NrDuCellId=0, Csirs4P8PAdaptiveSwThld=0;  
//Disabling CSI-RS for CM reuse 3  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_REUSE3_SW-0;  
//Disabling inter-cell CSI-RS for CM interference avoidance (by setting the NCellCsirsIntrfAvoid parameter to NO_AVOID)  
MOD NRDUCELLCSIRS: NrDuCellId=0, NCellCsirsIntrfAvoid=NO_AVOID;  
  
//Full-buffer UE-specific CSI-RS for CM  
//Deselecting the CSIRS_INCOMPATIBLE_UE_SW option  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_INCOMPATIBLE_UE_SW-0;  
  
//Optimized UE capability identification based on configurable CSI-RS for CM resources  
//((SA networking) Deselecting the CSIRS_UE_CAP_R16_SW option if optimized UE capability identification based on configurable CSI-RS for CM resources is not required  
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-0;  
//((NSA networking) Deselecting the CSIRS_UE_CAP_R16_SW option on the NR side and the R16_CSI_RS_UE_CAPB_QUERY_SW option on the LTE side if optimized UE capability identification based on configurable CSI-RS for CM resources is not required  
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-0;  
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=R16_CSI_RS_UE_CAPB_QUERY_SW-0;  
  
//Disabling UE capability adaptation based on configurable CSI-RS for CM resources  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CONFIG_CAP_ADAPTIVE_SW-0;  
//Disabling adaptation of CSI-RS for CM capabilities in codebooks  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CB_CAP_ADAPTIVE_SW-0;  
//Disabling SSB beam switching (by setting the SsbBeamSwitchingMode parameter to NO_SWITCHING)  
MOD NRDUCELLCSIRS: NrDuCellId=0, SsbBeamSwitchingMode=NO_SWITCHING;  
//Deselecting the CSIRS_QCL_SW option  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_QCL_SW-0;  
//Disabling interference measurement resource combination  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=IM_RESOURCE_MERGE_SW-0;  
//Disabling CSI-RS resource allocation optimization  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=CSIRS_ALLOC_OPT_SW-0;  
//Disabling UE-specific CSI-RS reconfiguration reduction  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=USER_CSI_RECFG_TIMES_RDC_SW-0;  
//Disabling CSI measurement optimization  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=CSI_MEAS_OPT_SW-0;
```

High-frequency TDD command examples

```
//Optimized UE capability identification based on configurable CSI-RS for CM resources  
//((SA networking) Deselecting the CSIRS_UE_CAP_R16_SW option if optimized UE capability identification based on configurable CSI-RS for CM resources is not required  
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-0;  
//((NSA networking) Deselecting the CSIRS_UE_CAP_R16_SW option on the NR side and the R16_CSI_RS_UE_CAPB_QUERY_SW option on the LTE side if optimized UE capability identification based on configurable CSI-RS for CM resources is not required  
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CSIRS_UE_CAP_R16_SW-0;  
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=R16_CSI_RS_UE_CAPB_QUERY_SW-0;
```

```
//Disabling UE-specific TRS resource management  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=USER_TRS_BEAM_SW-0;  
//Disabling interference measurement resource combination  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=IM_RESOURCE_MERGE_SW-0;  
//Disabling full-buffer UE-specific CSI-RS for BM  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=USER_BM_CSIRS_SW-0;  
//Disabling the function of ignoring CSI-RS capabilities  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_CAPB_IGNORE_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

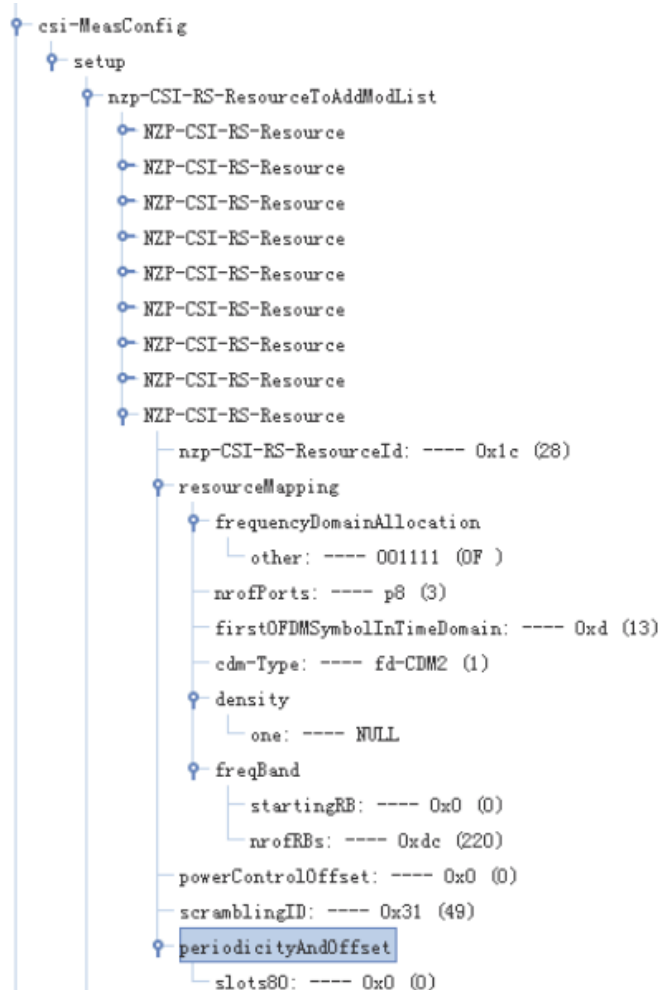
After periodic CSI-RS measurement is enabled, perform the following operations to verify the activation (using SA networking as an example):

- Step 1** Start Uu signaling tracing for cells enabled with periodic CSI-RS measurement as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** In SA networking, check the traced RRCReconfiguration message, as shown in [Figure 7-2](#). If each resource field in the *csi-MeasConfig* IE contains the value of **periodicityAndOffset**, periodic CSI-RS measurement has taken effect.

NOTE

In NSA networking, check the SgNB Addition Request Acknowledge message for the IEs described in SA networking.

Figure 7-2 RRCReconfiguration message tracing report (SA networking)



----End

After UE-specific TRS resource management is enabled, perform the following operations to verify the activation (using NSA networking as an example):

- Step 1** Enable a large-packet UE to access a cell.
- Step 2** Start X2 signaling tracing for cells enabled with UE-specific TRS resource management as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > X2 Interface Trace**.
- Step 3** Check the traced SGNB-MOD-REQUIRED message. If the value of **periodicityAndOffset** in the **TRS Resource** field in the *csi-MeasConfig* IE indicates the period and slot offset of UE-specific TRS resources, UE-specific TRS resources have been allocated to the UE. If the value of **periodicityAndOffset** indicates the period and slot offset of cell-specific TRS resources, cell-specific TRS resources have been allocated to the UE.

NOTE

In SA networking, check the UUAP_RRC_RECFG message for the IEs described in NSA networking.

----End

After full-buffer UE-specific CSI-RS for BM is enabled, perform the following operations to verify the activation (using NSA networking as an example):

- Step 1** Enable a full-buffer UE to access a cell.
- Step 2** Start X2 signaling tracing for cells enabled with full-buffer UE-specific CSI-RS for BM as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > X2 Interface Trace**.
- Step 3** Check the traced SGNB-MOD-REQUIRED message. If the value of *BM CSI-RS resource* in the *csi-MeasConfig* IE does not contain the **periodicityAndOffset** field, full-buffer UE-specific CSI-RS for BM has taken effect. Otherwise, the function has not taken effect.

 NOTE

In SA networking, check the UUAP_RRC_RECFG message for the IEs described in NSA networking.

----End

Observe the following performance indicators to check whether each sub-function of CSI-RS channel management has taken effect.

Table 7-4 Indicators related to the sub-functions involved in CSI-RS resource management

Indicator	Formula
Average uplink cell throughput	Cell Uplink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average uplink MCS index	$(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + 3 \times \text{N.ChMeas.PUSCH.MCS.3} + \dots + 27 \times \text{N.ChMeas.PUSCH.MCS.27} + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \text{N.ChMeas.PUSCH.MCS.3} + \dots + \text{N.ChMeas.PUSCH.MCS.27} + \text{N.ChMeas.PUSCH.MCS.28})$

Indicator	Formula
Average downlink MCS index	$\frac{(0 \times N.ChMeas.PDSCH.MCS.0 + 1 \times N.ChMeas.PDSCH.MCS.1 + 2 \times N.ChMeas.PDSCH.MCS.2 + 3 \times N.ChMeas.PDSCH.MCS.3 + \dots + 27 \times N.ChMeas.PDSCH.MCS.27 + 28 \times N.ChMeas.PDSCH.MCS.28)}{(N.ChMeas.PDSCH.MCS.0 + N.ChMeas.PDSCH.MCS.1 + N.ChMeas.PDSCH.MCS.2 + N.ChMeas.PDSCH.MCS.3 + \dots + N.ChMeas.PDSCH.MCS.27 + N.ChMeas.PDSCH.MCS.28)}$
Downlink IBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink IBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$
Uplink RBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$

7.4.3 Network Monitoring

None

8 SRS Resource Management

8.1 Principles

8.1.1 Overview

SRS refers to uplink sounding signals. In the current version, UEs send SRS within the active BWP. The gNodeB receives and processes SRS to obtain corresponding information such as SINR, RSRP, and PMI. The value of **usage** in SRS resource configurations determines which functions SRS is used for. (In the current version, **usage** can be set to **codebook** or **antennaSwitching** in NR TDD.) [Table 8-1](#) provides more details.

Table 8-1 Functions that SRS is used for

Value of usage	Function	Description
codebook	Uplink single-user multiple-input multiple-output (SU-MIMO)/multi-user multiple-input multiple-output (MU-MIMO)	• UEs perform PMI-based weighting on data to be transmitted. • The gNodeB performs uplink SRS-based link adaptation (LA) and sends the result to UEs to instruct them in data transmission.
	Uplink beam management	The optimal receive beams are selected based on SRS to serve uplink channels.
	Uplink timing	The gNodeB obtains timing alignment information based on SRS measurement and sends the timing advance command (TAC) to UEs.

Value of usage	Function	Description
antenna Switching	Downlink SU-MIMO/MU-MIMO	- In low frequency bands, the gNodeB performs SRS weight-based weighting on data to be transmitted. This is inapplicable to high frequency bands. - The gNodeB performs downlink SRS-based LA and sends data based on the LA result.
	Downlink beam management	In low frequency bands, the optimal transmit beams are selected based on SRS to serve downlink channels. This is inapplicable to high frequency bands.

NOTE

The SRS resources of each UE include multiple SRS sets (with the value of **codebook** or **antennaSwitching**). For more information on the SRS resource parameters included in each SRS set, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 (Release 15).

SRS-related concepts are denoted as follows:

- SRS slot period: indicates the SRS transmission period for a UE. If a UE sends SRS at an interval of X slots (or ms), X is the SRS slot period. The SRS period mentioned afterwards refers to the SRS slot period.
- SRS slot offset: indicates the time-domain position (slot number) for SRS transmission in each period.

For more information on these concepts, see section 6.2.1 "UE sounding procedure" in 3GPP TS 38.214 (Release 15).

SRS is classified into three types: periodic, semi-persistent, and aperiodic. SRS resource configurations are signaled to UEs by the *SRS-Config* IE in an RRCReconfiguration message. After receiving the resource configurations for periodic SRS, UEs periodically send SRS using the indicated time-frequency resources. After receiving the resource configurations for aperiodic SRS, UEs send SRS using the scheduled resources indicated in the DCI.

- Periodic SRS

After receiving resource configurations for periodic SRS, UEs periodically transmit SRS.

The transmission period for periodic SRS can be configured adaptively, adjusted based on UE characteristics, or configured statically.

- SRS period adaptation: If the **SRS_PERIOD_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is selected, the gNodeB adaptively adjusts the SRS period for UEs based on the number of UEs and SRS resource consumption of the given cell.
 - When the cell serves a small number of UEs and has sufficient SRS resources, a shorter SRS period is used for all UEs in the cell,

resulting in higher UE and cell throughputs compared to a fixed SRS period.

- When the cell serves a large number of UEs and has insufficient SRS resources, a longer SRS period is used for all UEs in the cell, resulting in higher cell throughput compared to a fixed SRS period.
- UE-characteristic-based SRS period adaptation: If the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is selected, the gNodeB adaptively adjusts the SRS period for UEs based on the number of UEs and SRS resource consumption of the given cell. This function takes effect only in low-speed low-frequency TDD cells (with **NRDUCell.HighSpeedFlag** set to **LOW_SPEED**), and cannot work with SRS period adaptation.
 - When the cell serves a small number of UEs and has sufficient SRS resources, a shorter SRS period is used for all UEs in the cell, and the SRS period used for large-packet UEs is shorter than that used for small-packet UEs. This allows the cell to serve more UEs for which a short SRS period is adaptively configured compared to SRS period adaptation. Therefore, when the cell serves a small number of UEs, this function further increases UE throughput compared to SRS period adaptation.
 - When the cell serves a large number of UEs and has insufficient SRS resources, a longer SRS period is used for all UEs in the cell, and the SRS period used for large-packet UEs is shorter than that used for small-packet UEs. This allows the cell to serve more UEs for which a short SRS period is adaptively configured compared to SRS period adaptation. Therefore, when the cell serves a large number of UEs, this function further increases UE and cell throughputs compared to SRS period adaptation.

When UE-characteristic-based SRS period adaptation is enabled, the QCI-specific minimum SRS periods for UEs can be specified by the **NRDUCellQciBearer.QciSpecificSrsPeriod** parameter. The SRS period adaptively configured for a UE based on its characteristics must be greater than or equal to the value of this parameter.

When UE-characteristic-based SRS period adaptation is enabled, the scenario-specific SRS resource allocation policy can be specified by the **NRDUCellSrs.ScenarioSrsResAllocPol** parameter. This parameter takes effect only when the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is selected and the **SRS_INTRF_COORD_S_SLOT_SW** or **SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter is selected.

- If this parameter is set to **CAPC_PRI**, SRS resources with the **usage** parameter set to **antennaSwitching** can be configured for a large number of antenna selection UEs, but the SRS period is relatively long.
- If this parameter is set to **PRFM_PRI**, SRS resources with the **usage** parameter set to **antennaSwitching** can be configured for a small number of antenna selection UEs, but the SRS period is relatively short.

- Static SRS period configuration: If the **SRS_PERIOD_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is deselected and the **USER_CHARACTER_SRS_ADAPT_SW** option of the same parameter is deselected, the SRS period can be specified by the **NRDUCellSrs.SrsPeriod** parameter.
- Semi-persistent SRS
After receiving resource configurations for semi-persistent SRS, UEs periodically transmit SRS only when the transmission is activated through the MAC CE.
- Aperiodic SRS
After receiving resource configurations for aperiodic SRS, UEs transmit SRS according to the DCI for uplink or downlink scheduling.

 NOTE

In the current version, the gNodeB only allocates periodic SRS resources for low-frequency NR cells.

In the current version, the gNodeB does not allocate semi-persistent SRS resources.

In the current version, the gNodeB only allocates aperiodic SRS resources for high-frequency NR cells.

SRS Interference Coordination (Low-Frequency TDD)

Downlink SU-MIMO and MU-MIMO scheduling in NR TDD requires favorable SRS SNR. Inter-cell intra-frequency SRS interference decreases the downlink UE rates in the cells. To address this issue, Huawei introduces the SRS interference coordination function. After UE-characteristic-based SRS period adaptation is enabled, SRS interference coordination can reduce intra-frequency SRS interference between cells. You are advised to enable SRS interference coordination for both the serving cell and neighboring cells when there are a large number of stationary UEs and inter-cell uplink SRS interference is strong. After this function takes effect, inter-cell uplink SRS interference can be reduced, thereby increasing the downlink cell throughput and CEU-perceived throughput. If this function is enabled in other scenarios, it will take effect, but may decrease the downlink cell throughput and user-perceived throughput.

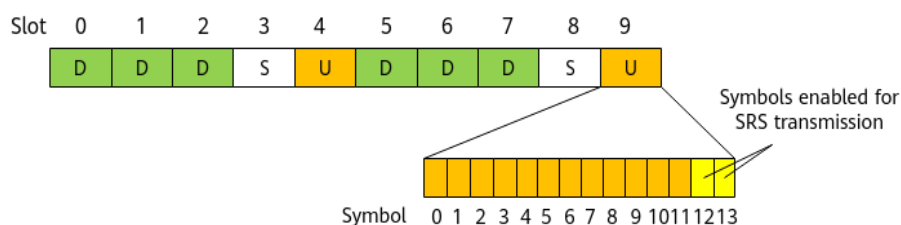
SRS interference coordination can be performed based on self-contained slots or both self-contained and uplink slots. The two sub-functions cannot be enabled together.

- SRS interference coordination based on self-contained slots
In low-frequency TDD, SRS interference coordination based on self-contained slots is controlled by the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. This function can be enabled only when UE-characteristic-based SRS period adaptation is enabled.
 - If this option is deselected, SRS interference coordination based on self-contained slots is disabled.
 - If this option is selected, SRS interference coordination based on self-contained slots is enabled. The gNodeB allocates staggered SRS time-domain resources in self-contained slots among cells in a coordinated manner.
- SRS interference coordination based on self-contained and uplink slots

In low-frequency TDD, SRS interference coordination based on self-contained and uplink slots is controlled by the **SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. This function can be enabled only when UE-characteristic-based SRS period adaptation is enabled.

- If this option is deselected, SRS interference coordination based on self-contained and uplink slots is disabled.
- If this option is selected, SRS interference coordination based on self-contained and uplink slots is enabled. The gNodeB enables some symbols in uplink slots to transmit SRS and adjusts the available time-domain positions for SRS transmission in a cell. **Figure 8-1** shows the enabled symbols with the 4:1 slot configuration. In addition, the gNodeB can allocate staggered SRS time-domain resources in self-contained and uplink slots among cells in a coordinated manner.

Figure 8-1 SRS interference coordination based on self-contained and uplink slots with the 4:1 slot configuration



SRS interference coordination based on self-contained and uplink slots improves downlink performance using uplink resources. However, it allows more UEs in a cell to use staggered time-domain resources than SRS interference coordination based on self-contained slots. As such, you are advised to enable SRS interference coordination based on self-contained and uplink slots when there are a large number of downlink services and a small number of uplink services (for example, the downlink PRB usage is greater than the uplink PRB usage) and the UBBPg provides contiguous coverage for hotspot areas. In other scenarios, you are advised to enable SRS interference coordination based on self-contained slots.

In addition to the preceding interference coordination functions, adaptive SRS interference coordination is also supported.

In low-frequency TDD, adaptive SRS interference coordination is controlled by the **ADAPT_SRS_INTRF_COORD_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. This function can be enabled only when SRS interference coordination based on self-contained slots is enabled.

- If this option is selected, adaptive SRS interference coordination is enabled. The gNodeB adaptively adjusts the SRS resource allocation mode of the cell based on the total amount of available SRS resources and SRS resources consumed by UEs in the cell.
- If this option is deselected, adaptive SRS interference coordination is disabled.

SRS Resource Set Adjustment (Low-Frequency TDD)

In low-frequency TDD, SRS resource set adjustment is supported when the transmission period for periodic SRS is adjusted based on UE characteristics. This function is controlled by the **SRS_RES_SET_ADJ_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter.

- If this option is deselected, the gNodeB does not adjust the SRS resource sets configured for UEs.
- If this option is selected, the cell serves a large number of RRC_CONNECTED UEs, and there is a large number of downlink paired layers, then the gNodeB identifies UEs for which a large number of SRS resource sets are configured in the cell and reduces the number of SRS resource sets configured for them, thereby increasing the downlink cell throughput.

UE Number Increase for SRS (Low-Frequency TDD)

UE number increase for SRS is supported when one of the following is true for the transmission period for periodic SRS: It is statically configured in high-speed low-frequency TDD cells. It is adjusted based on UE characteristics in low-speed low-frequency TDD cells. It is adaptively configured in SUL cells. This function is controlled by the **SPEC_INCREASE_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter.

- If this option is deselected, UE number increase for SRS is disabled.
- If this option is selected, UE number increase for SRS is enabled. The gNodeB uses fewer ports to receive SRS, allowing more RRC_CONNECTED UEs to be allocated SRS resources, and increasing the maximum number of RRC_CONNECTED UEs supported in a cell.

Coherent-Codebook-based Scheduling (Low-Frequency TDD)

In low-frequency TDD, coherent-codebook-based scheduling is supported and is controlled by the **COHERENT_CODEBOOK_SCH_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. When two SRS ports are used, a UE supports coherent-codebook-based scheduling if the value of *pusch-TransCoherence* in the *MIMO-ParametersPerBand* IE reported by the UE is **fullCoherent**. If this option is selected and the maximum number of uplink layers (carried in the *MIMO-LayersUL* IE) reported by a UE that supports coherent-codebook-based scheduling is one, the base station configures 2-port codebook-dedicated SRS resources for the UE. For more information, see descriptions of the *pusch-TransCoherence* and *MIMO-LayersUL* IEs in section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V17.4.0.

SRS Resource Adjustment for 4T UEs

In low frequency bands, SRS resource adjustment for 4T UEs is supported to adjust the SRS resource allocation mode for them. This function is controlled by the **UL_4T_SRS_RES_ADJ_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter, which is deselected by default.

- If this option is selected, the SRS period is shortened for 4T UEs that report the *supportedSRS-TxPortSwitch* IE with a value of **t4r4** or **t4r8**, and prolonged for other 4T UEs.
- If this option is deselected, the SRS resource allocation mode is not adjusted for 4T UEs.

8.1.2 Cell-specific SRS

A cell-specific SRS slot includes the time-domain resources for all UEs in a cell to transmit SRS. In NR, the gNodeB obtains the number and positions of slots

available for SRS transmissions in a radio frame and the number and positions of symbols for SRS in the SRS slots based on the duplex mode and slot configuration.

 NOTE

In NR TDD, SRS is configured only in the self-contained slots in a radio frame. For more information on the principles and settings of slot configuration, see *Standards Compliance*.

SUL Cell-specific SRS

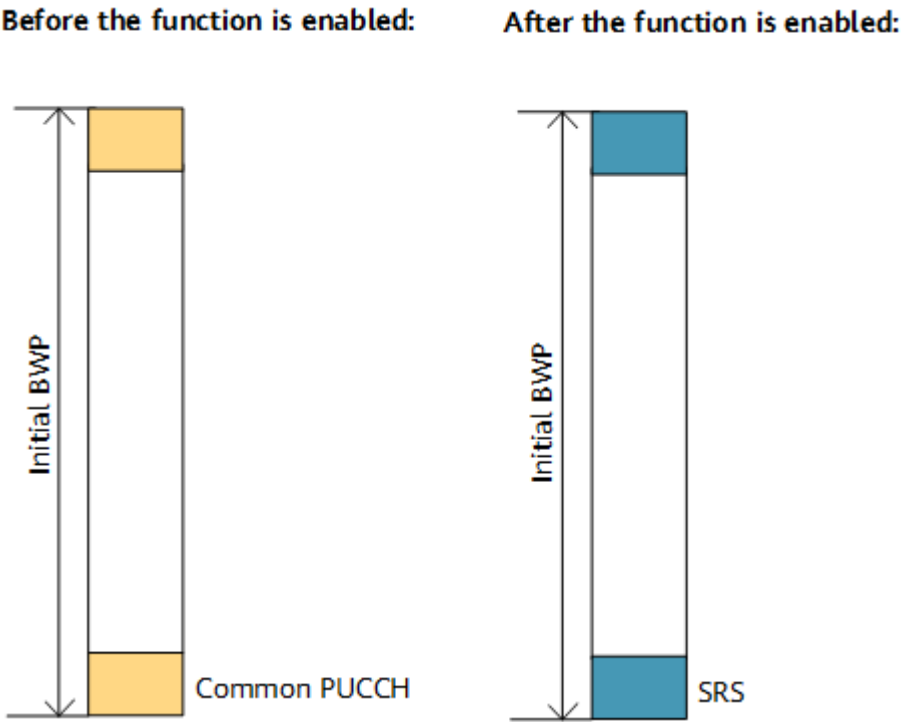
For an SUL cell, SRS time-domain resources are determined based on the **LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudSwitch** parameter. For details about this option, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

- If both options are deselected, the gNodeB can configure slot 5 in a radio frame for UEs to transmit SRS. SRS is transmitted in symbol 13 of this slot.
- If either option is selected, the gNodeB adaptively selects a slot in a radio frame for UEs to transmit SRS based on the dynamic negotiation results between NR and LTE. SRS is transmitted in symbol 13 of the selected slot.

SRS Transmission Using Common PUCCH Resources

In an SUL cell, SRS transmission using common PUCCH resources can be enabled by selecting the **COMMON_PUCCH_AVOID_SRS_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter. After this function is enabled, common PUCCH transmission will not occupy the symbols of SRS slots (symbol 13 of slots 0 and 5), and SRS transmission will preempt the frequency-domain resources for common PUCCH, as shown in [Figure 8-2](#). The common PUCCH is indicated by `pucch-ResourceCommon`. For details, see section 9.2.1 "PUCCH Resource Sets" in 3GPP TS 38.213 V15.5.0. You are advised to enable SRS transmission using common PUCCH resources when UE bandwidth adaptation (controlled by the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp.BwpConfigSwitch** parameter) is enabled. This function applies only to SUL cells served by macro base stations, but not `LampSite` base stations.

Figure 8-2 SRS transmission using common PUCCH resources



8.1.3 UE-specific SRS

8.1.3.1 UE-specific SRS Slot

A UE sends SRS in a UE-specific SRS slot. SRS slots are configured only for periodic SRS and semi-persistent SRS, not for aperiodic SRS. The slots for aperiodic SRS are determined during scheduling.

8.1.3.2 UE-specific SRS Bandwidth

NR allows UE-specific SRS bandwidths to be configured. A UE sends SRS using a UE-specific SRS bandwidth. 3GPP specifications define a maximum of four types of SRS bandwidths ($B_{SRS} = 0, 1, 2, \text{ or } 3$ in the SRS bandwidth configurations). [Table 8-2](#) describes a UE-specific SRS bandwidth configuration example. For more information, see section 6.4.1.4 "Sounding reference signal" in 3GPP TS 38.211 V15.7.0.

Table 8-2 UE-specific SRS bandwidth configurations

C_{SRS}	$B_{SRS} = 0$		$B_{SRS} = 1$		$B_{SRS} = 2$		$B_{SRS} = 3$	
	$m_{SRS,0}$	N_0	$m_{SRS,1}$	N_1	$m_{SRS,2}$	N_2	$m_{SRS,3}$	N_3
0	4	1	4	1	4	1	4	1
...	...	...	...	...	...	...	...	...

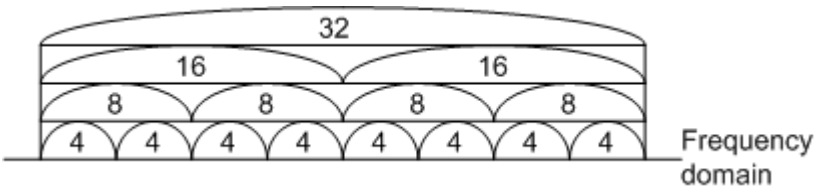
9	32	1	16	2	8	2	4	2
...	...	...	...	...	...	...	...	...
63	272	1	16	17	8	2	4	2

 NOTE

- C_{SRS} : indicates the SRS bandwidth index.
- B_{SRS} : indicates the bandwidth tree level.
- $m_{SRS,B}$: indicates the SRS bandwidth on a B_{SRS} level.
- N_B : indicates the number of leaf nodes on a B_{SRS} level.
 - When B_{SRS} is not 0, N_B is calculated as follows: SRS bandwidth corresponding to B_{SRS} minus 1/SRS bandwidth corresponding to B_{SRS} .
 - When B_{SRS} is 0, N_B is 1.

Figure 8-3 shows an SRS bandwidth configuration tree corresponding to C_{SRS} of 9, where a UE-specific SRS bandwidth can be 32, 16, 8, or 4 RBs.

Figure 8-3 SRS bandwidth configuration tree



For aperiodic SRS, the system determines C_{SRS} and B_{SRS} for SRS bandwidths based on the BWP.

The system separately determines C_{SRS} and B_{SRS} for broadband bandwidths and narrowband bandwidths of periodic SRS based on the BWP. To improve SRS measurement accuracy and coverage capability, SRS bandwidths for UEs are adaptively adjusted based on the channel quality.

8.1.4 SRS Multiplexing

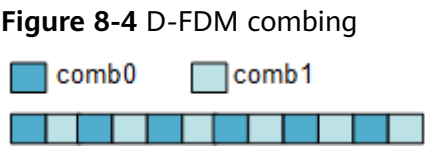
An SRS multiplexing mode determines how different UEs transmit SRS using time-frequency resources. In NR, frequency division multiplexing (FDM) and code division multiplexing (CDM) are applied.

FDM

FDM allows different UEs to transmit SRS using different frequencies. There are two types of FDM: localized-FDM (L-FDM) and distributed-FDM (D-FDM).

- L-FDM
Different UEs use different continuous frequency-domain resources to send SRS.

- **D-FDM**
Different UEs alternately send SRS over subcarriers. In the current version, the number of supported combs is 2, indicating that a bandwidth is divided into even and odd subcarriers. For example, one UE sends SRS on odd subcarriers, and another sends on even subcarriers, staggering SRS transmissions in the frequency domain. Also, the SRS transmission period for a single UE is combed in the frequency domain, also known as D-FDM combing. **Figure 8-4** shows D-FDM combing. **comb0** indicates the subcarriers used to transmit SRS of one UE, and **comb1** indicates the subcarriers used to transmit SRS of another UE.



CDM

CDM allows SRS with the same symbol position, bandwidth, and comb (for example, the SRS corresponding to two ports on a 2T4R UE) to occupy the same time-frequency position. Codewords use cyclic shift sequences generated from the same ZC sequence. In the current version, CDM allows for a maximum of four cyclic shifts.

8.1.5 SRS Frequency Hopping

In low frequency bands, SRS frequency hopping enables a UE to transmit SRS over a small bandwidth in one symbol and over another bandwidth in the subsequent symbol, thereby achieving channel sounding across the entire system bandwidth. Frequency hopping offers channel quality measurement across the entire bandwidth using a small sounding bandwidth. Only inter-slot frequency hopping for periodic SRS is supported in this version.

8.2 Network Analysis

8.2.1 Benefits

SRS resource management improves network performance by properly utilizing SRS resources.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

Aperiodic SRS does not involve network impacts as it currently does not require parameter configuration.

[Table 8-3](#) describes the network impacts of each sub-function for periodic SRS. For the definitions of related indicators, see [8.4.2 Activation Verification](#).

Table 8-3 Network impacts of the sub-functions involved in SRS resource management

Sub-Function Name	Network Impacts
SRS period adaptation	Enabling this function (by selecting the SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) causes the cell to restart and all UEs served by the cell to re-access the network. After the cell is restarted, a long initial SRS period is configured for UEs in the cell to ensure that they are all allocated SRS resources. If the number of UEs changes significantly after this function is enabled, the system adaptively changes the SRS period for UEs, leading to frequent RRC reconfigurations. To prevent this from happening, the hysteresis for shortening the cell-level SRS period is always 10 minutes. Moreover, enabling this function also affects the number of RRC reconfigurations. Specifically, the number of RRC reconfigurations for triggering SRS bandwidth adaptation decreases if the wideband SRS bandwidth is insufficient; the number of RRC reconfigurations for triggering SRS resource allocation for a longer period increases if SRS resources are insufficient. After this function is enabled, its scenario-specific impacts on performance are as follows: • When SRS resources are sufficient, the gNodeB individually configures shorter SRS periods for newly admitted UEs in a cell. This improves user experience for these UEs, but decreases the maximum number of UEs supported in the cell. In the case of small inter-site distances or strong inter-cell interference, for example, for CEUs, a short SRS period increases SRS interference among UEs served by different cells. This decreases the average downlink and uplink cell throughputs and the average downlink and uplink UE throughputs. (A UE is considered a CEU if the difference between the measured SRS RSRP of its serving cell and that of a neighboring cell is 3 dB.) • When SRS resources are slightly insufficient, the gNodeB prolongs SRS periods for newly admitted UEs in a cell when allocating SRS resources. This allows more UEs to access the cell but may cause user experience to deteriorate. • When SRS resources are severely or extremely insufficient, the gNodeB individually configures longer SRS periods for newly admitted UEs in a cell. This further increases the number of UEs that can access the cell and achieves a tradeoff between cell capacity and performance. However, for UEs moving at a speed of approximately 30 km/h, a long SRS period decreases the average downlink UE throughput.

Sub-Function Name	Network Impacts
UE-characteristic-based SRS period adaptation	In low-frequency TDD, enabling this function (by selecting the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) causes the cell to restart and all UEs served by the cell to re-access the network. After the cell is restarted, a long initial SRS period is configured for UEs in the cell to ensure that they are all allocated SRS resources. If the number of UEs changes significantly after this function is enabled, the system adaptively changes the SRS period for UEs, leading to frequent RRC reconfigurations. To prevent this from happening, the hysteresis for shortening the cell-level SRS period is always 10 minutes. After this function is enabled, it will affect the number of RRC reconfigurations and the average CQI compared to static SRS period configuration as follows: • For a cell serving a majority of small-packet UEs, the number of RRC reconfigurations decreases. This is because UE-specific SRS bandwidth adaptation (leading to RRC reconfiguration) does not take effect for small-packet UEs after UE-characteristic-based SRS period adaptation is enabled, even though RRC reconfiguration is triggered in the case of a service change. • For a cell serving a majority of large-packet UEs, the number of RRC reconfigurations increases. This is because RRC reconfiguration is triggered in the case of a service change after UE-characteristic-based SRS period adaptation is enabled. In addition, if the average RI measured and reported by UEs increases or remains unchanged, the average CQI may decrease. If the average RI measured and reported by UEs decreases, the average CQI may increase. After this function is enabled, it will affect the uplink IBLER and uplink PRB usage of a cell compared to static SRS period configuration as follows: • If a UE is performing uplink large-packet services, it will be configured with a long SRS period. As a result, the uplink IBLER and uplink PRB usage of the cell may increase. • If a UE is performing downlink large-packet services, there is no impact. This function results in more RRC reconfigurations than SRS period adaptation because RRC reconfiguration is triggered in the case of a service change. After this function is enabled, its scenario-specific impacts on performance are as follows: If all UEs in a cell are small-packet UEs, the number of UEs supported by the cell increases and SRS interference among UEs served by different cells decreases. However, the average downlink and uplink cell throughputs and the average downlink and uplink UE throughputs decrease.

Sub-Function Name	Network Impacts
	If all UEs in a cell are large-packet UEs or there are both small-packet and large-packet UEs in a cell, the accuracy of channel measurement results of large-packet UEs increases. This ensures the performance of features that depend on SRS measurement results. However, both SRS interference among UEs served by different cells and SRS resource consumption increase. • If SRS interference exists among UEs served by different cells, the average downlink and uplink cell throughputs and the average downlink and uplink UE throughputs decrease. • If there is no SRS interference among UEs served by different cells, the average number of downlink MU-MIMO layers and the average downlink rank may decrease in the cell, whereas the average downlink and uplink cell throughputs increase. Furthermore, the average downlink and uplink UE throughputs are affected as follows: - When the traffic volume proportion of downlink small-packet services in the cell exceeds 90%, the average downlink and uplink UE throughputs may decrease. - When the traffic volume proportion of downlink small-packet services in the cell is less than or equal to 90%, the average downlink and uplink UE throughputs may increase. If the downlink cell performance deteriorates due to excessively high beamforming gains provided by SRS-based weights, the average downlink cell throughput and average downlink UE throughput may decrease when the weights in use change from PMI-based to SRS-based. After this function is enabled in a heavy-load cell (where there are a large number of UEs and the PRB usage is high), the average downlink MCS index in the cell will change, affecting the average uplink CCE aggregation level and downlink BLER of the cell as follows: • A larger average downlink MCS index causes a lower average uplink CCE aggregation level and a possible increase in the downlink BLER. • A smaller average downlink MCS index causes a higher average uplink CCE aggregation level and a possible decrease in the downlink BLER. After this function is enabled in a light-load cell (where there are a small number of UEs and the PRB usage is low), the average uplink CCE aggregation level in the cell may decrease. Configuring the minimum SRS period for UE-characteristic-based SRS period adaptation using the NRDUCellQciBearer.QciSpecificSrsPeriod parameter results in the following scenario-specific impacts on network performance:

Sub-Function Name	Network Impacts
	• For FWA or static eMBB UEs: – A smaller minimum SRS period results in lower average downlink cell throughput and lower average downlink UE throughput. – A larger minimum SRS period results in higher average downlink cell throughput and higher average downlink UE throughput. • For moving eMBB UEs: – A smaller minimum SRS period results in higher average downlink cell throughput and higher average downlink UE throughput. – A larger minimum SRS period results in lower average downlink cell throughput and lower average downlink UE throughput.
Static SRS period configuration	If the gNodeB always uses the SRS period specified by the NRDUCellSrs.SrsPeriod parameter: • A smaller value of this parameter increases the accuracy of SRS-related channel measurement and guarantees the performance of features that depend on SRS measurement. However, it also results in more SRS resource consumption and fewer RRC_CONNECTED UEs that can be allocated SRS resources. • A larger value of this parameter results in less SRS resource consumption and more RRC_CONNECTED UEs that can be allocated SRS resources. However, it also decreases the accuracy of SRS-related channel measurement and negatively affects the performance of features that depend on SRS measurement. If a fixed SRS period is used, the number of RRC_CONNECTED UEs that can be allocated SRS resources and cell performance may be difficult to balance. Therefore, you are advised to enable UE-characteristic-based SRS period adaptation by selecting the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter or SRS period adaptation by selecting the SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter.

Sub-Function Name	Network Impacts
SRS interference coordination (including SRS interference coordination based on self-contained slots and SRS interference coordination based on self-contained and uplink slots)	In low-frequency TDD, enabling either of the sub-functions of SRS interference coordination can reduce inter-cell uplink SRS interference, thereby increasing the CEU throughput and average downlink cell throughput. You are advised to enable either of the sub-functions for both the serving cell and neighboring cells when there are a large number of stationary UEs and inter-cell uplink SRS interference is strong, thereby ensuring the gains mentioned earlier. If either of them is enabled in other scenarios, it will take effect, but may decrease the user-perceived throughput and average downlink cell throughput. In addition, when SRS interference coordination based on self-contained and uplink slots is enabled, the average number of available PUSCH symbols in the cell decreases; the average uplink cell throughput and average uplink UE throughput may decrease; the uplink PRB usage of the cell may increase. As such, you are advised to enable SRS interference coordination based on self-contained and uplink slots when there are a large number of downlink services and a small number of uplink services (for example, the downlink PRB usage is greater than the uplink PRB usage) and contiguous coverage is provided in hotspot areas.
Adaptive SRS interference coordination	After this function is enabled in a low-frequency TDD cell where there are a small number of UEs, most UEs are stationary, and SRS interference is strong, the average downlink cell throughput increases and the service drop rate decreases.
SRS resource set adjustment	After this function is enabled in a low-frequency TDD cell with a consistently large number of RRC_CONNECTED UEs and a consistently large number of downlink paired layers, the average downlink cell throughput increases. If the number of RRC_CONNECTED UEs or the number of downlink paired layers suddenly decreases or significantly fluctuates in the cell with this function enabled, the average downlink cell throughput may decrease.
UE number increase for SRS	After this function is enabled, its scenario-specific impacts are as follows: - When a cell is under light or medium load and there are a small number of UEs accessing the cell per second, the average downlink and uplink cell throughputs decrease because the number of ports where UEs are allocated SRS resources is reduced. - When a cell is under heavy load or there are a large number of UEs accessing the cell per second, the service drop rate decreases and the average downlink and uplink cell throughputs increase because more UEs can be allocated SRS resources in the cell.

Sub-Function Name	Network Impacts
Coherent-codebook-based scheduling	In low-frequency TDD, after this function takes effect, the average uplink throughput of CEUs increases and the average downlink UE throughput may fluctuate.
SRS resource adjustment for 4T UEs	After this function takes effect, the SRS period for 4T UEs changes, causing fluctuations in the following indicators, which depend on the number of 4T4R and 4T8R UEs: • Average SRS interference • Average uplink and downlink UE throughputs • Uplink and downlink MCS indexes • Uplink and downlink IBLERs • Uplink and downlink RBLERs
SRS transmission using common PUCCH resources	After this function is enabled in an SUL cell, common PUCCH and SRS will not interfere with each other. SRS can be configured on the full band, increasing the average uplink cell throughput and average uplink UE throughput, but prolonging the access delay on average by 0.2 slots. A maximum of 600 UEs can access a 5 MHz cell.

8.3 Requirements

8.3.1 Licenses

There are no license requirements for basic functions.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	UE-characteristic-based SRS period adaptation must be enabled before the minimum SRS period for UE-characteristic-based SRS period adaptation (specified by the NRDUCellQciBearer.QciSpecificSrsPeriod parameter) is configured. In addition, if the NRDUCellQciBearer.QciSpecificSrsPeriod parameter is set to SL160 or SL320, the SRS_INTRF_COORD_S_SLOT_SW or SRS_INTRF_COORD_S_USLOT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter must be selected and the NRDUCellSrs.ScenarioSrsResAllocPol parameter must be set to CAPC_PRI.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	UE-characteristic-based SRS period adaptation must be enabled before SRS resource set adjustment (controlled by the SRS_RES_SET_ADJ_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) is enabled.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	User-characteristic-based SRS period adaptation must be enabled before SRS interference coordination based on self-contained slots (controlled by the SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	User-characteristic-based SRS period adaptation must be enabled before SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_UL_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	For low-speed low-frequency TDD cells (for which NRDUCell.HighSpeedFlag is set to LOW_SPEED), UE-characteristic-based SRS period adaptation must be enabled before UE number increase for SRS (controlled by the SPEC_INCREASE_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled. This requirement does not apply to high-speed low-frequency TDD cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Scheduling optimization for HARQ-ACK resource set 0	HARQ_ACK_RES_SET0_SCH_OPT_SW option of the NRDUCellPucchPerformanceSw parameter	<i>Channel Management</i>	After SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_UL_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the increase in the downlink spectral efficiency may be insignificant due to insufficient uplink scheduling resources. Therefore, scheduling optimization for HARQ-ACK resource set 0 must be enabled before SRS interference coordination based on self-contained and uplink slots is enabled. After scheduling optimization for HARQ-ACK resource set 0 is enabled, the gNodeB can schedule more resources in HARQ-ACK resource set 0, removing limitations to PUCCH format-1 HARQ-ACK resource allocation in some scenarios.
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Downlink MU-MIMO must be enabled before SRS resource set adjustment (controlled by the SRS_RES_SET_ADJ_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS Short Resource Allocate Policy	NRDUCellSrs.SrsShortResAllocPol	None	SRS period adaptation must be enabled in a cell with the NRDUCell.UlBandwidth parameter set to CELL_BW_90M or the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter selected before the NRDUCellSrs.SrsShortResAllocPol parameter is set to a value other than OFF . Therefore, the NRDUCellSrs.SrsShortResAllocPol parameter must be set to OFF before SRS period adaptation is disabled.
High-frequency TDD	None	None	None	None

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	SRS period adaptation (controlled by the SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_US LOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_US LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with positioning-dedicated SRS transmission in uplink slots.
Low-frequency TDD	U-TDOA- and AoA-based high-precision positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_US LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with U-TDOA- and AoA-based high-precision positioning.
Low-frequency TDD	SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	SRS period adaptation (controlled by the SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) cannot work with SRS field strength-based positioning.
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with SRS period adaptation (controlled by the SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter).
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with coherent-codebook-based scheduling (controlled by the COHERENT_CODEBOOK_SCH_SW option of the NRDUCellChnCoverageAlgorithm.UlCoverageAlgorithmSwitch parameter).
Low-frequency TDD	SRS period adaptation	SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter	<i>Channel Management</i>	SRS period adaptation cannot work with UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter).
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgorithmExtSwitch parameter	<i>Channel Management</i>	When SRS interference coordination based on self-contained slots (controlled by the SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgorithmExtSwitch parameter) is enabled, SRS interference coordination based on self-contained and uplink slots cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	SRS interference coordination based on self-contained slots must be enabled before adaptive SRS interference coordination (controlled by the ADAPT_SRS_INTRF_COORD_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled.
Low-frequency TDD	Configuration of the maximum port number used for CSI-RS for CM	NRDUCellCsirs.FR1MaxCellCsirsPortNum	<i>Channel Management</i>	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter (specifying the maximum number of ports used to transmit CSI-RS for CM) cannot be set to 16PORT .
Low-frequency TDD	Cell-level PUCCH configuration	NRDUCellPucch.StructureType	<i>Channel Management</i>	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellPucch.StructureType parameter must be set to LONG_STRUCTURE .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3CC aggregation	NRDUCellCarrMgmt.CaDIMaxCcNum set to DL3CC	<i>CA Experience Improvement</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_S LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with 3CC aggregation (with NRDUCellCarrMgmt.CaDIMaxCcNum set to DL3CC).
Low-frequency TDD	Massive CA	NRDUCellCarrMgmt.CaDIMaxCcNum set to DL4CC or DL5CC	<i>CA Experience Improvement</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_S LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with massive CA.
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_S LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) can be enabled only when SRS carrier switching is disabled.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_S LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot work in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot work with Cell Combination.
Low-frequency TDD	Configuration of NUL 1TX-to-SUL 1TX switching time	NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime	<i>UL and DL Decoupling</i>	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime parameter cannot be set to MICROSECOND140 .
Low-frequency TDD	LTE FDD and NR Uplink Spectrum Sharing	LTE_NR_UL_SPECTRUM_SHARING_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>	In SUL cells, LTE FDD and NR Uplink Spectrum Sharing cannot work with SRS transmission using common PUCCH resources (controlled by the COMMON_PUCCH_AVOID_SRS_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter).
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot work with proactive avoidance of remote interference.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecodingMimoSrsPeriodShortSw parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter) cannot work with SRS period shortening for downlink 1T4R UEs.
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecodingMimoSrsPeriodShortSw parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter) cannot work with SRS period shortening for downlink 2T4R UEs.
Low-frequency TDD	SRS period shortening for downlink 1T2R UEs	DL_1T2R_SRS_PRD_SHORTEN_SW option of the NRDUCellPdschPrecodingMimoSrsPeriodShortSw parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter) cannot work with SRS period shortening for downlink 1T2R UEs.
Low-frequency TDD	PUSCH scheduling in self-contained slots	PUSCH_SCH_IN_S_SLOT_SW option of the NRDUCellPdschULPuschAlgorithmSwitch parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter) cannot work with PUSCH scheduling in self-contained slots.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Policy for PUCCH resource sharing with the PUSCH	NRDUCellPusch.PuschSharePucchResPol	None	When SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) is enabled, the NRDUCellPusch.PuschSharePucchResPol parameter can only be set to NOT_SHARED .
Low-frequency TDD	SRS resource allocation for UEs performing downlink large-packet services	DL_BIG_PKT_SRS_RES_ALLOC_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot work with SRS resource allocation for UEs performing downlink large-packet services.
Low-frequency TDD	Uplink data convergence in symbols	UL_SYMBOL_CONVER_SW option of the NRDUCellServExp.PowerSavingOptSwitch parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot work with uplink data convergence in symbols.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain reservation mode	NRDUCellTimeIntrf.TimeDomainRsvMode	None	When the NRDUCellTimeIntrf.TimeDomainRsvMode parameter is set to TIME_DOMAIN_RESERVED_MODE_2 , SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled.
Low-frequency TDD	Common PUCCH RB resource optimization	COMM_PUCCH_RB_RES_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with common PUCCH RB resource optimization.
Low-frequency TDD	Fast response to fallback DCI	FALLBACK_DCI_FAST_RSP_SW option of the NRDUCellAlgoSwitch.CpLatencySwitch parameter	None	SRS interference coordination based on self-contained and uplink slots (controlled by the SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) cannot be enabled together with fast response to fallback DCI.
Low-frequency TDD	Antenna selection SRS impact on PUSCH optimization	AS_SRS_IMPACT_ON_PUSCH_OPT_SW option of the NRDUCellUIMc.UlAmcPerformanceSwitch parameter	None	UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) cannot be enabled together with antenna selection SRS impact on PUSCH optimization.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	None	None	None	None

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	In high-speed scenarios, UEs move rapidly. A long SRS period adversely affects UE performance.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	DRX	BASIC_DRX_S W option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	For low-frequency cells: - Periodic SRS: The gNodeB does not receive or measure SRS from UEs in DRX sleep time. Therefore, the quantity of valid SRS received by the gNodeB decreases. - Aperiodic SRS: The gNodeB does not schedule SRS from UEs in DRX sleep time. Therefore, the quantity of valid SRS received by the gNodeB decreases. - If UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) is enabled, the DRX sleep time will be shortened to improve the SRS-based weight performance. For high-frequency cells: Aperiodic SRS: The base station performs scheduling only within the length of the Inactivity Timer triggered by common data transmission or the On Duration Timer. Therefore, the quantity of valid SRS received by the gNodeB decreases and the measurement interval increases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Cell Combination</i>	When a cell (such as a common cell, a hyper cell, or a combined cell) is set up on a base station, the SRS period adaptation scheme for all cells served by the base station must be the same (either SRS period adaptation or UE-characteristic-based SRS period adaptation), and a combination of the adaptation scheme and static SRS period configuration is supported. Specifically, When there are no other activated cells on the base station, the new cell can use static SRS period configuration, SRS period adaptation, or UE-characteristic-based SRS period adaptation. When there are activated cells using SRS period adaptation on the base station, UE-characteristic-based SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. In this case, in order to enable UE-characteristic-based SRS period adaptation for the new cell, you must deactivate all activated cells using SRS period adaptation on the base station. Then, enable UE-characteristic-based SRS period adaptation or static SRS period configuration for the deactivated cells, and reactivate these cells. Lastly, enable UE-characteristic-based SRS period adaptation for the new cell.

RAT	Function Name	Function Switch	Reference	Description
				When there are activated cells using UE-characteristic-based SRS period adaptation on the base station, SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. In this case, in order to enable SRS period adaptation for the new cell, you must deactivate all activated cells using UE-characteristic-based SRS period adaptation on the base station. Then, enable SRS period adaptation or static SRS period configuration for the deactivated cells, and reactivate these cells. Lastly, enable SRS period adaptation for the new cell.
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When downlink intra-band CA is enabled and the SRS period for an antenna-selection UE in a PCell (which works on the primary component carrier) is 5 ms or 10 ms, SRS resources in an SCell (which works on the secondary component carrier) may fail to be allocated, affecting user experience.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When downlink intra-FR inter-band CA is enabled and the SRS period for an antenna-selection UE in a PCell (which works on the primary component carrier) is 5 ms or 10 ms, SRS resources in an SCell (which works on the secondary component carrier) may fail to be allocated, affecting user experience.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	After UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter) is enabled for a cell, UEs in the cell can be allocated SRS resources at a shorter interval, increasing the gains of SRS carrier switching.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When SRS interference coordination based on self-contained and uplink slots is enabled together with intra-base-station UL CoMP: • If the UBBPfw is used, only SRS interference coordination based on self-contained and uplink slots takes effect, and intra-base-station UL CoMP does not. • If the UBBPg is used, both intra-base-station UL CoMP and SRS interference coordination based on self-contained and uplink slots can take effect, but the number of UEs for which intra-base-station UL CoMP takes effect may decrease in the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Device-pipe identification	- SA networking: SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter - NSA networking: SpecUeIdentifySwitch option of the UeCooperationPara.SpecUserCooperationSwitch parameter on the eNodeB side (Device-pipe identification is generally controlled by the eNodeB.)	<i>Device-Pipe Synergy</i>	When device-pipe identification is enabled, the gNodeB can identify UE types on the network. In this case, after SRS resource set adjustment (controlled by the SRS_RES_SET_ADJ_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) takes effect, the impacts on the antenna selection capability of UEs for which SRS resource sets have been adjusted are as follows: - There is no impact on Huawei UEs supporting antenna selection. - Non-Huawei UEs supporting antenna selection no longer support antenna selection. When device-pipe identification is disabled, the gNodeB cannot identify UE types on the network. In this case, after SRS resource set adjustment (controlled by the SRS_RES_SET_ADJ_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) takes effect, antenna-selection UEs for which SRS resource sets have been adjusted no longer support antenna selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Full-power dual transmission	UL_RANK1_FULL_POWER_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	For a UE that supports only FullPowerMode1 of full-power dual transmission and supports coherent-codebook-based scheduling, when coherent-codebook-based scheduling (controlled by the COHERENT_CODEBOOK_SCH_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter) is enabled, only coherent-codebook-based scheduling takes effect for the UE.
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode	<i>MIMO (TDD)</i>	It is recommended that MetaAAU beam sensing be enabled together with UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellsRS.SrsAlgoSwitch parameter).
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDirRankSelAlgoSw parameter	<i>FWA</i>	When downlink 8-layer transmission for a single FWA user takes effect, SRS resource set adjustment (controlled by the SRS_RES_SET_ADJ_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SuMimoSrsPeriodShortSw parameter	None	For a cell serving a large number of UEs, SRS period shortening for downlink 1T4R UEs can be enabled but may not take effect. In this case, enabling UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter) can increase the probability that SRS period shortening for downlink 1T4R UEs takes effect. However, the service drop rate in the cell may increase if some UEs have compatibility issues after SRS period shortening for downlink 1T4R UEs takes effect.
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SuMimoSrsPeriodShortSw parameter	None	For a cell serving a large number of UEs, SRS period shortening for downlink 2T4R UEs can be enabled but may not take effect. In this case, enabling UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter) can increase the probability that SRS period shortening for downlink 2T4R UEs takes effect. However, the service drop rate in the cell may increase if some UEs have compatibility issues after SRS period shortening for downlink 2T4R UEs takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T2R UEs	DL_1T2R_SRS_PRD_SHORTEN_SW option of the NRDUCellPdschPrecodingMimoSrsPeriodShortSw parameter	None	For a cell serving a large number of UEs, SRS period shortening for downlink 1T2R UEs can be enabled but may not take effect. In this case, enabling UE-characteristic-based SRS period adaptation (controlled by the USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrsSrsAlgorithm parameter) can increase the probability that SRS period shortening for downlink 1T2R UEs takes effect. However, the service drop rate in the cell may increase if some UEs have compatibility issues after SRS period shortening for downlink 1T2R UEs takes effect.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

 NOTE

3900 series base stations do not support high-frequency TDD functions.

Boards

SRS interference coordination based on self-contained and uplink slots: All NR-capable main control boards as well as the UBBPfw and UBBPg series and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Other sub-functions of SRS resource management have the following board requirements:

Except for the RAT, the sub-functions of this function have no other requirements on main control boards and baseband processing units. All main control boards and baseband processing units of the corresponding RAT support these sub-functions. For details about the RATs supported by different main control boards and baseband processing units, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

SRS interference coordination based on self-contained and uplink slots: All NR TDD-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Other sub-functions of SRS resource management have the following RF module requirements:

Except for the RAT, the sub-functions of this function have no other requirements on RF modules. All RF modules of the corresponding RAT support these sub-functions. For details about the RATs supported by different RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Cells

In low-frequency TDD, cells established on the same base station must use the same SRS period adaptation scheme (either SRS period adaptation or UE-characteristic-based SRS period adaptation), and they support a combination of the adaptation scheme and the static configuration. Specifically,

- To specify an SRS period configuration for a newly deployed cell (for example, a common cell, a hyper cell, or a combined cell), the following requirements must be met:
 - When there are no other activated cells on the base station, the new cell can use static SRS period configuration, SRS period adaptation, or UE-characteristic-based SRS period adaptation.
 - When there are activated cells using SRS period adaptation on the base station, UE-characteristic-based SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. In this case, in order to enable UE-characteristic-based SRS period adaptation for the new cell, you must deactivate all activated cells using SRS period adaptation on the base station. Then, enable UE-characteristic-based SRS period adaptation or static SRS period configuration for the deactivated cells, and reactivate these cells. Lastly, enable UE-characteristic-based SRS period adaptation for the new cell.
 - When there are activated cells using UE-characteristic-based SRS period adaptation on the base station, SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. In this case, in order to enable SRS period adaptation for the new cell, you must deactivate all activated cells using UE-characteristic-based SRS period adaptation on the base station. Then, enable SRS period adaptation or static SRS period

- configuration for the deactivated cells, and reactivate these cells. Lastly, enable SRS period adaptation for the new cell.
- To change the SRS period configuration of a cell, the following requirements must be met:
 - When there are no other activated cells on the base station, the SRS period configuration of the current cell can be changed to static SRS period configuration, SRS period adaptation, or UE-characteristic-based SRS period adaptation.
 - When there are activated cells using SRS period adaptation on the base station, the SRS period configuration of the current cell cannot be directly changed to UE-characteristic-based SRS period adaptation. In this case, in order to change the SRS period configuration of the current cell to UE-characteristic-based SRS period adaptation, you must deactivate all activated cells using SRS period adaptation on the base station. Then, enable UE-characteristic-based SRS period adaptation or static SRS period configuration for the deactivated cells, and reactivate these cells. Lastly, enable UE-characteristic-based SRS period adaptation for the current cell.
 - When there are activated cells using UE-characteristic-based SRS period adaptation on the base station, the SRS period configuration of the current cell cannot be directly changed to SRS period adaptation. In this case, in order to change the SRS period configuration of the current cell to SRS period adaptation, you must deactivate all activated cells using UE-characteristic-based SRS period adaptation on the base station. Then, enable SRS period adaptation or static SRS period configuration for the deactivated cells, and reactivate these cells. Lastly, enable SRS period adaptation for the current cell.

8.3.5 Others

(TDD) SRS interference coordination based on self-contained and uplink slots:

- The cell bandwidth must range from 60 MHz to 100 MHz.
- The slot configuration must be 4:1, single-period 8:2, or dual-period 8:2. That is, the **NRDUCell.SlotAssignment** parameter must be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.
- As the 4.9 GHz frequency band is not supported, the cell frequency band cannot be n79.

(TDD) SRS interference coordination based on self-contained slots: As this function allocates SRS resources among different cells in a coordinated manner, PCIs need to be properly planned based on their mod 3 values for the cells to be enabled with this function. Otherwise, the average downlink and uplink UE throughputs (**User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**) may decrease, and the service drop rate (**SgNB-Triggered SgNB Abnormal Release Rate (CU)**) may increase after this function is enabled. As such, neighboring cells to be enabled with this function must be configured with different PCIs and have different PCI mod 3 values.

(TDD) Coherent-codebook-based scheduling: When two SRS ports are used, the value of *pusch-TransCoherence* in the *MIMO-ParametersPerBand* IE reported by UEs must be **fullCoherent**.

SRS resource adjustment for 4T UEs requires that UEs support 4T. Specifically, the value of the *maxNumberMIMO-LayersCB-PUSCH* IE reported by UEs must be

fourLayers. For details about the *maxNumberMIMO-LayersCB-PUSCH* IE, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.306 V15.3.0.

Other functions: none

8.4 Operation and Maintenance

Periodic SRS involves data configuration, activation verification, and network monitoring. Aperiodic SRS does not involve data configuration, activation verification, or network monitoring.

8.4.1 Data Configuration

8.4.1.1 Data Preparation

This function takes effect by default and no parameters are used for activation. [Table 8-4](#) describes the parameters used for function optimization.

Table 8-4 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Switch	NRDUCellSrsSrsAlgoSwitch	SRS_PERIOD_ADAPT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The SRS_PERIOD_ADAPT_SW option of this parameter specifies whether to enable SRS period adaptation. You are advised to use the recommended value.
SRS Algorithm Switch	NRDUCellSrsSrsAlgoSwitch	USER_CHARACTER_SRS_ADAPT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The USER_CHARACTER_SRS_ADAPT_SW option of this parameter specifies whether to enable UE-characteristic-based SRS period adaptation. The default setting is recommended. If UE-characteristic-based SRS period adaptation is enabled, modify the settings of related parameters as described in Setting Notes for UE-Characteristic-based SRS Period Adaptation .

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Switch	NRDUCellSrsSrsAlgoSwitch	COMMON_PUCCH_AVOID_SRS_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The COMMON_PUCCH_AVOID_SRS_SW option of this parameter specifies whether to enable SRS transmission using common PUCCH resources. You are advised to use the recommended value.
SRS Period	NRDUCellSrsSrsPeriod	None	Use the recommended value. If a short SRS period is configured, the number of UEs transmitting SRS in a cell may not reach the maximum. If a long SRS period is configured, the number of UEs transmitting SRS in a cell increases, but the performance of the features that depend on SRS measurement deteriorates.
Qci-Specific SRS Period	NRDUCellQciBearer.QciSpecificSrsPeriod	None	Use the recommended value. Scenario-specific recommended values are as follows: - For FWA or static eMBB UEs: SL80 - For moving eMBB UEs: SL10
SRS Algorithm Extension Switch	NRDUCellSrsSrsAlgoExtSwitch	SRS_INTRF_COORD_S_SLOT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The SRS_INTRF_COORD_S_SLOT_SW option of this parameter specifies whether to enable SRS interference coordination based on self-contained slots. Set this option to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Extension Switch	NRDUCellSrsSrsAlgoExtSwitch	SRS_INTRF_COORD_S_U_SLOT_SW	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. The SRS_INTRF_COORD_S_U_SLOT_SW option of this parameter specifies whether to enable SRS interference coordination based on self-contained and uplink slots. Set this option to its recommended value.
SRS Algorithm Extension Switch	NRDUCellSrsSrsAlgoExtSwitch	SRS_RES_SET_ADJ_SW	The SRS_RES_SET_ADJ_SW option of this parameter specifies whether to enable SRS resource set adjustment. The default setting is recommended.
SRS Algorithm Extension Switch	NRDUCellSrsSrsAlgoExtSwitch	SPEC_INCREASE_SW	The SPEC_INCREASE_SW option of this parameter specifies whether to enable UE number increase for SRS. Set this option to its recommended value.
SRS Algorithm Extension Switch	NRDUCellSrsSrsAlgoExtSwitch	ADAPT_SRS_INTRF_COORD_SW	The ADAPT_SRS_INTRF_COORD_SW option of this parameter specifies whether to enable adaptive SRS interference coordination. Set this option to its recommended value.
Scenario-specific SRS Resource Alloc Policy	NRDUCellSrsScenarioSrsResAllocPol	None	Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference. Use the recommended value.
UL Coverage Algorithm Switch	NRDUCellCnCovAlgo.UlCoverageAlgoSwitch	COHERENT_CODEBOOK_SCH_SW	The COHERENT_CODEBOOK_SCH_SW option of this parameter specifies whether to enable coherent-codebook-based scheduling. Set this option to its recommended value.
SRS Algorithm Switch	NRDUCellSrsSrsAlgoSwitch	UL_4T_SRS_RES_ADJ_SW	The UL_4T_SRS_RES_ADJ_SW option of this parameter specifies whether to enable SRS resource adjustment for 4T UEs. Set this option to its recommended value.

Setting Notes for UE-Characteristic-based SRS Period Adaptation

If UE-characteristic-based SRS period adaptation needs to be enabled, the settings of the following parameters must be modified to ensure cell performance. The corresponding MML command examples are provided in [8.4.1.2 Using MML Commands](#).

You are advised to set the following parameters to their default values in all scenarios:

- **NRDUCellPdsch.MaxMimoLayerNum**, specifying the maximum layer quota for downlink MIMO
- **NRDUCellPusch.InitULSinrAdjust**, specifying the initial uplink SINR adjustment value
- **gNodeBParam.CompatibilityAlgoSwitch** with the **SRS_RECONFIG_METHOD_SW** option deselected, indicating that the SRS full configuration switch is turned off

You are advised to apply the following parameter settings in all scenarios:

- Select the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter to enable SRS interference coordination based on self-contained slots.
- Select the **ADAPT_SRS_INTRF_COORD_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter to enable adaptive SRS interference coordination.
- Set the **NRDUCellPdschPrecode.CorrCalcSrsWeightValidPrd** parameter (specifying the validity period of SRS-based weights for correlation calculation) to **25**.
- Select the **SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW** option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter to enable initial MCS adjustment value optimization for SRS-based weights.
- Set the **NRDUCellDLAmc.SrsWeightInitialMcsOffset** parameter (specifying the initial MCS offset for SRS-based weights) to **0**.
- Select the **DL_MCS_ADJ_OPT_WT_CHANGE_SW** option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter to enable MCS adjustment value optimization after downlink weight type switching.
- Set the **NRDUCellDLRank.DIRankDetectCorrThld** parameter (specifying the correlation coefficient threshold for downlink rank detection) to **100**.
- Select the **MU_AMC_OPT_SW** option of the **NRDUCellULAmc.UlAmcAlgoSw** parameter to enable AMC optimization for MU-MIMO.
- Select the **SRS_SINR_AMP_OPT_SW** option of the **NRDUCellULAmc.UlAmcAlgoSw** parameter to enable SRS SINR amplitude optimization.
- Set the **NRDUCellULAmc.PuschMcsAdjVal** parameter (specifying the MCS adjustment value for the PUSCH) to **1**.

You are also advised to set additional parameters for specific scenarios as follows:

- In coverage-limited scenarios, set the **NRDUCellSrs.SrsWideBandIndexCsrs** parameter (specifying the SRS wideband index Csrs) to **63**.

Coverage is limited if the proportion of RRC_CONNECTED UEs in the overlapping area of a cell is 30% or less and the proportion of CQIs less than or equal to 6 is 30% or more.

Proportion of RRC_CONNECTED UEs in the overlapping area of a cell = $\frac{N.User.RRCCConn.Avg.CellOverlap}{N.User.RRCCConn.Avg}$

Proportion of CQIs less than or equal to 6 = $\frac{\text{Sum of } N.ChMeas.CQI.SingleCW.0 \text{ through } N.ChMeas.CQI.SingleCW.6}{\text{Sum of } N.ChMeas.CQI.SingleCW.0 \text{ through } N.ChMeas.CQI.SingleCW.15}$

- In single-user peak rate scenarios, select the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDLRank.DIRankSelAlgoSw** parameter to enable rank adaptation for CEUs performing downlink small-packet services.
For details about this function, see *Downlink Scheduling*.
- If the cell bandwidth is 100 MHz, set the **NRDUCellDLMimo.DIMuMimoSrsPreSnrThld** parameter (specifying the threshold of the SRS SINR before equalization for UEs to enter downlink MU-MIMO) to **-75**.

In addition to the preceding setting notes, the **DL_WEIGHT_RANK_MCS_FAST_ADJ_SW** option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter always takes effect regardless of the parameter setting as long as UE-characteristic-based SRS period adaptation is enabled.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Activating and deactivating SRS period adaptation/UE-characteristic-based SRS period adaptation/SRS interference coordination based on self-contained slots/SRS interference coordination based on self-contained and uplink slots will cause cell reestablishment.

Activation Command Examples

The following provides activation command examples for the sub-functions of SRS resource management.

Low-frequency TDD command examples

```
//Enabling adaptive configuration, UE-characteristic-based adjustment, or static configuration of the
transmission period for periodic SRS as required
//MML command examples for adaptive configuration of the transmission period for periodic SRS:
//Enabling SRS period adaptation (assuming that there are no other activated cells on the base station, or
there are other activated cells on the base station but no cells using UE-characteristic-based SRS period
adaptation)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-1;
//Enabling SRS period adaptation for the cell with NrDuCellId set to 0 (assuming that there is an activated
cell with NrDuCellId set to 1 using UE-characteristic-based SRS period adaptation on the base station,
which must be deactivated beforehand)
DEA NRCELL: NrCellId=1;
MOD NRDUCELLSRS: NrDuCellId=1, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-1;
ACT NRCELL: NrCellId=1;
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-1;

//MML command examples for UE-characteristic-based adjustment of the transmission period for periodic
SRS:
//Enabling UE-characteristic-based SRS period adaptation (assuming that there are no other activated cells
on the base station, or there are other activated cells on the base station but no cells using SRS period
adaptation)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=USER_CHARACTER_SRS_ADAPT_SW-1;
//Enabling UE-characteristic-based SRS period adaptation for the cell with NrDuCellId set to 0 (assuming
that there is an activated cell with NrDuCellId set to 1 using SRS period adaptation on the base station,
which must be deactivated beforehand)
DEA NRCELL: NrCellId=1;
MOD NRDUCELLSRS: NrDuCellId=1, SrsAlgoSwitch=USER_CHARACTER_SRS_ADAPT_SW-1;
ACT NRCELL: NrCellId=1;
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=USER_CHARACTER_SRS_ADAPT_SW-1;

//MML command examples for static configuration of the transmission period for periodic SRS:
//Enabling static SRS period configuration, that is, disabling SRS period adaptation for the cell and using a
fixed SRS period
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-0;
```

High-frequency TDD command examples

In the current version, high-frequency TDD supports only aperiodic SRS, which does not involve MML commands.

Optimization Command Examples

Low-frequency TDD command examples

```
//Adjusting the following parameter settings to ensure cell performance when UE-characteristic-based SRS
period adaptation is enabled
//Setting the MaxMimoLayerNum parameter to its default value in all scenarios
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_DEFAULT;
//Setting the InitUISinrAdjust parameter to its default value in all scenarios
MOD NRDUCELLPUSCH: NrDuCellId=0, InitUISinrAdjust=-8;
//Turning off the SRS full configuration switch in all scenarios
MOD GNODEBPARAM: CompatibilityAlgoSwitch=SRS_RECONFIG_METHOD_SW-0;
//Enabling SRS interference coordination based on self-contained slots in all scenarios
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_INTRF_COORD_S_SLOT_SW-1;
//Enabling adaptive SRS interference coordination in all scenarios
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=ADAPT_SRS_INTRF_COORD_SW-1;
//Changing the value of CorrCalcSrsWeightValidPrd to 25 in all scenarios
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, CorrCalcSrsWeightValidPrd=25;
//Turning on SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW in all scenarios
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW-1;
//Changing the value of SrsWeightInitialMcsOffset to 0 in all scenarios
MOD NRDUCELLDLAMC: NrDuCellId=0, SrsWeightInitialMcsOffset=0;
//Turning on DL_MCS_ADJ_OPT_WT_CHANGE_SW in all scenarios
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=DL_MCS_ADJ_OPT_WT_CHANGE_SW-1;
//Changing the value of DlRankDetectCorrThld to 100 in all scenarios
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankDetectCorrThld=100;
//Turning on MU_AMC_OPT_SW and SRS_SINR_AMP_OPT_SW, and changing the value of PuschMcsAdjVal
to 1 in all scenarios
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=MU_AMC_OPT_SW-1&SRS_SINR_AMP_OPT_SW-1,
```

```

PuschMcsAdjVal=1;
//Changing the value of SrsWideBandIndexCsrs to 63 in coverage-limited scenarios
MOD NRDUCELLSRS: NrDuCellId=0, SrsWideBandIndexCsrs=63;
//Enabling rank adaptation for CEUs performing downlink small-packet services in single-user peak rate scenarios
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SMALL_PKT_CEU_RANK_ADAPT_SW-1;
//Changing the value of DlMuMimoSrsPreSinrThld to -75 if the cell bandwidth is 100 MHz
MOD NRDUCELLDLMIMO: NrDuCellId=1, DlMuMimoSrsPreSinrThld=-75;

//Optimization command examples when the transmission period for periodic SRS is adjusted based on UE characteristics:
//Enabling UE number increase for SRS
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SPEC_INCREASE_SW-1;
//Modifying the QCI-specific SRS period
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, QciSpecificSrsPeriod=SL10;
//Modifying the scenario-specific SRS resource allocation policy
MOD NRDUCELLSRS: NrDuCellId=0, ScenarioSrsResAllocPol=CAPC_PRI;
//Enabling SRS interference coordination based on self-contained and uplink slots (only when SRS interference coordination based on self-contained slots is disabled)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_INTRF_COORD_S_U_SLOT_SW-1;
//Enabling SRS resource set adjustment
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_RES_SET_ADJ_SW-1;
//Enabling UE number increase for SRS (applicable only to a low-speed low-frequency TDD cell when UE-characteristic-based SRS period adaptation is enabled for the cell)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SPEC_INCREASE_SW-1;
//Enabling coherent-codebook-based scheduling
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=COHERENT_CODEBOOK_SCH_SW-1;
//Enabling SRS resource adjustment for 4T UEs
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=UL_4T_SRS_RES_ADJ_SW-1;

//Optimization command examples when the transmission period for periodic SRS is configured statically (only after both SRS period adaptation and UE-characteristic-based SRS period adaptation are disabled):
//Modifying the SRS period based on the network plan
MOD NRDUCELLSRS: NrDuCellId=0, SrsPeriod=SL160;
//Enabling UE number increase for SRS (applicable only to a high-speed low-frequency TDD cell when static SRS period configuration is enabled for the cell)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SPEC_INCREASE_SW-1;

```

High-frequency TDD command examples

In the current version, high-frequency TDD supports only aperiodic SRS, which does not involve MML commands.

Deactivation Command Examples

The following provides only deactivation command examples for the sub-functions. You can determine whether to restore the settings of other parameters based on actual network conditions.

Low-frequency TDD command examples

```

//Disabling SRS period adaptation for the cell
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_PERIOD_ADAPT_SW-0;
//Disabling UE-characteristic-based SRS period adaptation for the cell (only after disabling SRS resource set adjustment, SRS resource optimization, adaptive SRS interference coordination, SRS interference coordination based on self-contained slots, and SRS interference coordination based on self-contained and uplink slots for the cell, and additionally disabling UE number increase for SRS if the cell is a low-speed low-frequency TDD cell)
//Disabling SRS resource set adjustment
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_RES_SET_ADJ_SW-0;
//Disabling SRS resource optimization
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_RES_OPT_SW-0;
//Disabling adaptive SRS interference coordination
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=ADAPT_SRS_INTRF_COORD_SW-0;
//Disabling SRS interference coordination based on self-contained slots
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_INTRF_COORD_S_SLOT_SW-0;
//Disabling SRS interference coordination based on self-contained and uplink slots

```

```
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_INTRF_COORD_S_U_SLOT_SW-0;  
//Disabling UE number increase for SRS  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SPEC_INCREASE_SW-0;  
//Disabling UE-characteristic-based SRS period adaptation for the cell  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=USER_CHARACTER_SRS_ADAPT_SW-0;  
//Disabling coherent-codebook-based scheduling  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=COHERENT_CODEBOOK_SCH_SW-0;  
//Disabling SRS resource adjustment for 4T UEs  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=UL_4T_SRS_RES_ADJ_SW-0;
```

High-frequency TDD command examples

In the current version, high-frequency TDD supports only aperiodic SRS, which does not involve MML commands.

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

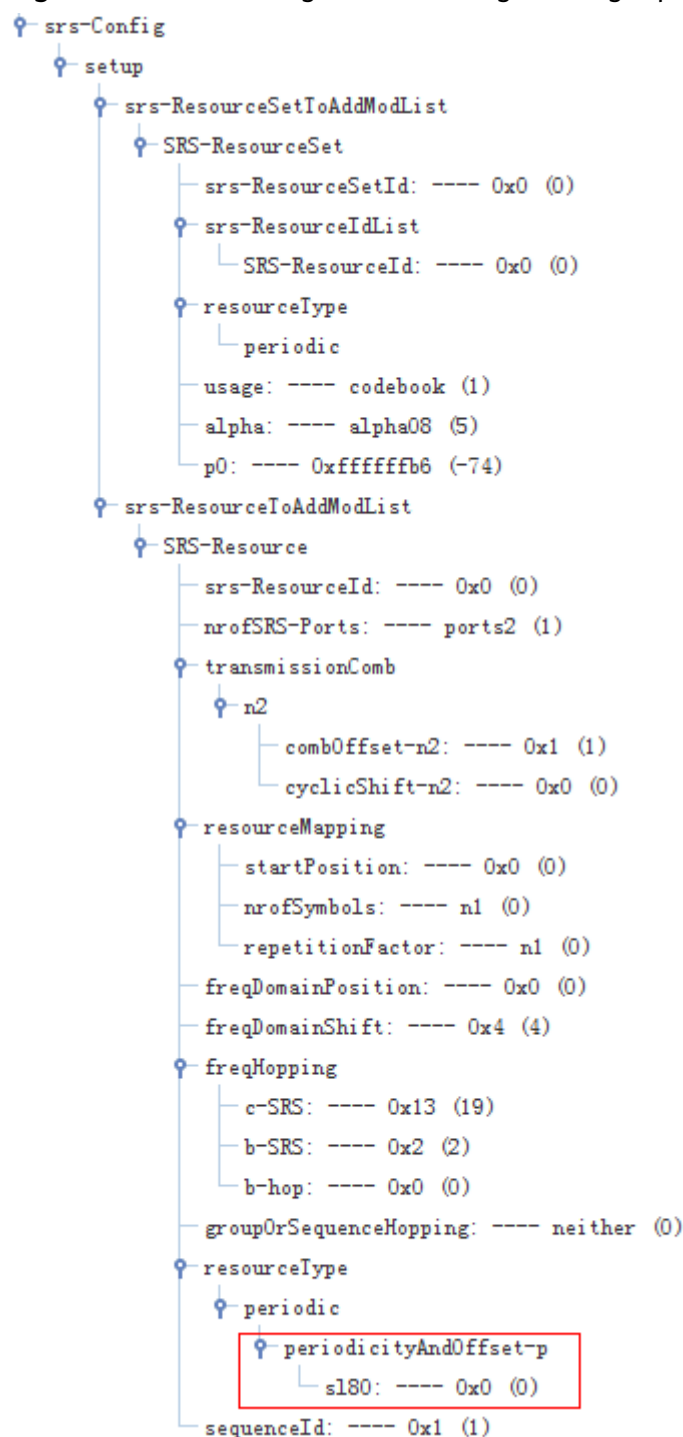
After SRS period adaptation or UE-characteristic-based SRS period adaptation is enabled, perform the following operations for activation verification:

- Step 1** Start Uu signaling tracing for cells enabled with SRS period adaptation as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** In SA networking, check the traced RRCReconfiguration message, as shown in [Figure 8-5](#). If the value of periodicityAndOffset-p180 in the *srs-Config* IE varies with different SRS periods, this function has taken effect.

NOTE

In NSA networking, check the SgNB Addition Request Acknowledge message for the IEs described in SA networking.

Figure 8-5 RRCReconfiguration message tracing report (SA networking)



----End

In TDD, after SRS interference coordination based on self-contained slots or SRS interference coordination based on self-contained and uplink slots is enabled, perform the following operations for activation verification:

- Step 1** Start Uu signaling tracing for cells enabled with SRS interference coordination based on self-contained slots or SRS interference coordination based on self-contained and uplink slots as follows: Log in to the MAE-Access and choose

Monitor > Signaling Trace > Signaling Trace Management. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace.**

Step 2 In SA networking, check the traced RRCReconfiguration message for the SRS time-domain positions indicated in the *SRS-Resource* IE of the *SRS-Config* IE in the message.

- If the SRS time-domain positions are in the self-contained slots, SRS interference coordination based on self-contained slots has taken effect.
- If the SRS time-domain positions of a cell are in the self-contained slots and those of another cell are in the uplink slots, SRS interference coordination based on self-contained and uplink slots has taken effect.

Figure 8-6 shows the *SRS-Resource* IE, in which "startPositon" indicates the SRS symbol position and "periodicityAndOffset-p" indicates the SRS slot position.

Figure 8-6 *SRS-Resource* IE

SRS-Resource ::=	SEQUENCE {
srs-ResourceId	SRS-ResourceId,
nrofSRS-Ports	ENUMERATED {port1, ports2, ports4},
ptrs-PortIndex	ENUMERATED {n0, n1 }
transmissionComb	CHOICE {
n2	SEQUENCE {
combOffset-n2	INTEGER (0..1),
cyclicShift-n2	INTEGER (0..7),
},	}
n4	SEQUENCE {
combOffset-n4	INTEGER (0..3),
cyclicShift-n4	INTEGER (0..11),
},	}
},	}
resourceMapping	SEQUENCE {
startPosition	INTEGER (0..5),
nrofSymbols	ENUMERATED {n1, n2, n4},
repetitionFactor	ENUMERATED {n1, n2, n4},
},	}
freqDomainPosition	INTEGER (0..67),
freqDomainShift	INTEGER (0..268),
freqHopping	SEQUENCE {
c-SRS	INTEGER (0..63),
b-SRS	INTEGER (0..3),
},	}
b-hop	INTEGER (0..3),
},	}
groupOrSequenceHopping	ENUMERATED { neither, groupHopping, sequ
resourceType	CHOICE {
aperiodic	SEQUENCE {
...	...
},	}
semi-persistent	SEQUENCE {
periodicityAndOffset-sp	SRS-PeriodicityAndOffset,
...	...
},	}
periodic	SEQUENCE {
periodicityAndOffset-p	SRS-PeriodicityAndOffset,
...	...
},	}
},	}

----End

Observe the following performance indicators to check whether each sub-function of SRS channel management has taken effect.

Table 8-5 Indicators related to the sub-functions involved in SRS resource management

Indicator	Formula
Average uplink cell throughput	Cell Uplink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average SRS interference	N.SRS.NI.Avg
Uplink IBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$
Downlink IBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink RBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$
Downlink RBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink PRB usage	N.PRB.UL.Used.Avg/N.PRB.UL.Avail.Avg
Downlink PRB usage	N.PRB.DL.Used.Avg/N.PRB.DL.Avail.Avg

Indicator	Formula
Average number of downlink MU-MIMO layers	$N.ChMeas.MIMO.DL.Pair.Layer / N.ChMeas.MIMO.DL.Pair.PRB$
Average downlink rank	$SUM(k \times N.PDSCH.InitTbDL.Rank.k) / SUM(N.PDSCH.InitTbDL.Rank.k)$ $k = 1 \text{ to } 4$
Traffic volume proportion of downlink small-packet services	$N.ThpVol.DL.LastSlot / N.ThpVol.DL$
Average uplink CCE aggregation level	$(1 \times N.CCE.UL.AggLvl1Num + 2 \times N.CCE.UL.AggLvl2Num + 4 \times N.CCE.UL.AggLvl4Num + 8 \times N.CCE.UL.AggLvl8Num + 16 \times N.CCE.UL.AggLvl16Num) / (N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)$
Service drop rate	$N.QosFlow.AbnormRel / (N.QosFlow.AbnormRel + N.QosFlow.NormRel) \times 100\%$
Average uplink MCS index	$(0 \times N.ChMeas.PUSCH.MCS.0 + 1 \times N.ChMeas.PUSCH.MCS.1 + 2 \times N.ChMeas.PUSCH.MCS.2 + 3 \times N.ChMeas.PUSCH.MCS.3 + \dots + 27 \times N.ChMeas.PUSCH.MCS.27 + 28 \times N.ChMeas.PUSCH.MCS.28) / (N.ChMeas.PUSCH.MCS.0 + N.ChMeas.PUSCH.MCS.1 + N.ChMeas.PUSCH.MCS.2 + N.ChMeas.PUSCH.MCS.3 + \dots + N.ChMeas.PUSCH.MCS.27 + N.ChMeas.PUSCH.MCS.28)$
Average downlink MCS index	$(0 \times N.ChMeas.PDSCH.MCS.0 + 1 \times N.ChMeas.PDSCH.MCS.1 + 2 \times N.ChMeas.PDSCH.MCS.2 + 3 \times N.ChMeas.PDSCH.MCS.3 + \dots + 27 \times N.ChMeas.PDSCH.MCS.27 + 28 \times N.ChMeas.PDSCH.MCS.28) / (N.ChMeas.PDSCH.MCS.0 + N.ChMeas.PDSCH.MCS.1 + N.ChMeas.PDSCH.MCS.2 + N.ChMeas.PDSCH.MCS.3 + \dots + N.ChMeas.PDSCH.MCS.27 + N.ChMeas.PDSCH.MCS.28)$
Uplink cell-level data volume	$N.ThpVol.UL$
Downlink cell-level data volume	$N.ThpVol.DL$
Average uplink throughput of CEUs	$(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$

8.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

9 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- *CoMP*
- *Co-MM (Low-Frequency TDD)*
- *DRX*
- *FWA*
- *Hyper Cell*
- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *MIMO (TDD)*
- *mmWave Beam Management (High-Frequency TDD)*
- *RedCap*
- *URLLC*
- *5G Networking and Signaling*
- *Beam Management*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Positioning*
- *Device-Pipe Synergy*
- *Multi-Frequency Smart Aggregation*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *High Speed Mobility*
- *Interference Avoidance*

- *Power Control*
- *Energy Conservation and Emission Reduction*
- *Scalable Bandwidth*
- *3D Networking Experience Improvement*
- *Standards Compliance*
- *Cell Combination*
- *Cell Management*
- *Remote Interference Management (Low-Frequency TDD)*
- *UE Power Saving*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Turbo Uplink (Low-Frequency TDD)*
- *Fusion Cell (Low-Frequency TDD)*
- *UL and DL Decoupling*
- *VoNR*
- *Service-differentiated Experience Guarantee*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Emergency Communications Solution (Low-Frequency TDD)*
- *Super Uplink*
- *mmWave Super Coverage (High-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*